

Manual

SMA-CMS Customizing Suite for SMA SMA-CMS 8.1

Version 1.0.23049

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1 SMA-CMS Customizing Suite for SMA

Purpose

The development license in version 8.1 is the basis for the custom development of SMA applications:

- You can create applications according to the user's requirements
- You can integrate a user-specific vocabulary
- You can configure your own input dialogs

2 SMA Installation

2.1 Notes on installation instructions

This document is only suitable for trained experts.

2.2 Further applicable documents

Installation instructions - Standard HYDRA MW 3.0:

Inst_hydra_mw30.pdf

Installation instructions - Standard HYDRA MW 2.1:

Inst_hydra_mw21.pdf

Installation instructions - MOC 3.0 services with MW 2.0/2.1:

inst_moc30-mw20-21.pdf

2.3 Requirements

2.3.1 SMA

- The **HYDRA 8 (MW 3.x, MW 4.0pe SP15)** system must be installed and running (alternative: MW2.1 with corresponding requirements see "2.4.7.2 Applications for MW2.1")
- **Windows Server 2008 R2 / Windows Server 2012 Standard / Windows Server 2016 Standard / Windows Server 2019 Standard** as SMA host
(a joint installation with HYDRA 8 on one server is possible)
- **IIS 7.5 (or higher)**
(Internet Information Services 7.5 is included in Windows Server 2008 R2)
- **.NET Framework 4.5.2 (or higher)**
You can find the installation package at <http://support.microsoft.com/kb/2901907> (or - if available – on the HYDRA8 server
[%HYDRADIR%\admttools\dotnet\dotNetFx452.exe](#))
- **asp.net** server pages are enabled
- SMA installation package **SMA_x64.zip**



SMA checks licenses. You must have purchased additional products in order to use the SMA functions.

- **Language and culture settings**

Smooth server operation can only be guaranteed with the following language and culture settings (e.g. SMA does not support Turkish as the software, L&L, cannot cope with this language and does not work as expected):

- 1) Language *German* with culture *German*
- 2) Language *English* with culture *English/USA*
- Other languages/cultures have not yet been released.

- **LICENSE with MW2.1**

Important: With MW 2.1 every SMA client occupies a HYD-KE license (console license). If you use SMA as a time and attendance client, a corresponding number of licenses should be available.

- You can use an HTML5 capable browser as a client (IE >= 10). We generally **recommend using a current browser**.
 - We recommend using the current Google Chrome browser due to its excellent performance.

2.3.2 Reporting

- We have converted the time sheet to **List&Label** (no separate installation necessary).

2.4 Installation

2.4.1 .NET Framework 4.5.2

Microsoft .NET Framework 4.5.2 is a highly compatible, direct update of Microsoft .NET Framework 4 and Microsoft .NET Framework 4.5. You can use the offline package in cases when the web installation is not possible due to a missing internet connection.

You can download the offline package from:

<http://support.microsoft.com/kb/2901907>

If available, you can find the offline package on the HYDRA 8 server:

[%HYDRADIR%\admttools\dotnet\dotNetFx452.exe\)](#)

2.4.2 IIS

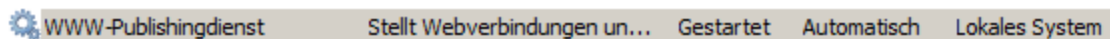
2.4.2.1 Windows 2008 R2

Check if the Microsoft Internet Information Services (IIS) are installed and started:

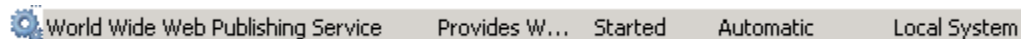
 **Start – Administration – Services**

The service "**WWW-Publishingdienst**" (German Windows) or "**World Wide Web Publishing Service**" (English Windows) must be available and started:

German:



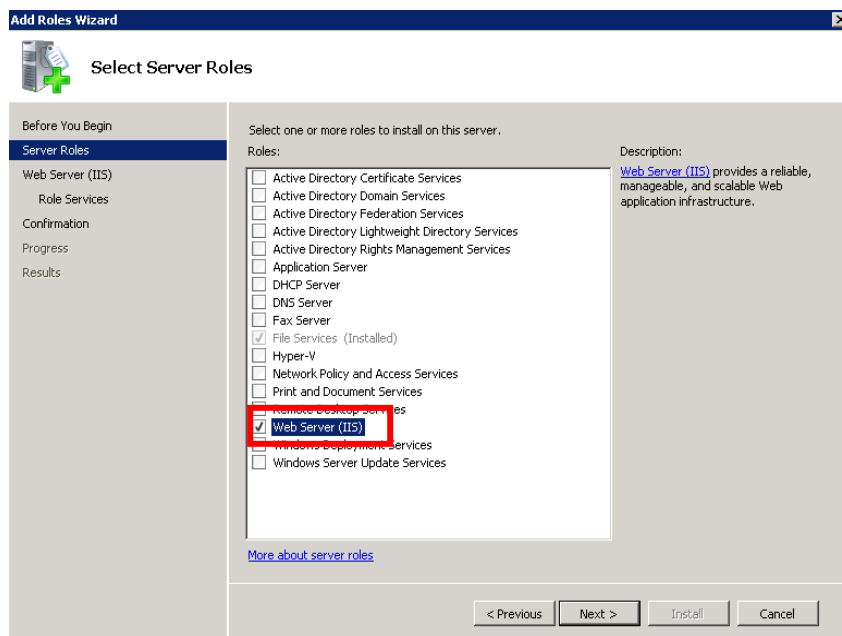
English:



If IIS is not available, proceed with the installation as follows:

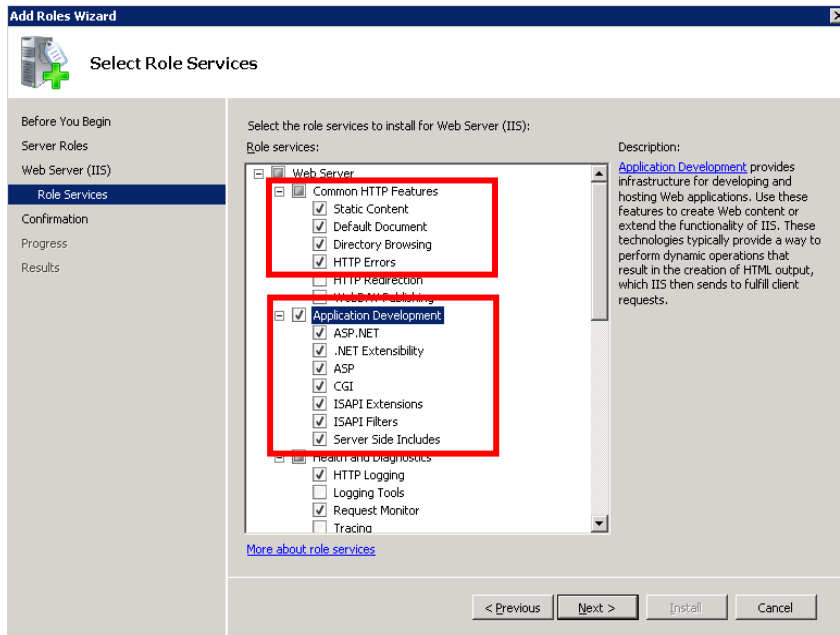
 **Start – Administration – Server Manager – Roles – Add roles**

Select server role "**Web server (IIS)**":

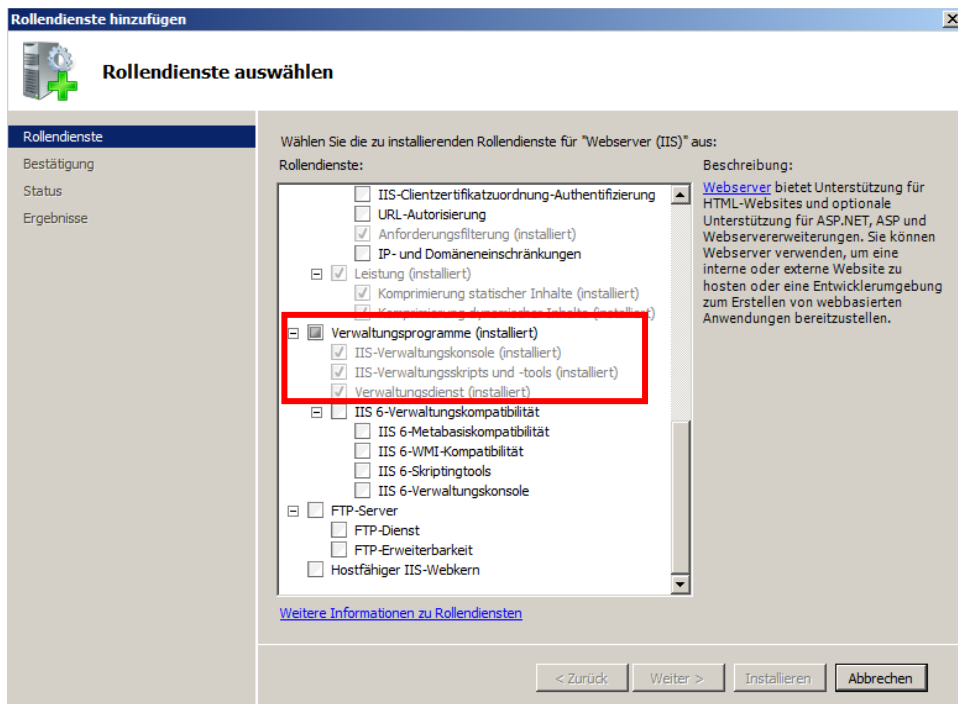


Next

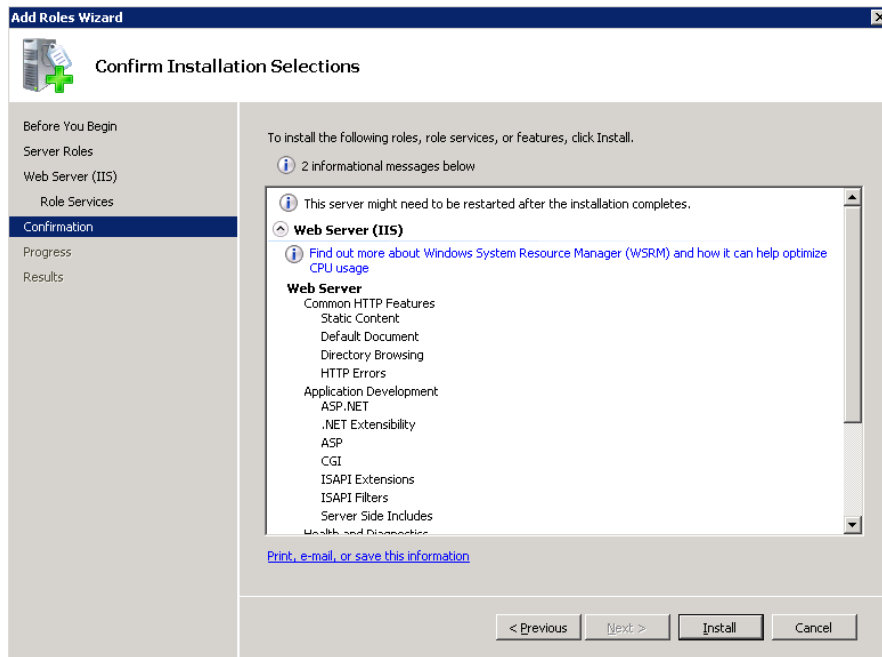
Select the role services "**Application development**":



Administration tools must be installed.



Next



Install

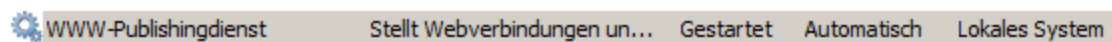
2.4.2.2 Windows Server 2012 Standard

Check if Microsoft Internet Information Services (IIS) are installed and started:
Go to *Administration*:

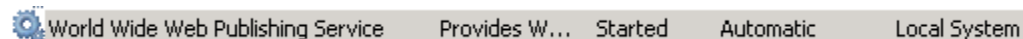
Start → Administration → Services

The service "**WWW-Publishingdienst**" (German Windows) or "**World Wide Web Publishing Service**" (English Windows) must be available and started.

German:



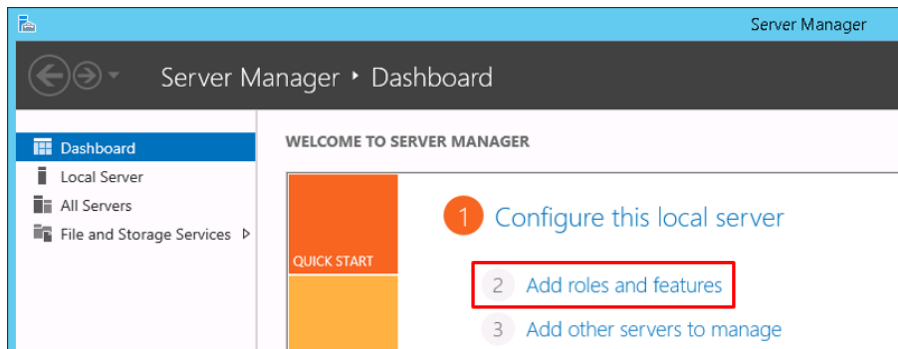
English:



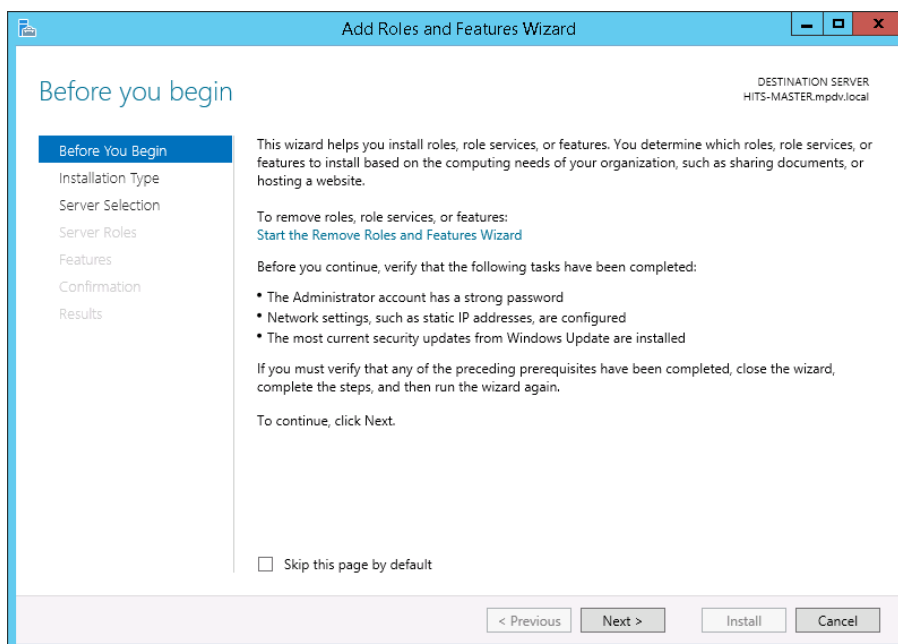
If IIS is not available, install it as described below:

Start → Administration → Server Manager

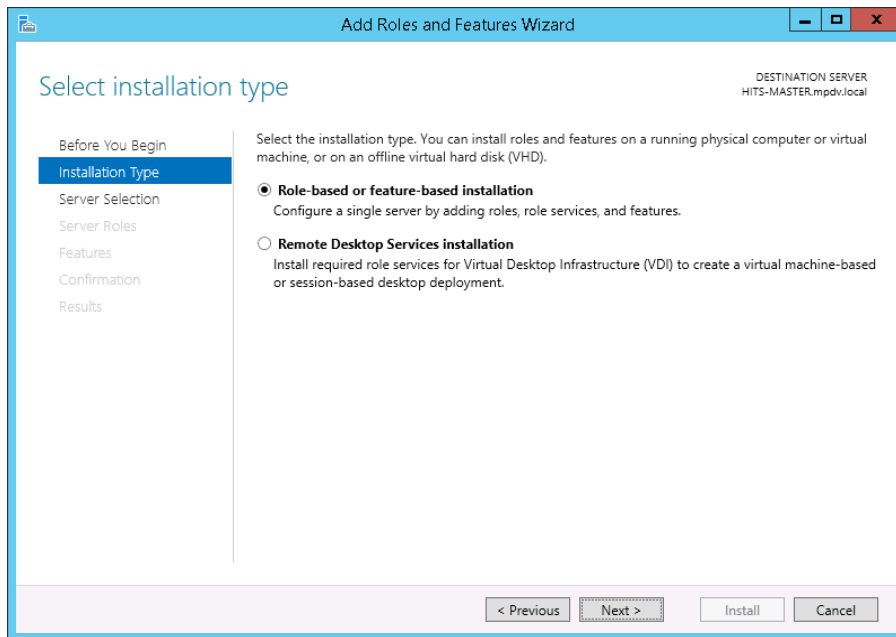
Select the option: "Add roles and features":



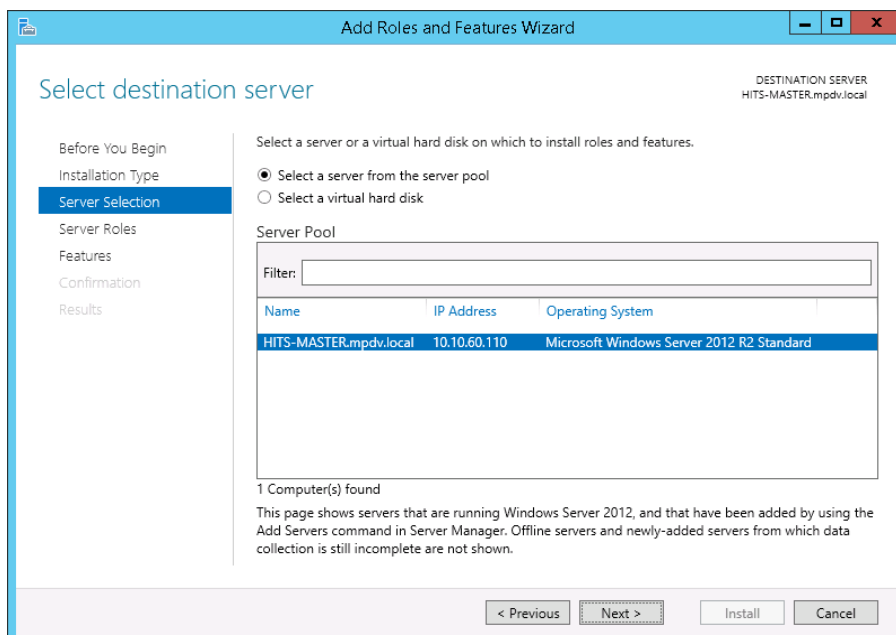
A wizard opens to add roles and features:



Next



Next

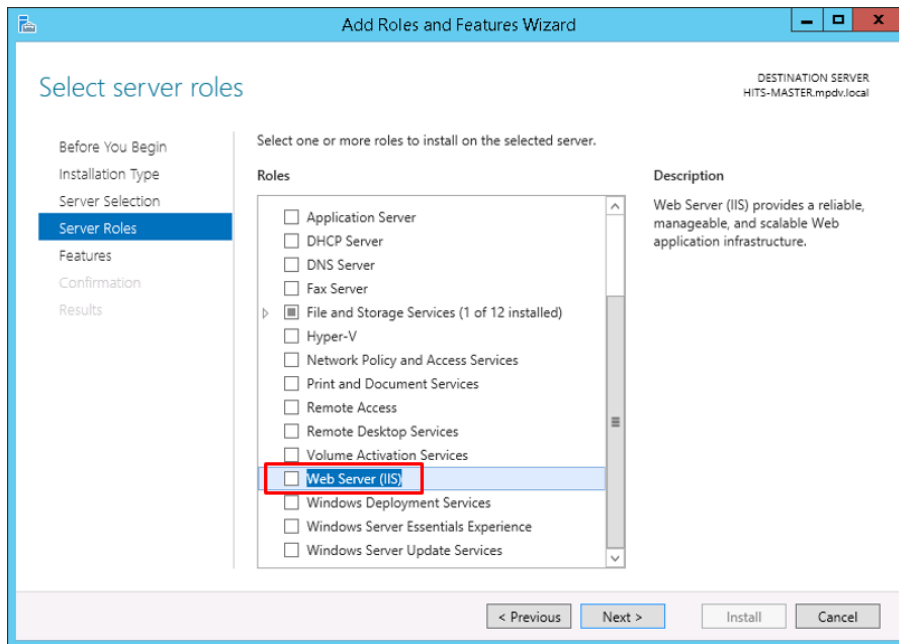


Select

your

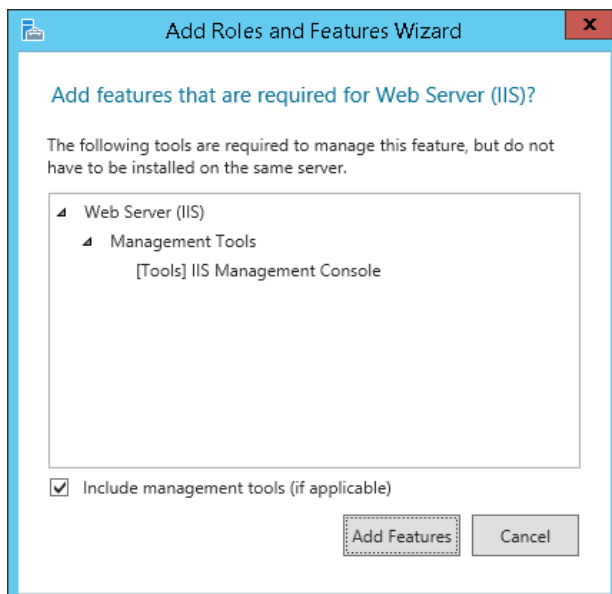
server.

Next



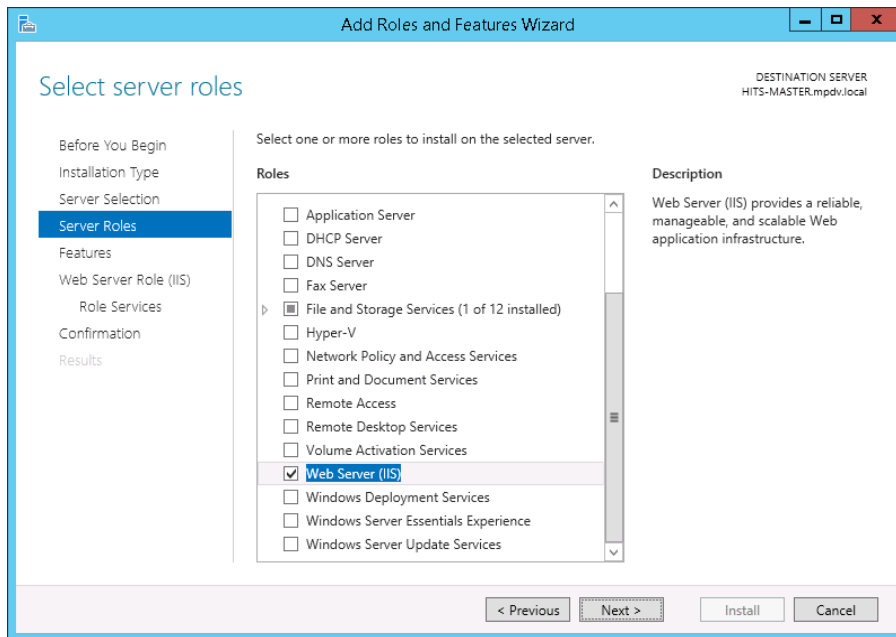
Select server role "Web server (IIS)".

A new window opens:



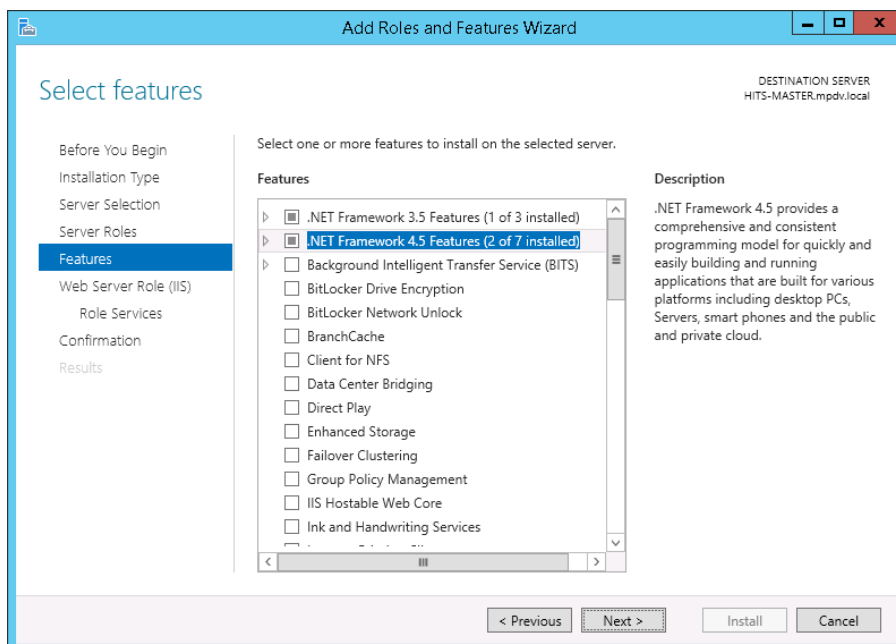
Check: "Include management tools (if applicable)"

Add features

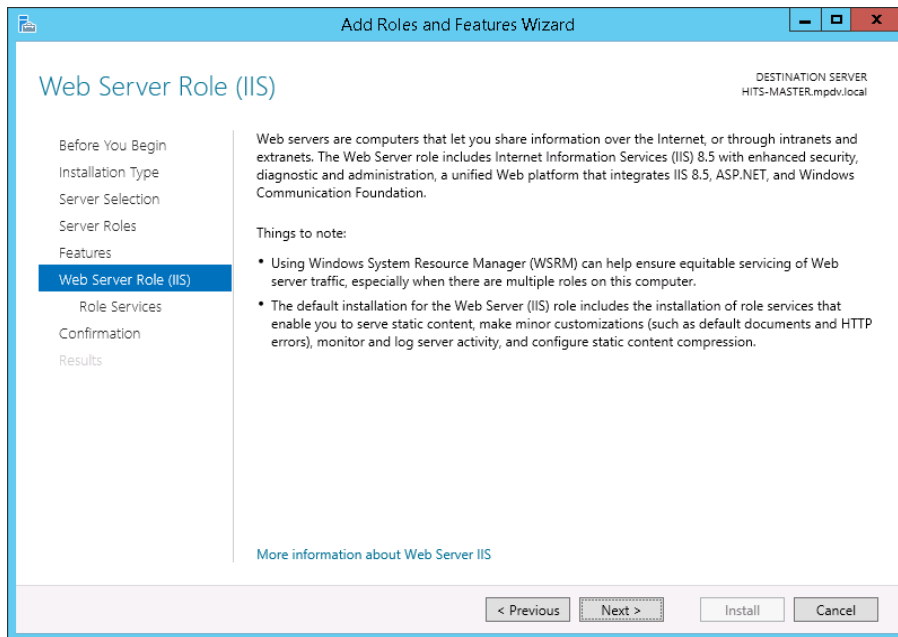


Next

Make sure the .NET Frameworks are installed:

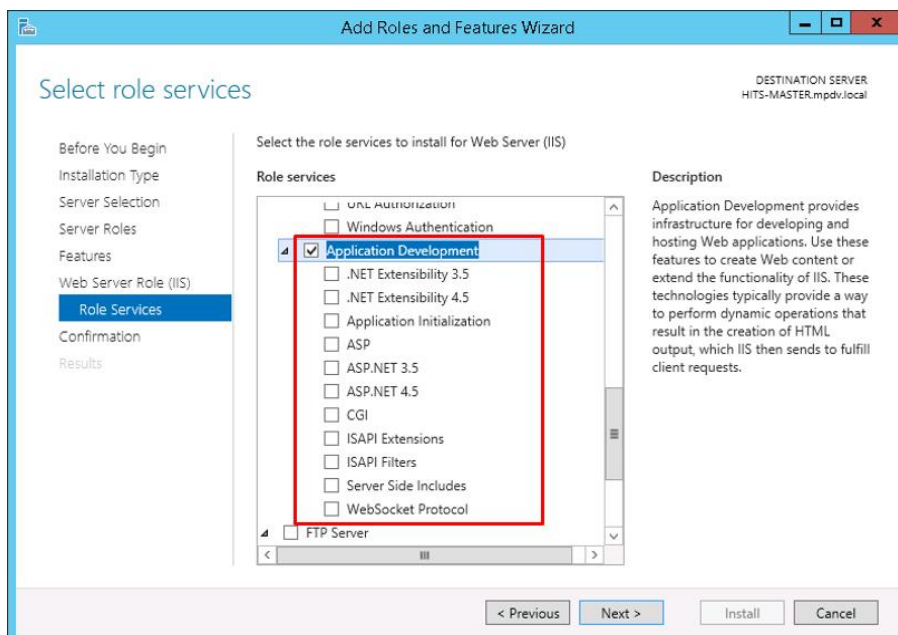


Next



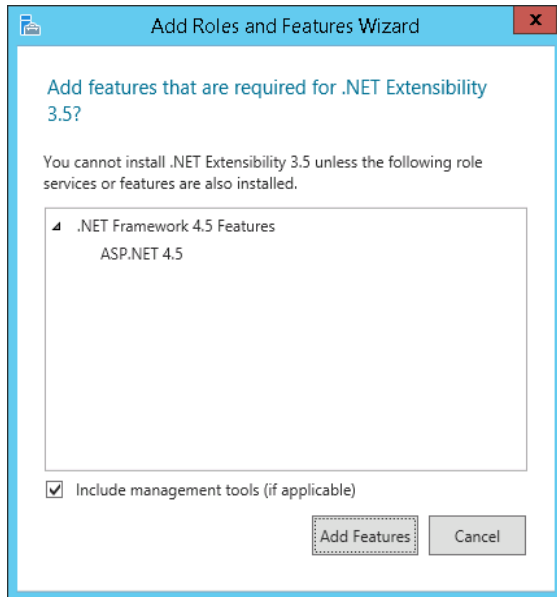
Next

Select "**Application development**" including all sub-options in addition to already selected services:

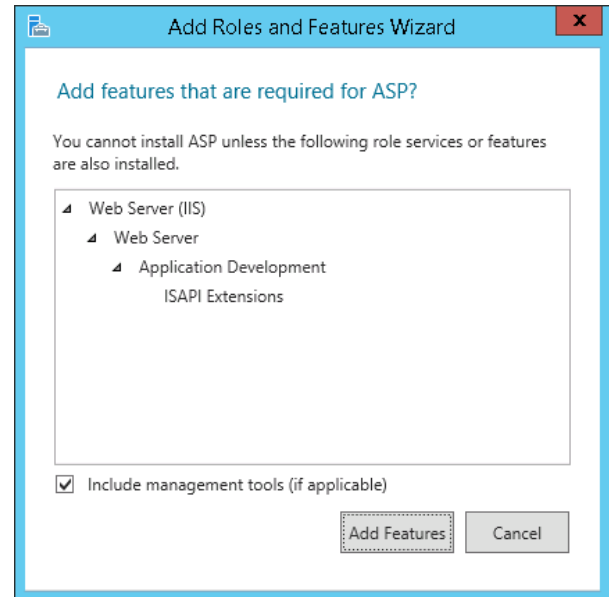


Add the following features when checking individual sub-options:

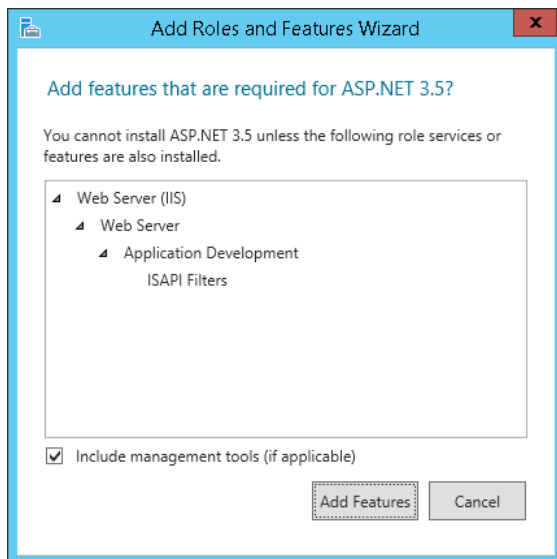
When prompted, add the following features:



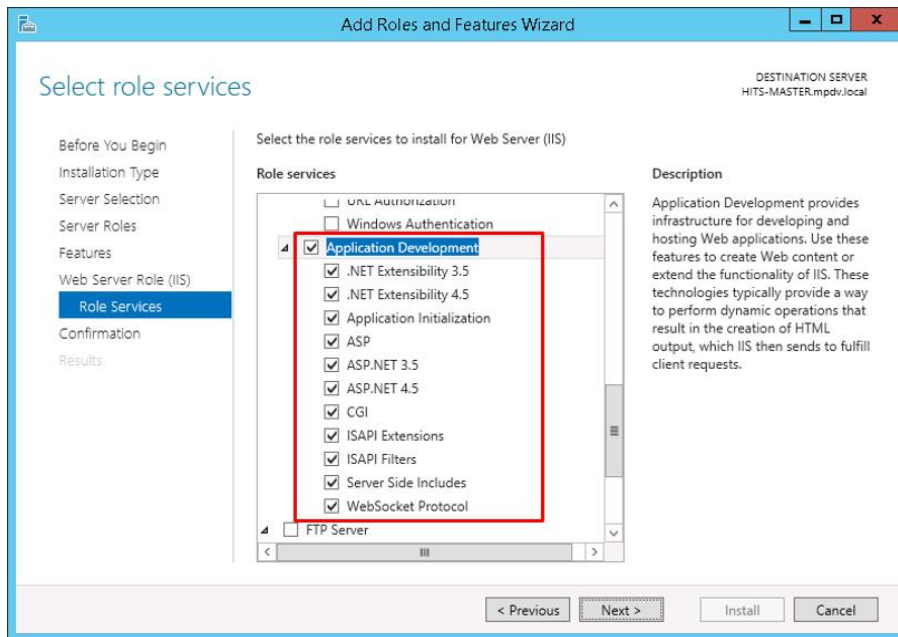
Add features



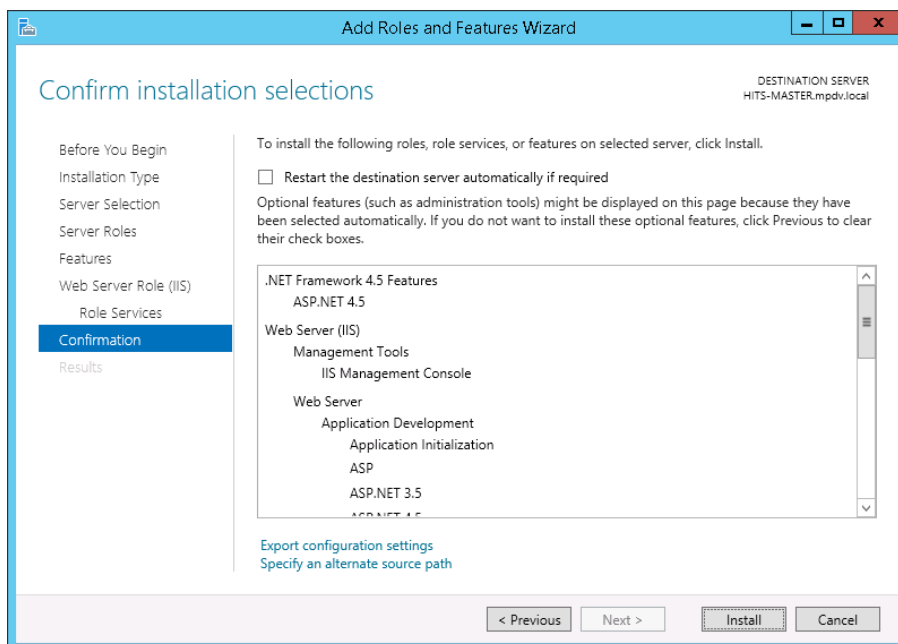
Add features



Add features

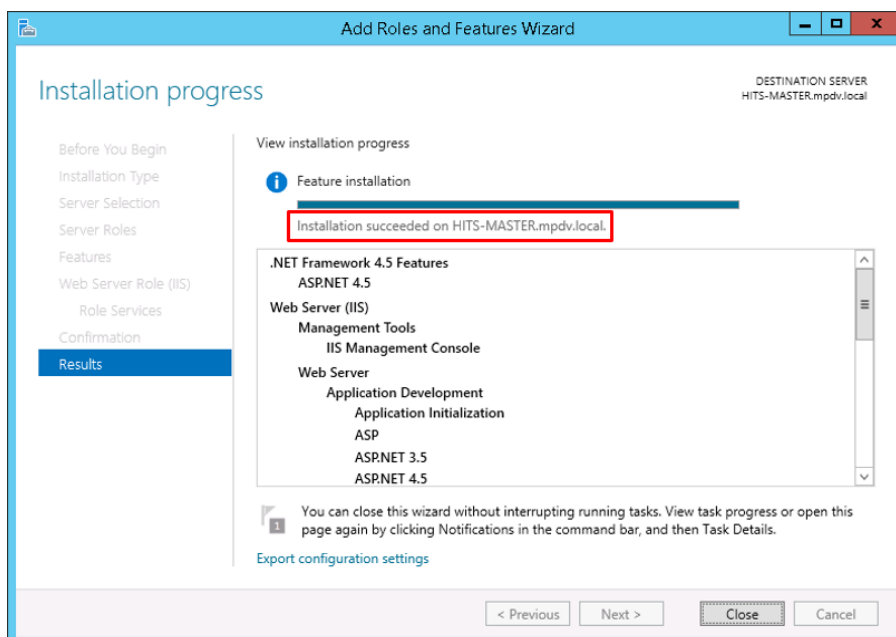
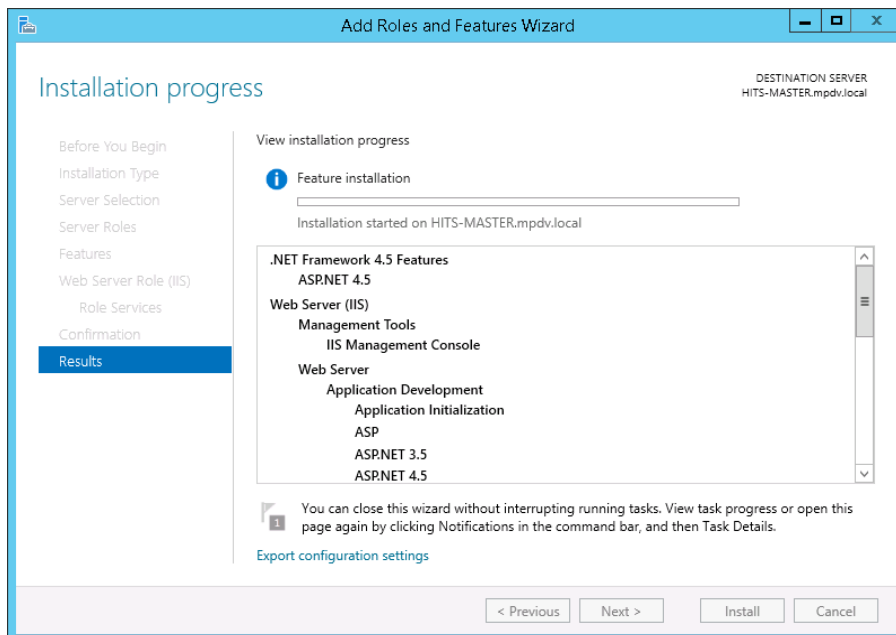


Next



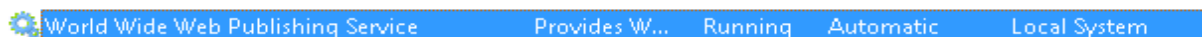
Install

Installation in progress:



Close (after successful installation)

The following Windows service should now be available:



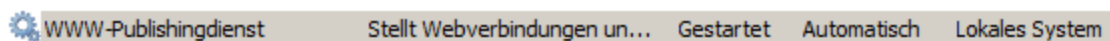
2.4.2.3 Windows Server 2016 Standard

Check if Microsoft Internet Information Services (IIS) are installed and started:
Go to *Administration*:

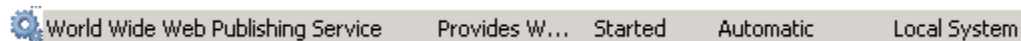
Start → Services

The service "**WWW-Publishingdienst**" (German Windows) or "**World Wide Web Publishing Service**" (English Windows) must be available and started.

German:



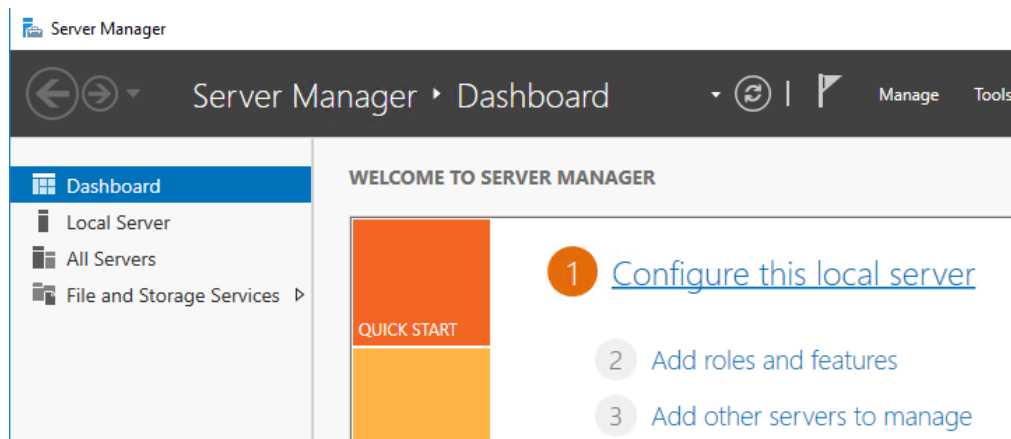
English:



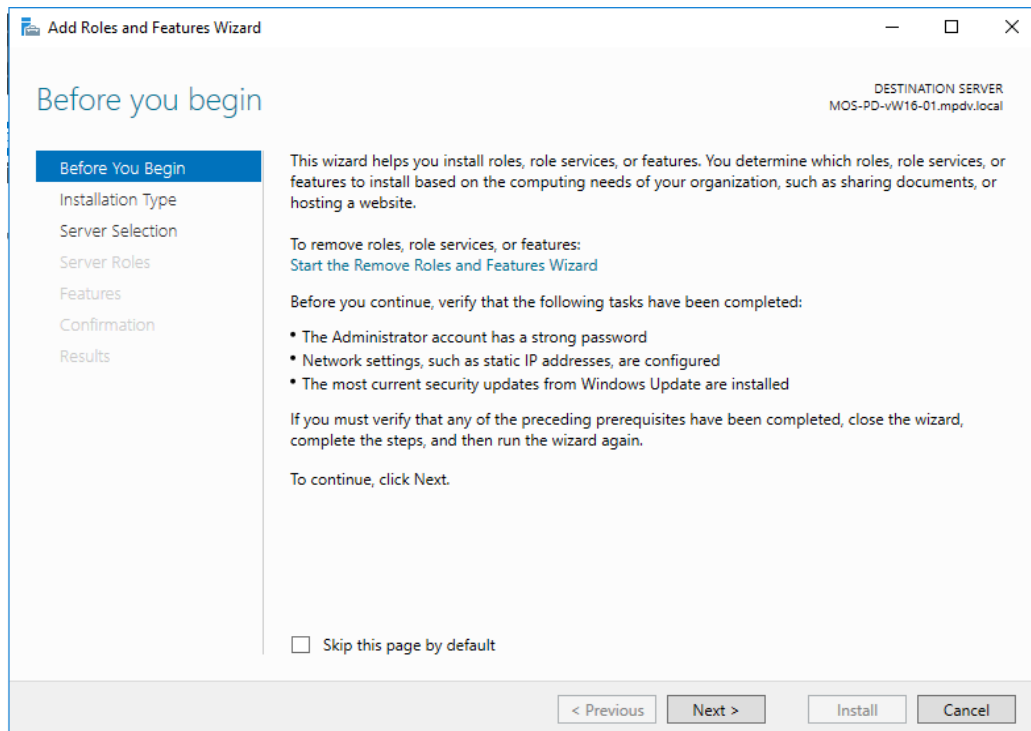
If IIS is not available, install it as described below:

Start → Server manager

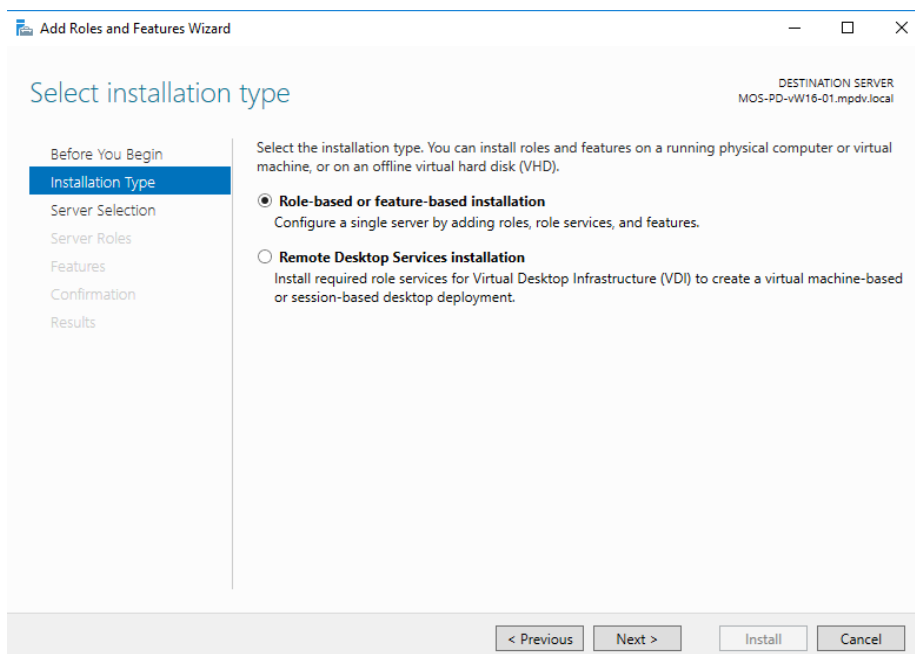
Select the option: "Add roles and features":



A wizard opens to add roles and features:



Next



Next

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
MOS-PD-vW16-01.mpdv.local

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
MOS-PD-vW16-01.mpdv...	10.10.60.128	Microsoft Windows Server 2016 Standard

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous Next > Install Cancel

Select your server.

Next

Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
MOS-PD-vW16-01.mpdv.local

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☐ DNS Server
- ☐ Fax Server
- ☒ File and Storage Services (2 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V
- ☐ MultiPoint Services
- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☒ Web Server (IIS)
- ☐ Windows Deployment Services
- ☐ Windows Server Essentials Experience
- ☐ Windows Server Update Services

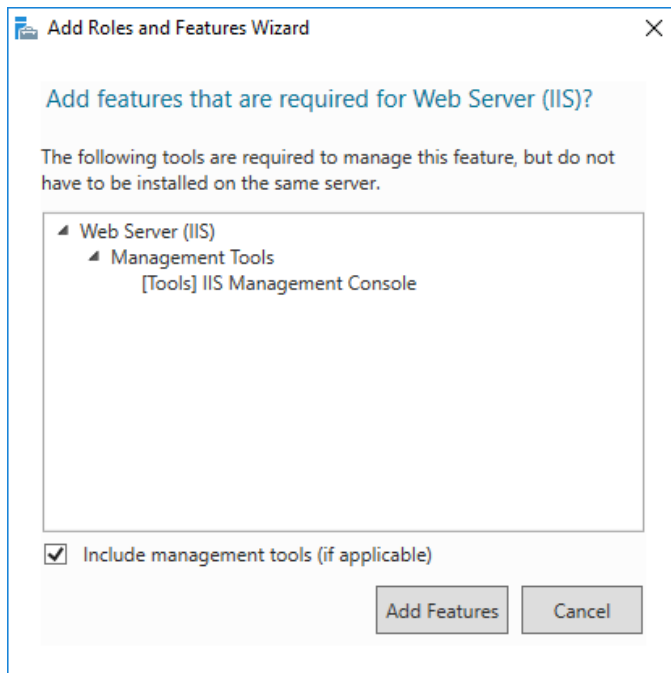
Description

Web Server (IIS) provides a reliable, manageable, and scalable Web application infrastructure.

< Previous Next > Install Cancel

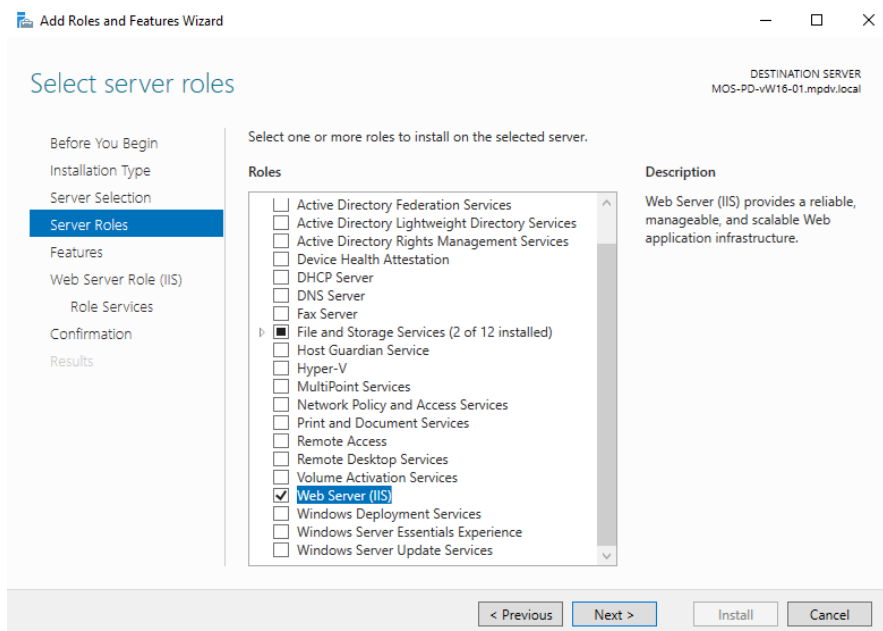
Select server role "Web server (IIS)".

A new window opens:



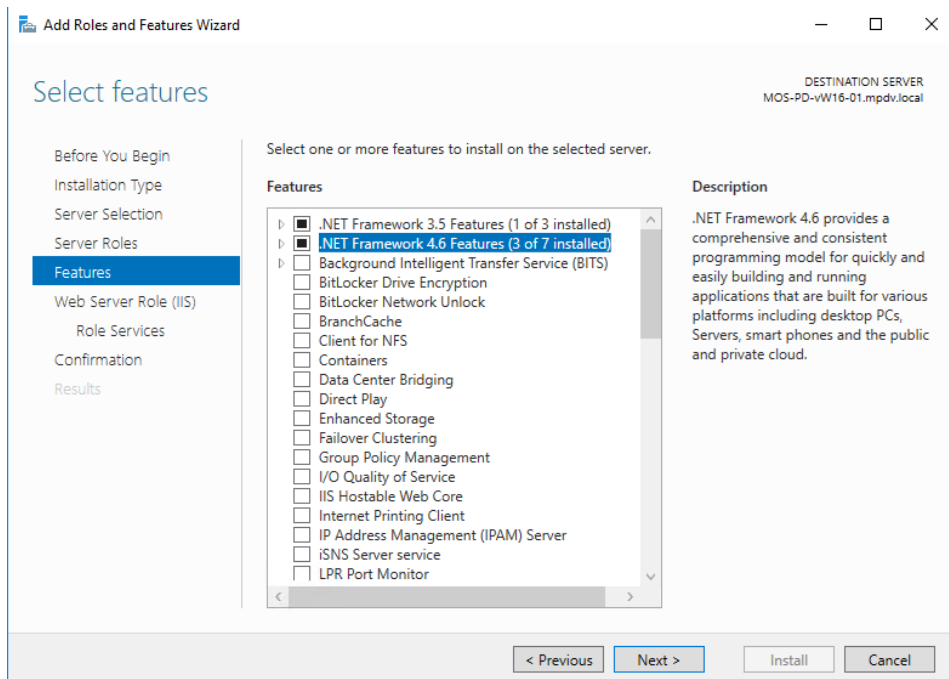
Check: "Include management tools (if applicable)"

Add features

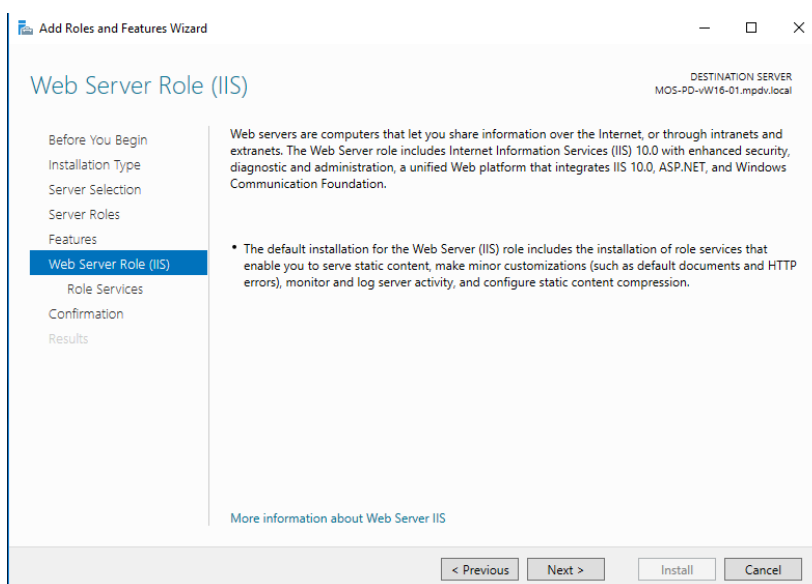


Next

Make sure the .NET Frameworks are installed:

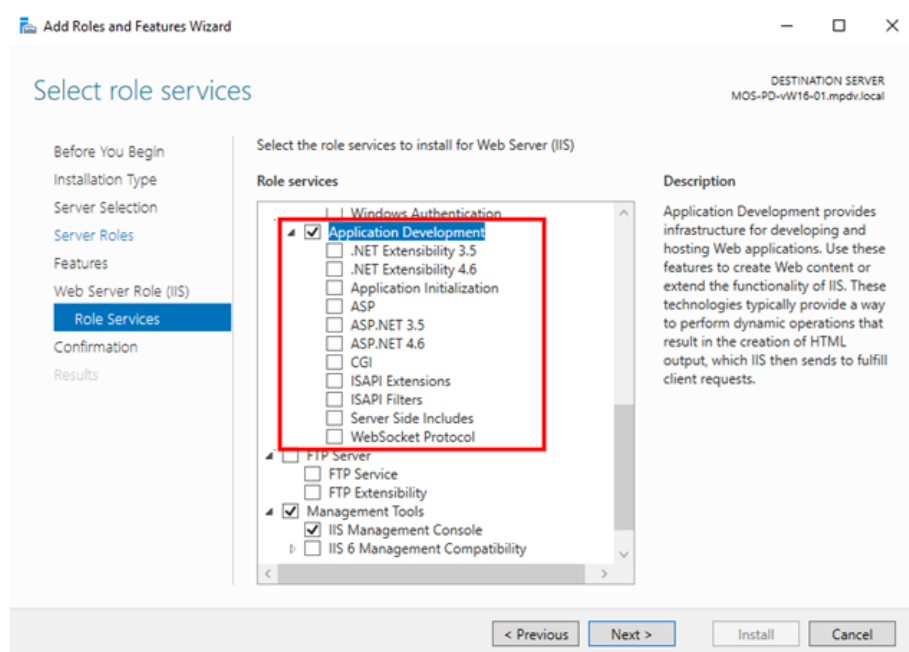


Next



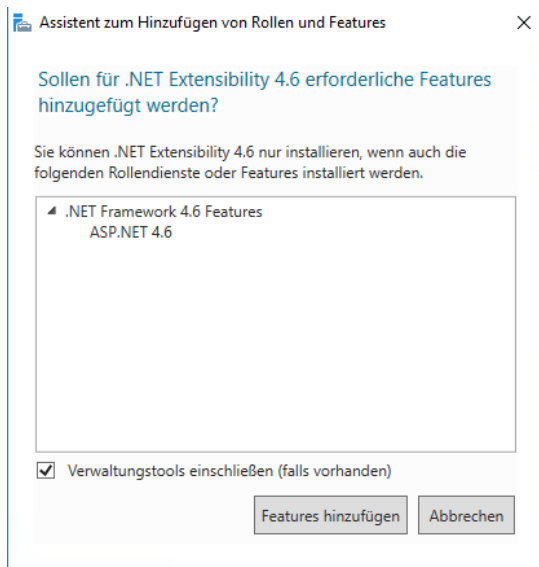
Next

Select "**Application development**" including all sub-options in addition to already selected services:

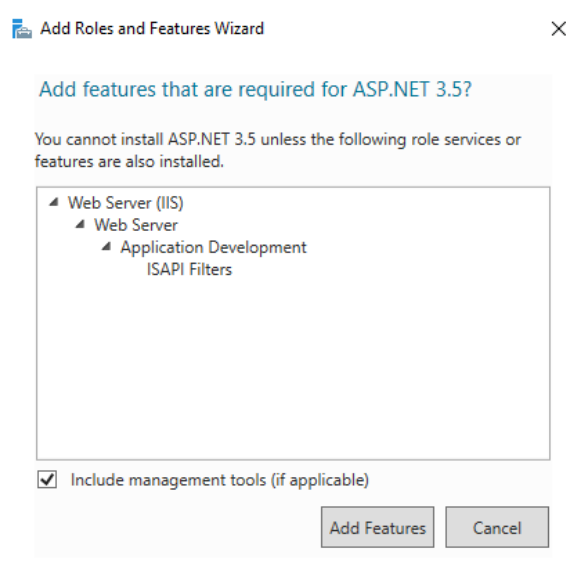


Add the following features, when checking individual sub-options:

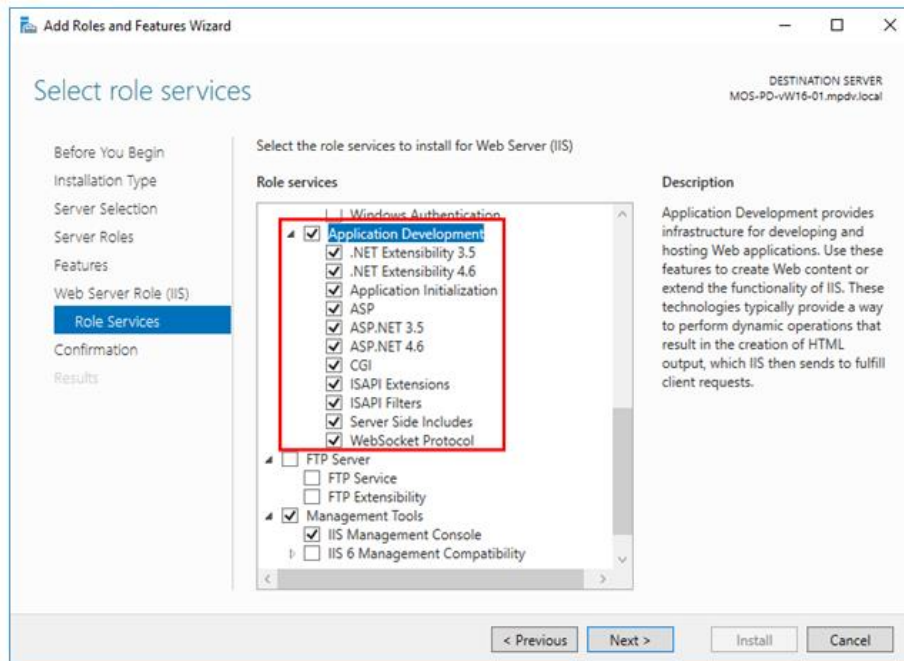
When prompted, add the following features: Example:



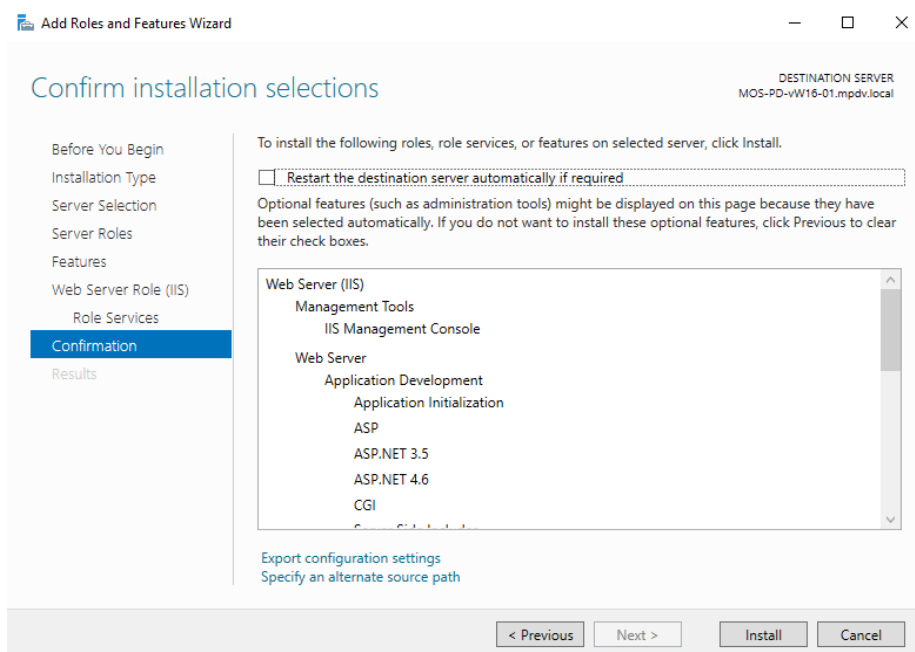
Add features



Add features

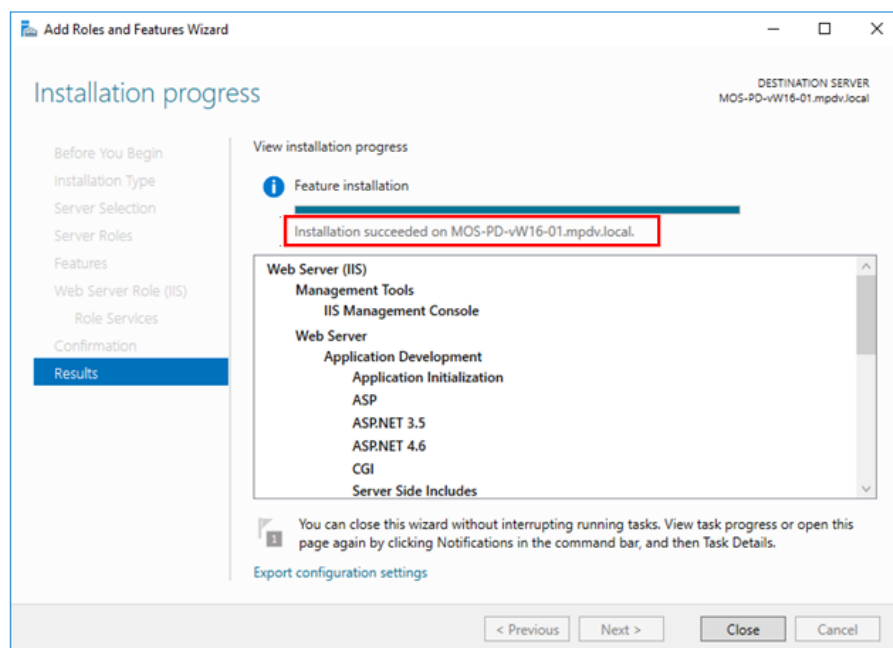
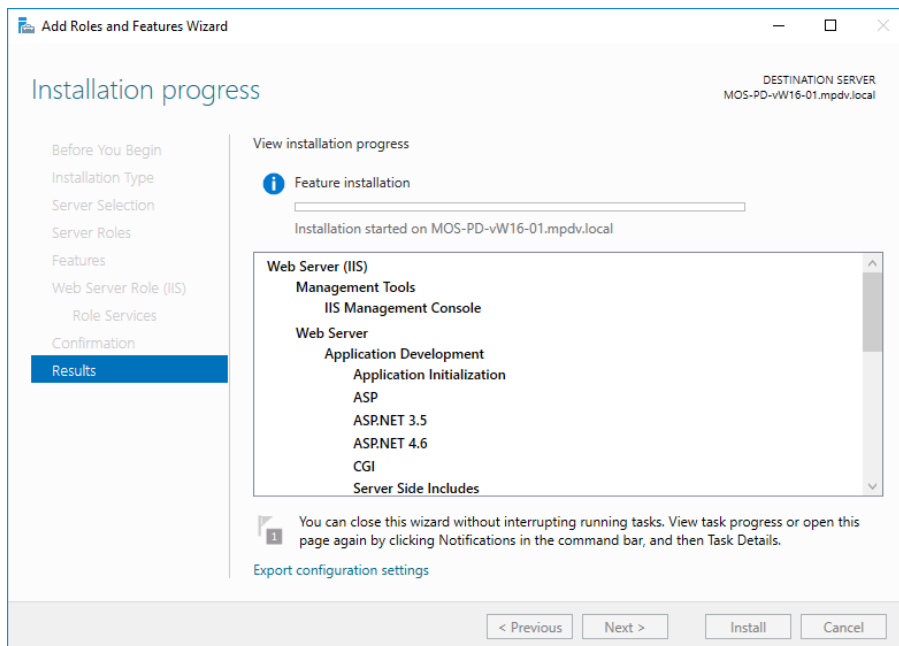


Next



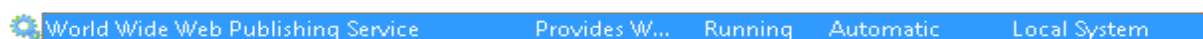
Install

Installation in progress:



Close (after successful installation)

The following Windows service should now be available:



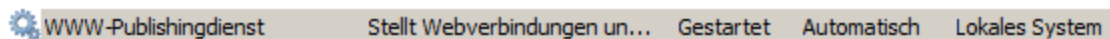
2.4.2.4 Windows Server 2019 Standard

Check if Microsoft Internet Information Services (IIS) are installed and started:
Go to *Administration*:

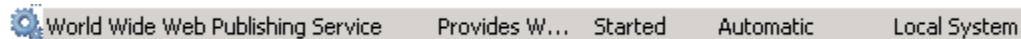
Start → Services

The service "**WWW-Publishingdienst**" (German Windows) or "**World Wide Web Publishing Service**" (English Windows) must be available and started.

German:



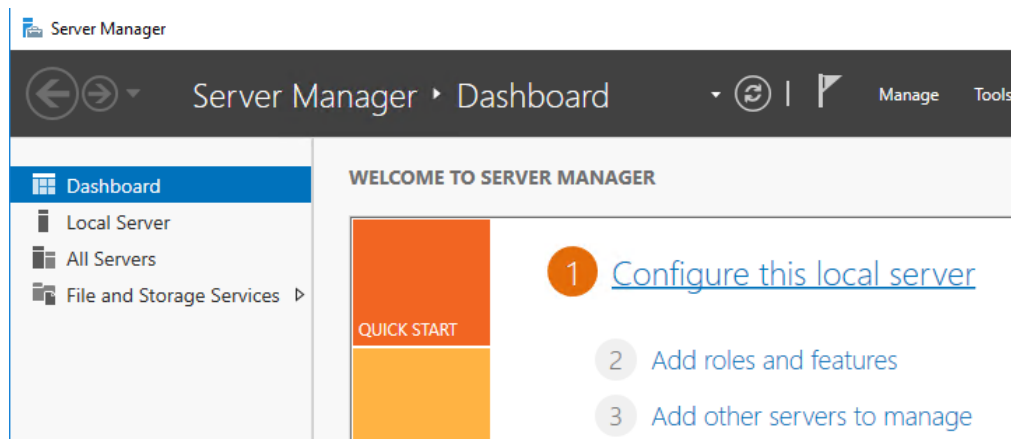
English:



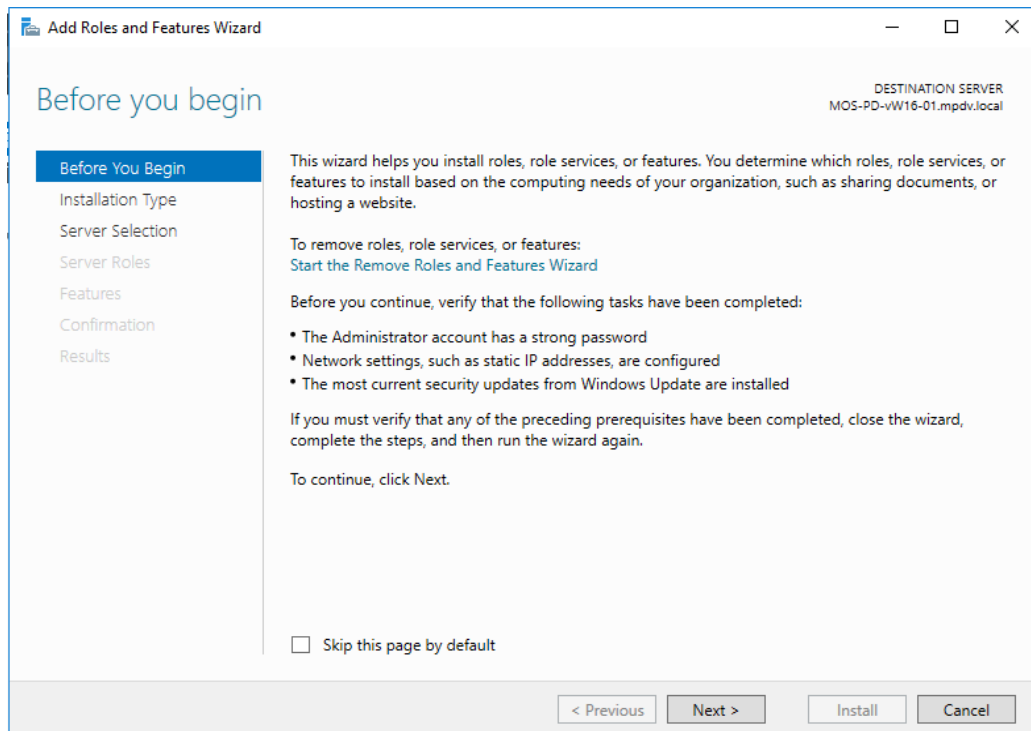
If IIS is not available, install it as described below:

Start → Server manager

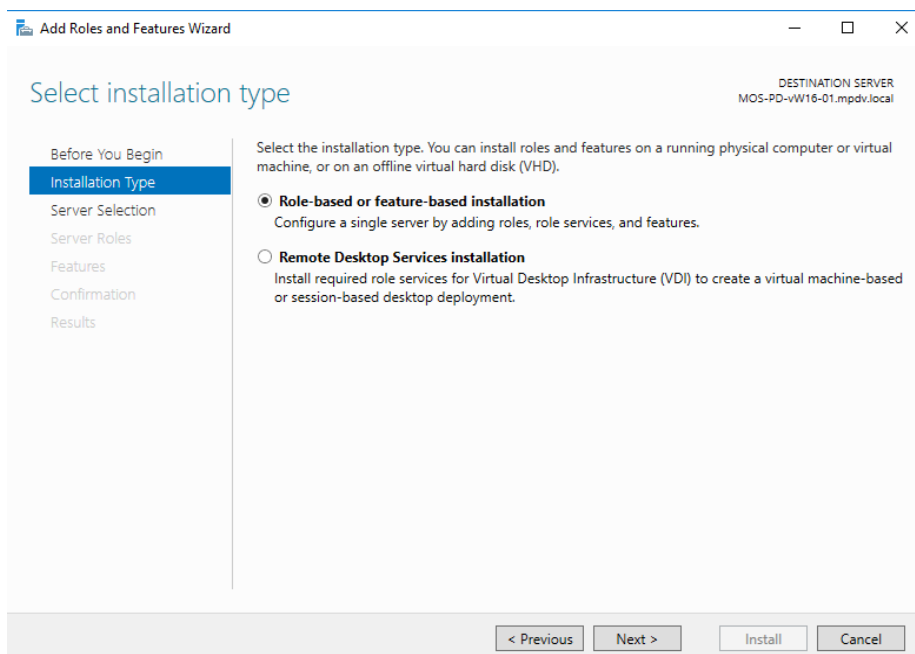
Select the option: "Add roles and features":



A wizard opens to add roles and features:



Next



Next

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
MOS-PD-vW16-01.mpdv.local

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
MOS-PD-vW16-01.mpd...	10.10.60.128	Microsoft Windows Server 2016 Standard

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous Next > Install Cancel

Select your server.

Next

Add Roles and Features Wizard

Select server roles

DESTINATION SERVER
MOS-PD-vW16-01.mpdv.local

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☐ DNS Server
- ☐ Fax Server
- ☒ File and Storage Services (2 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V
- ☐ MultiPoint Services
- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☒ Web Server (IIS)
- ☐ Windows Deployment Services
- ☐ Windows Server Essentials Experience
- ☐ Windows Server Update Services

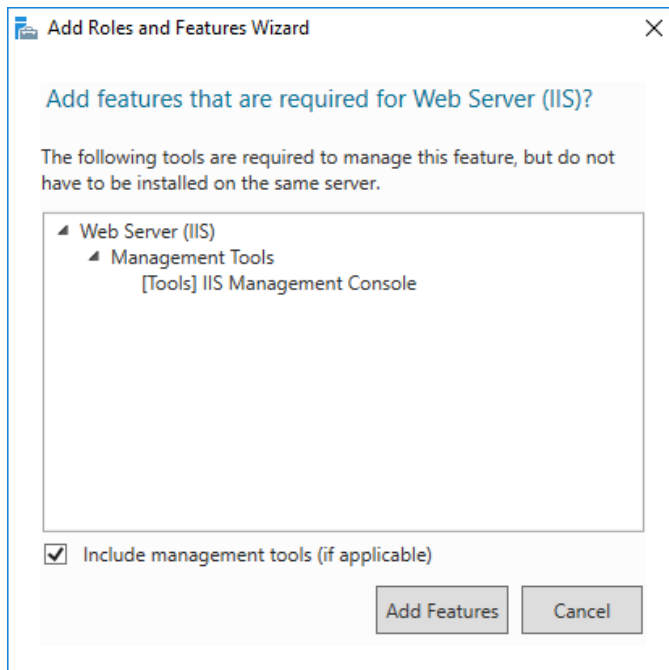
Description

Web Server (IIS) provides a reliable, manageable, and scalable Web application infrastructure.

< Previous Next > Install Cancel

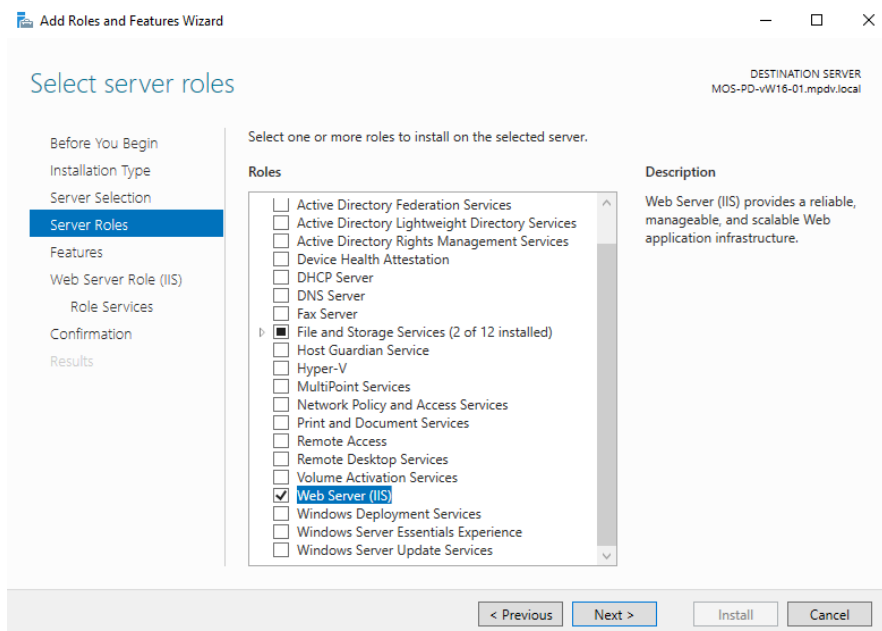
Select server role "Web server (IIS)".

A new window opens:



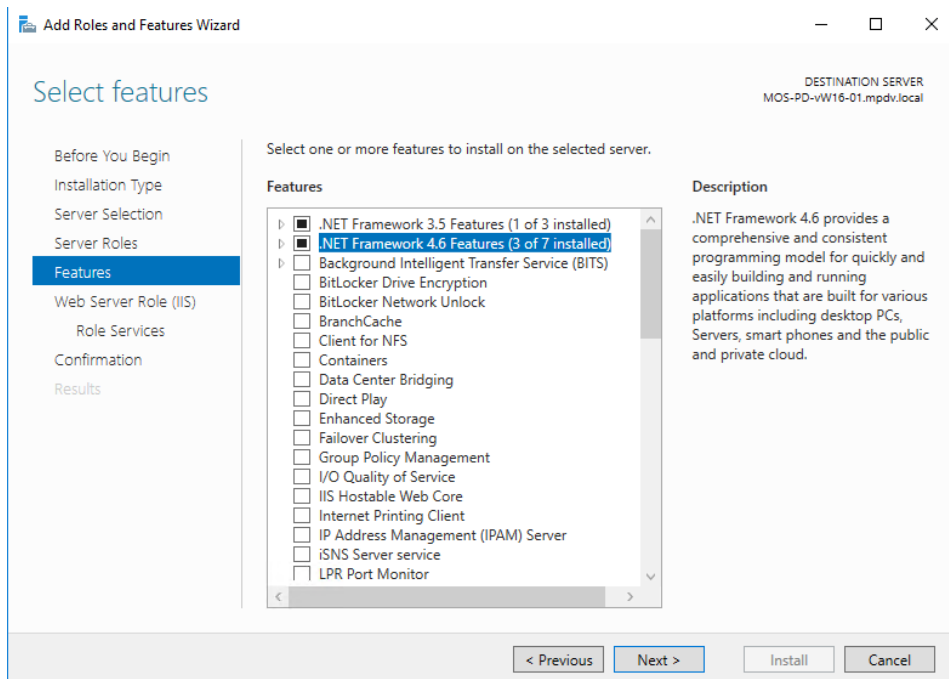
Check: "Include management tools (if applicable)"

Add features

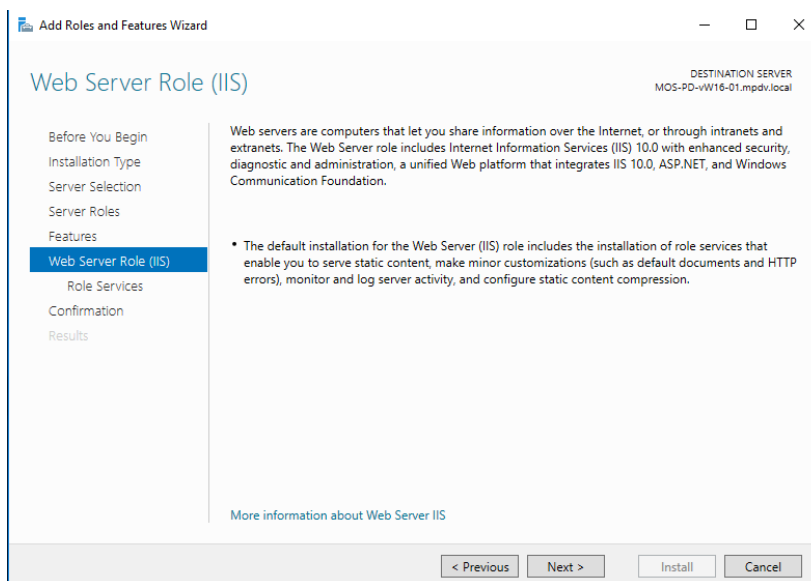


Next

Make sure the .NET Frameworks are installed:

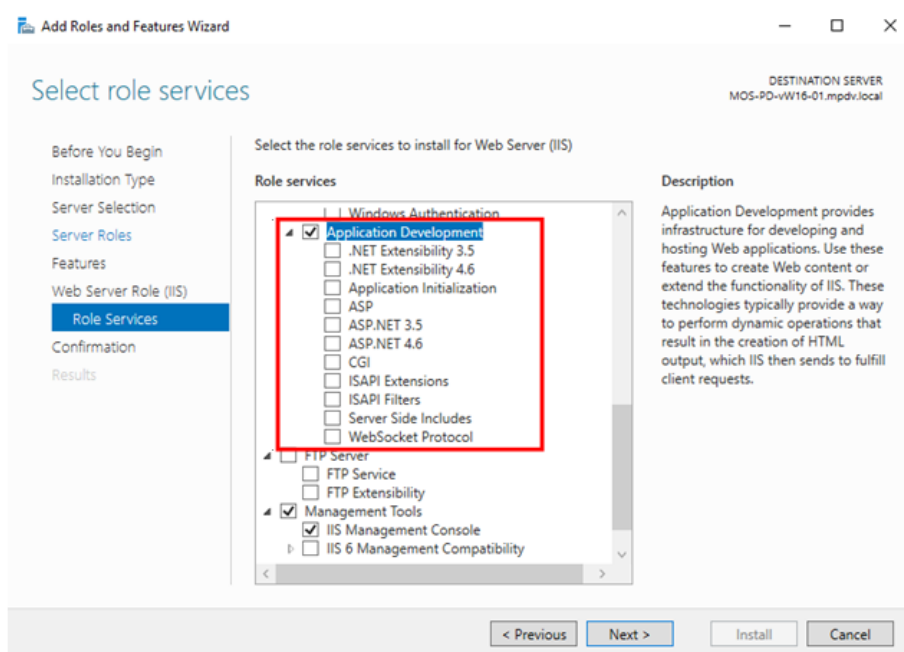


Next



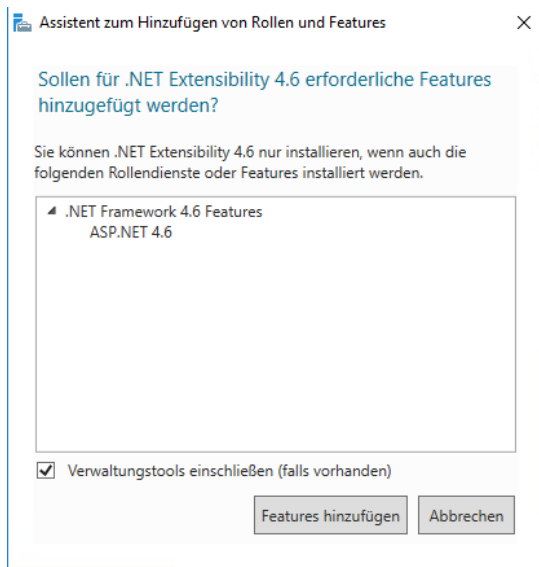
Next

Select "**Application development**" including all sub-options in addition to already selected services:

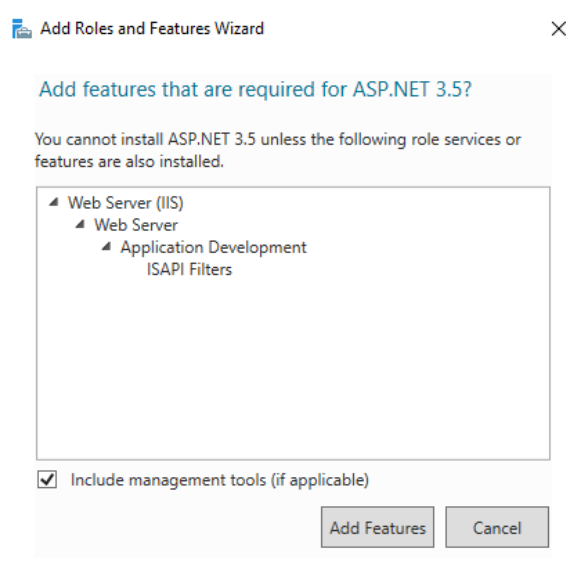


Add the following features, when checking individual sub-options:

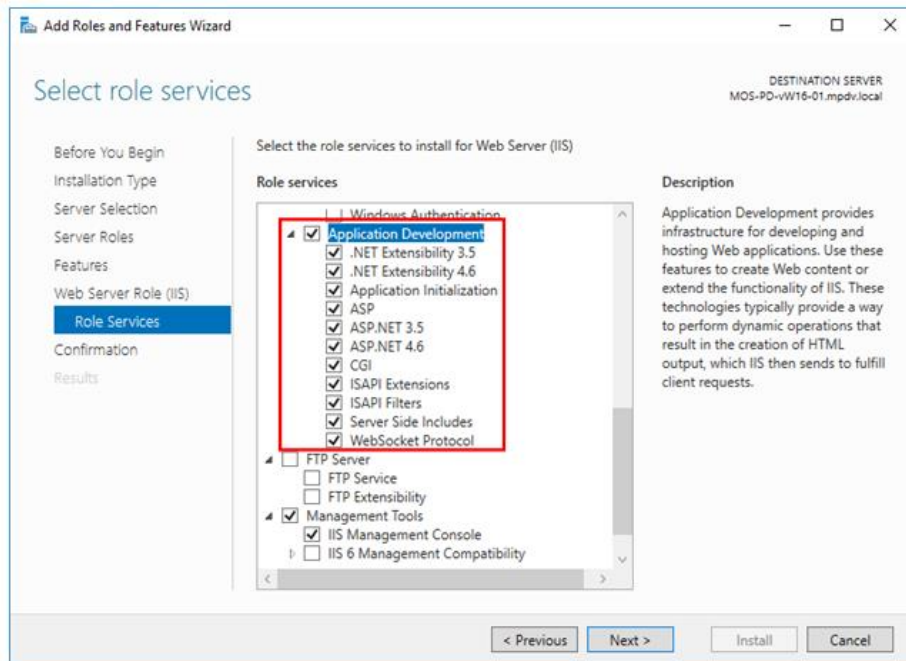
When prompted, add the following features: Example:



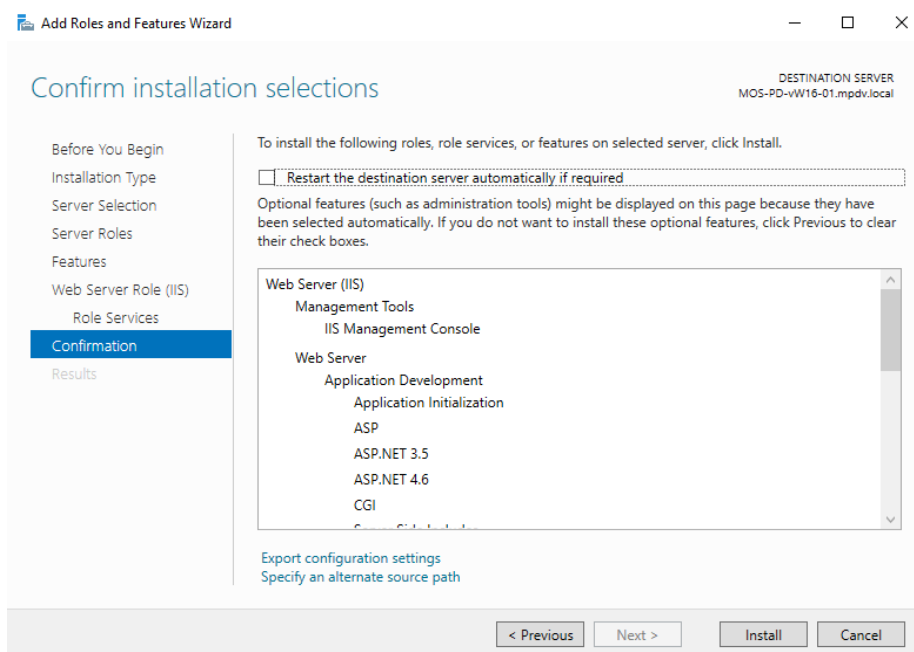
Add features



Add features

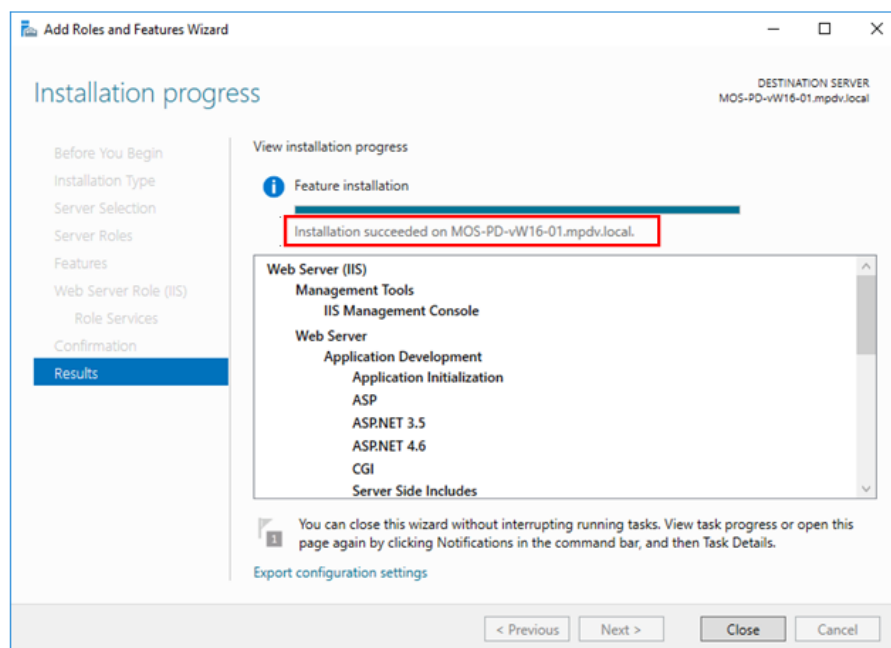
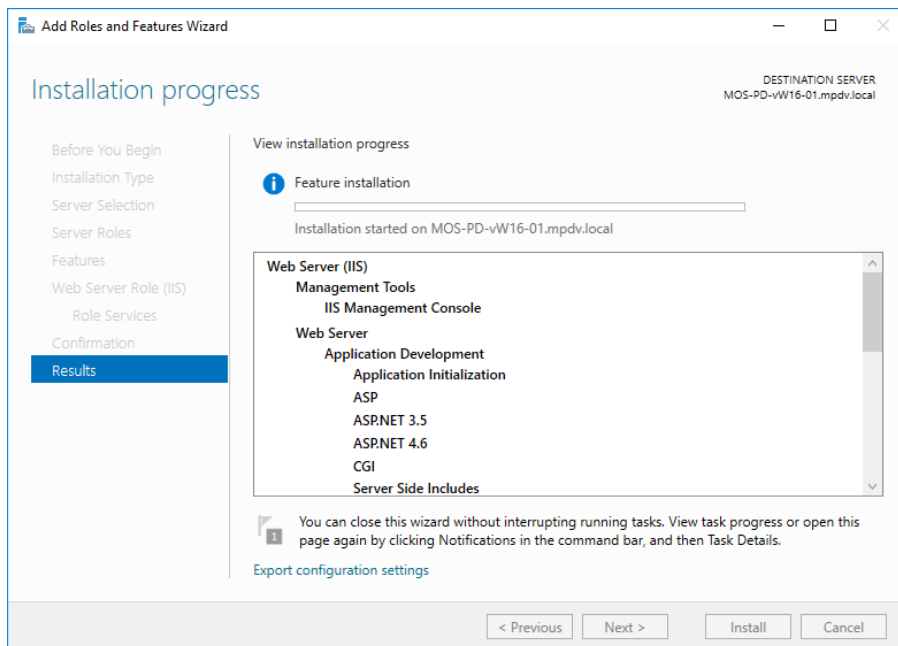


Next



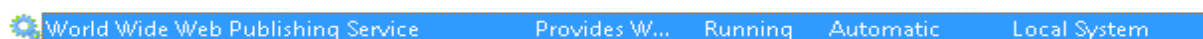
Install

Installation in progress:



Close (after successful installation)

The following Windows service should now be available:

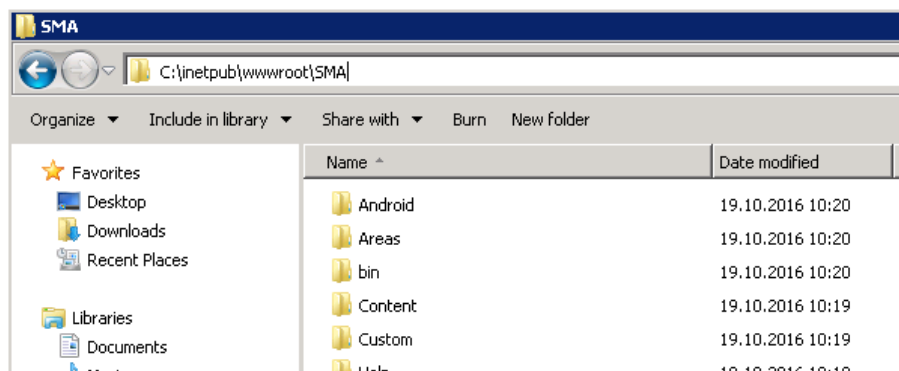


2.4.3 Create SMA site in IIS

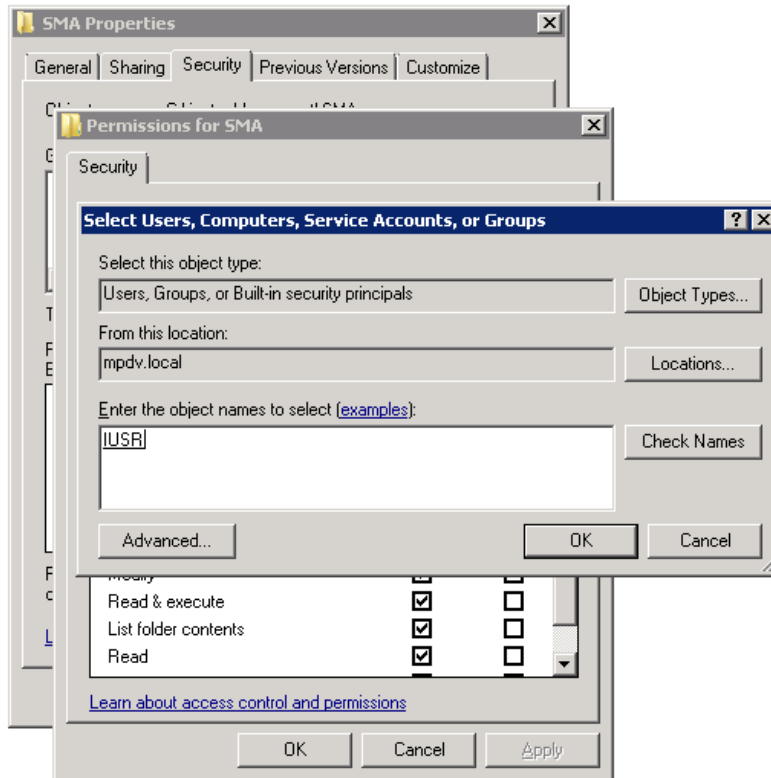
2.4.3.1 Windows 2008 R2 (IIS 7.5)

- Create a new directory "SMA" in the root directory of the IIS:
e.g.: `c:\inetpub\wwwroot\SMA`
- Unpack the SMA installation package **SMA_x64.zip** and copy the content to `c:\inetpub\wwwroot\SMA`

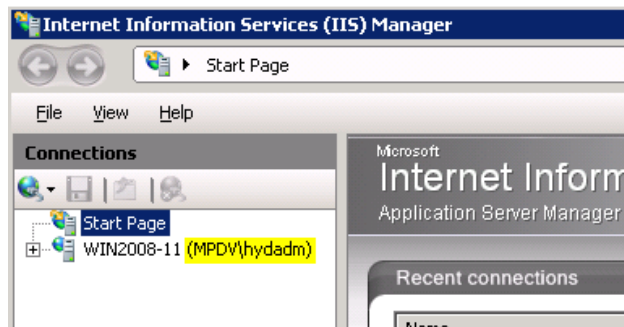
→ **Warning:** Check if the package is "locked". If so, then "unlock" the package by right-clicking (properties).



- Assign the following users to the folder:
 - IUSR
 - Request folder *properties*.
 - Go to the *Security* tab and click the *Edit* button.
 - Click *Add* in the dialog that opens.
 - Add the relevant user.

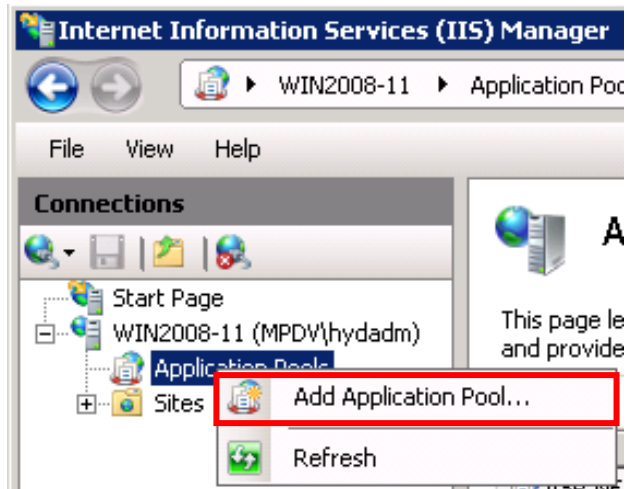


- Proceed as described above to add the user who is entered in the IIS administration GUI. You can check the person to be entered in the *Internet Information Services (IIS) Manager*:



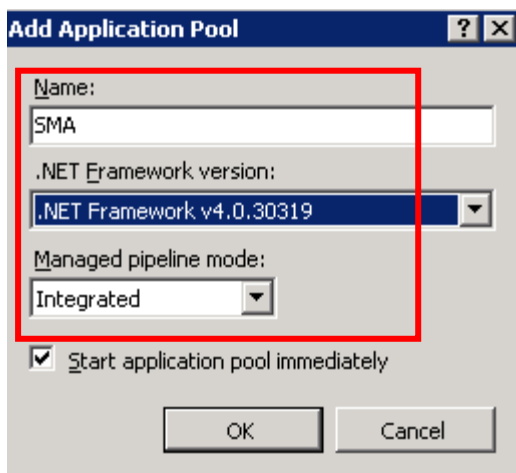
- Start the *Internet Information Services (IIS) Manager*:
 **Start – Administration – Internet Information Services (IIS) Manager**

- Right-click "Application pools"



Add application pool

- Name: SMA
.NET Framework version: .NET Framework v4.0.30319
Managed pipeline mode: Integrated

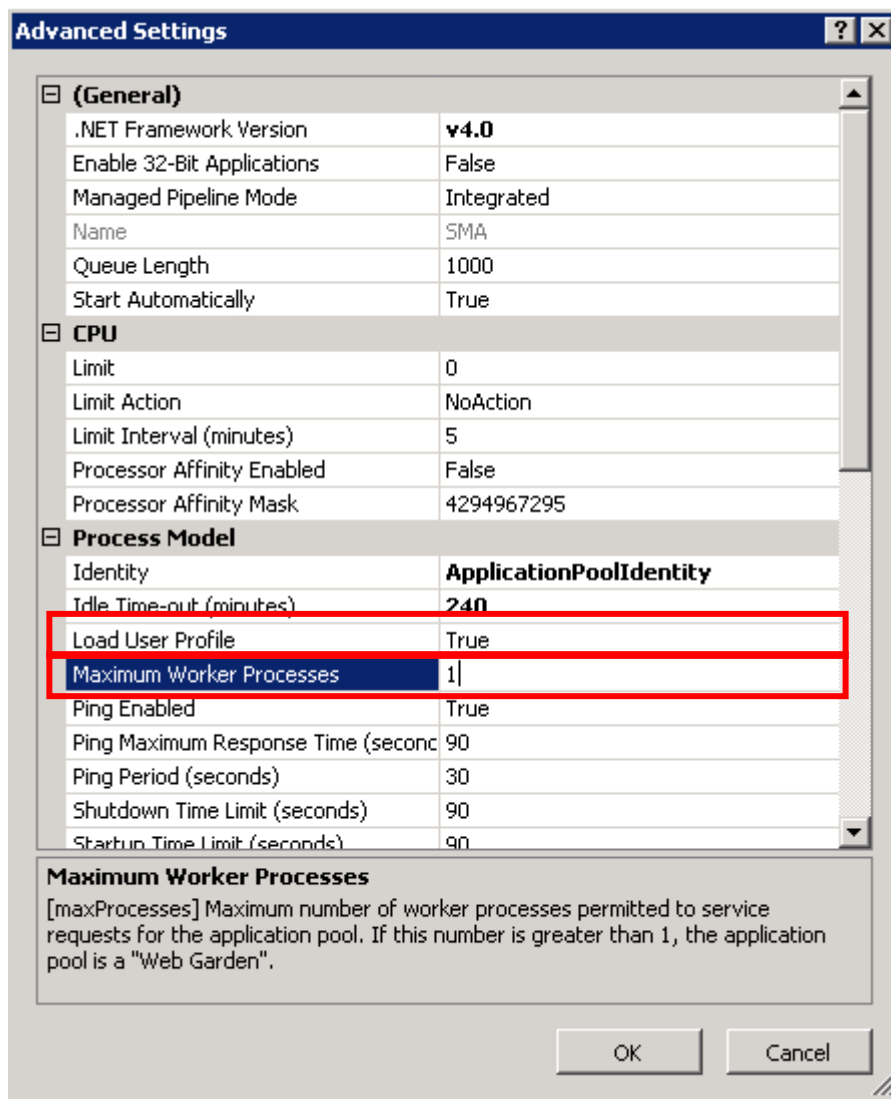


OK

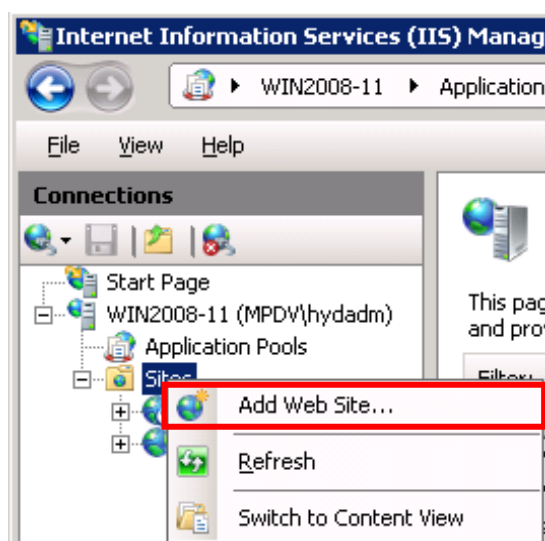
If the required .NET Framework version is not included in the selection list, you first have to register asp.net in IIS.

See [section 1.4.4 asp.net](#)

- Verify the following in the advanced settings of the application pool: **The entry "load user profile" must be set to "True"! Set the maximum number of worker processes to 1 or leave it at 1!**



- Right-click "Sites":



Add website

- Site name: SMA
Application pool: SMA
Physical path: c:\inetpub\wwwroot\SMA
Port: 8082

The screenshot shows the 'Add Web Site' dialog box. The 'Site name' field contains 'SMA' and the 'Application pool' dropdown is set to 'SMA'. The 'Physical path' field contains 'C:\inetpub\wwwroot\SMA'. The 'Binding' section shows 'Type' as 'http', 'IP address' as 'All Unassigned', and 'Port' as '8082'. The 'Start Web site immediately' checkbox is checked. The 'OK' button is highlighted.

OK

You can choose any port (here: 8082).

It all depends on the assigned ports or if ports should be reserved for future installations.



Check in any case if the ports 8080, 8081, 8082, etc. are already occupied by WSP installations in case of a HYDRA multi-system installation.

- **Permit APK download:** Allow the APK download in order to be able to download the client installation package directly from Android devices. To do so, select the server connection on the



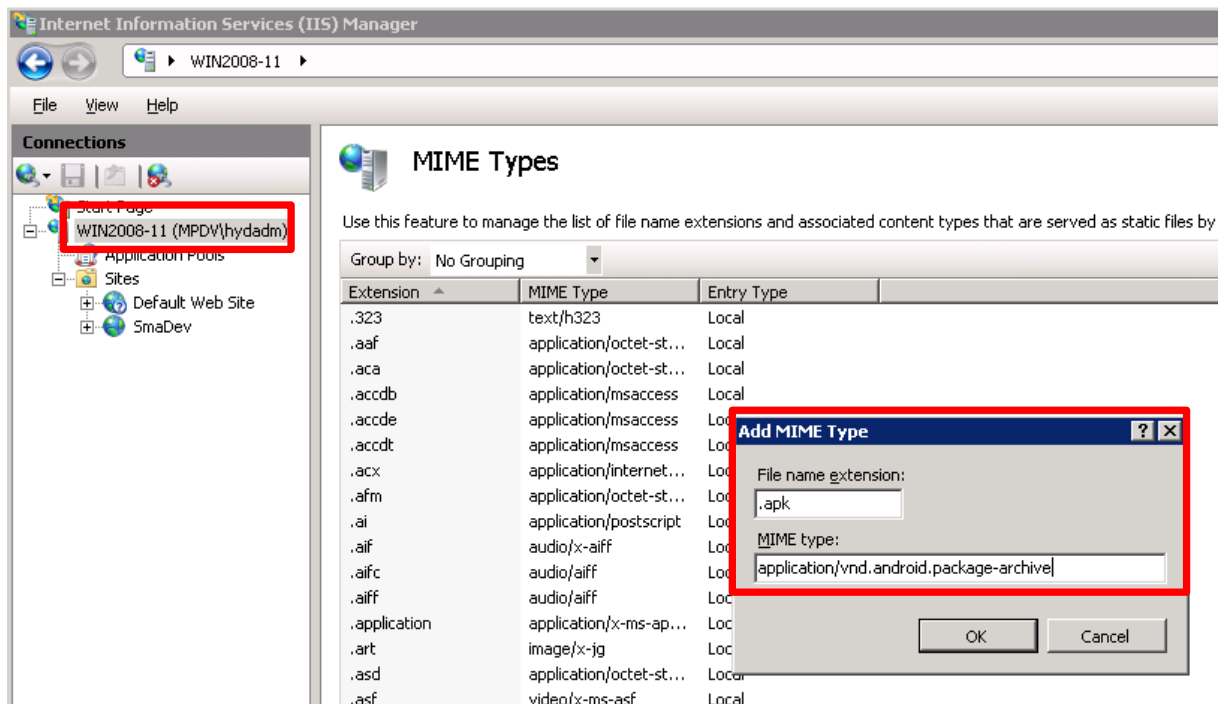
MIME-Typ

left hand side of the IIS Manager. In the center, you will find an icon called MIME type:

. A list opens. Click the "add" button on the right hand side to add a new MIME type.

File name extension: **.apk** (you must state the dot).

MIME type: **application/vnd.android.package-archive**



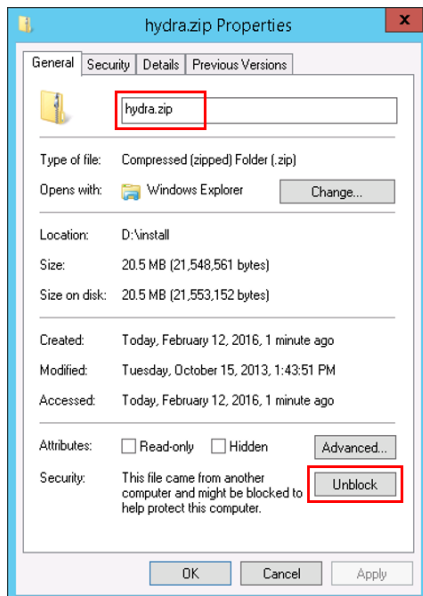
Then you have to restart the server.

2.4.3.2 Windows Server 2012 (IIS 8.5)

Create a new directory "SMA" in the IIS root directory:

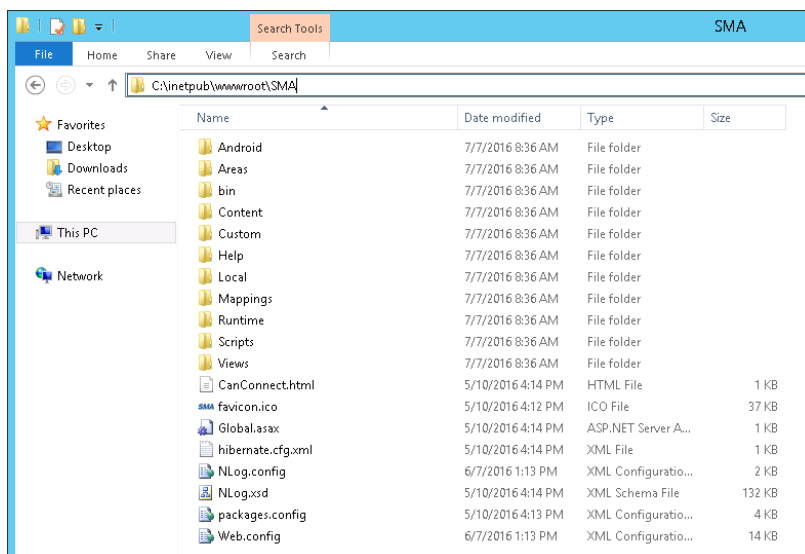
e.g.: **c:\inetpub\wwwroot\SMA**

Check the file properties of the SMA installation package **SMA_x64.zip** and make sure it is not blocked for security reasons, e.g.:

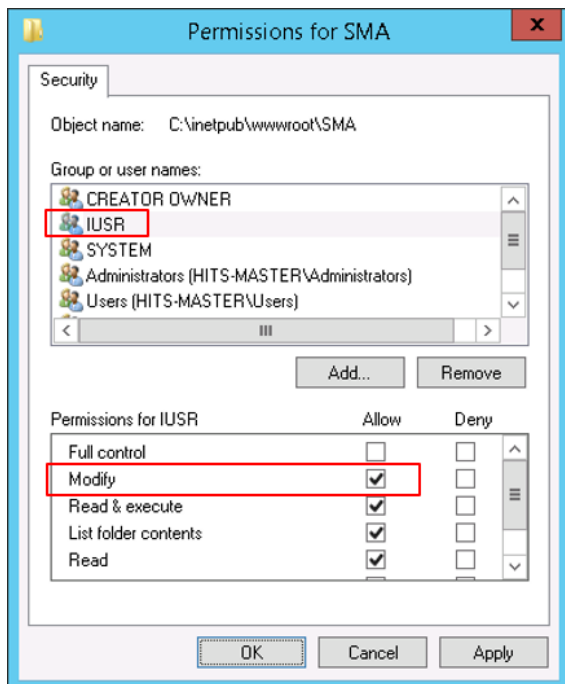


If it is blocked, then unblock it.

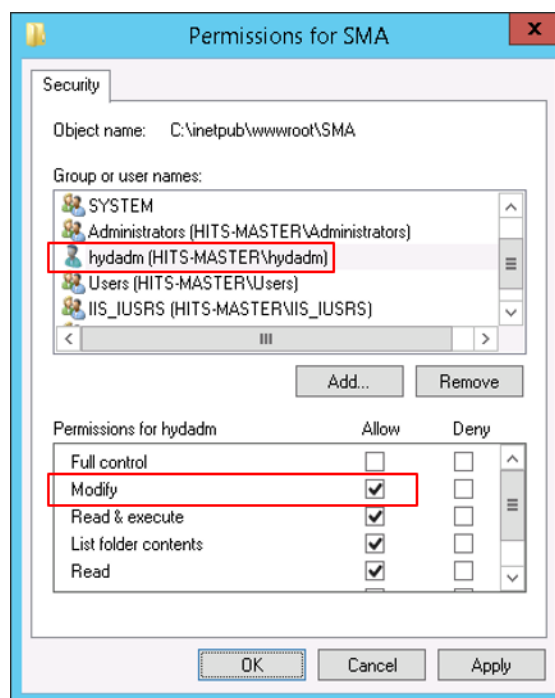
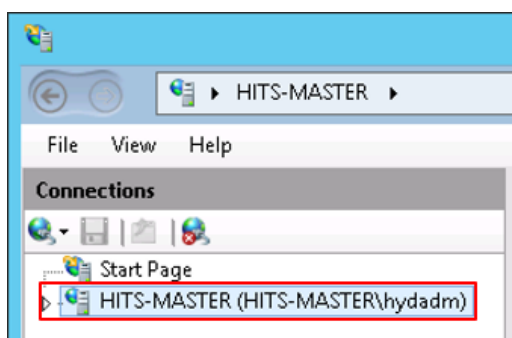
Unpack the SMA installation package **SMA_x64.zip** and copy the content to **c:\inetpub\wwwroot\SMA**



Assign the user "IUSR" to the SMA directory, e.g. `c:\inetpub\wwwroot\SMA`



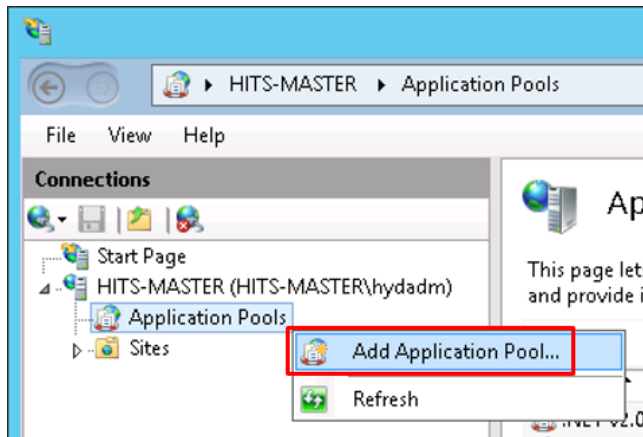
Assign the user displayed in the "Internet Information Services (IIS) Manager", e.g. "hydadm", to the SMA directory, e.g.: `c:\inetpub\wwwroot\SMA`.



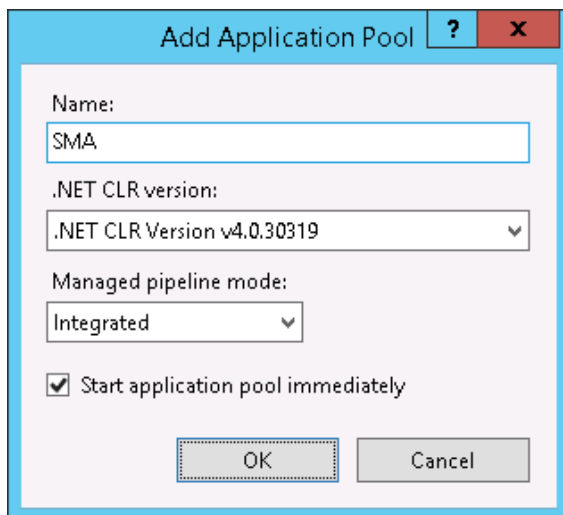
Start the "Internet Information Services (IIS) Manager":

Start → Administration → Internet Information Services (IIS) Manager

Right-click "Application pools"



Add application pool...

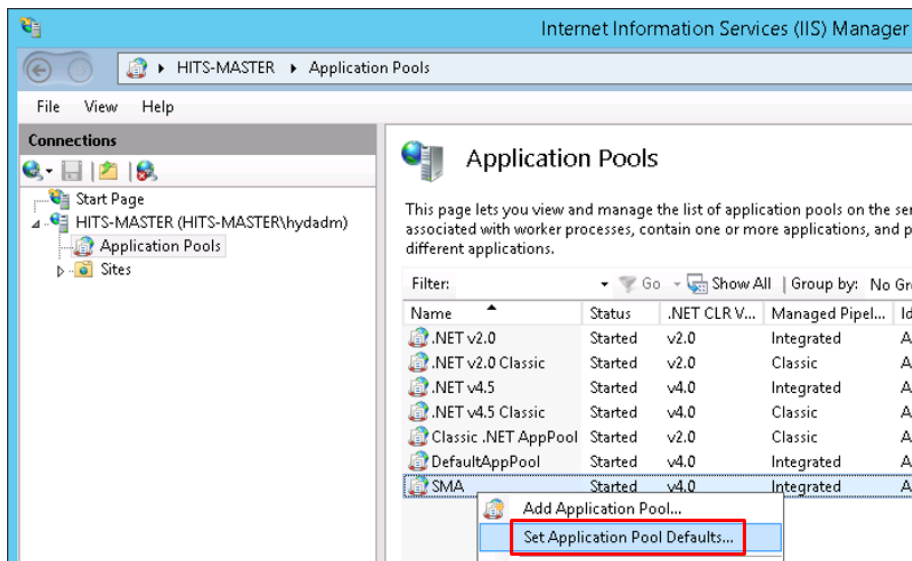


Name: SMA
.NET CLR version: .NET CLR version v4.0.30319
Managed pipeline mode: Integrated
OK

If the required .NET CLR version v4.0.30319 is not included in the selection list, you first have to register asp.net in IIS (see section [1.4.4. asp.net](#)).

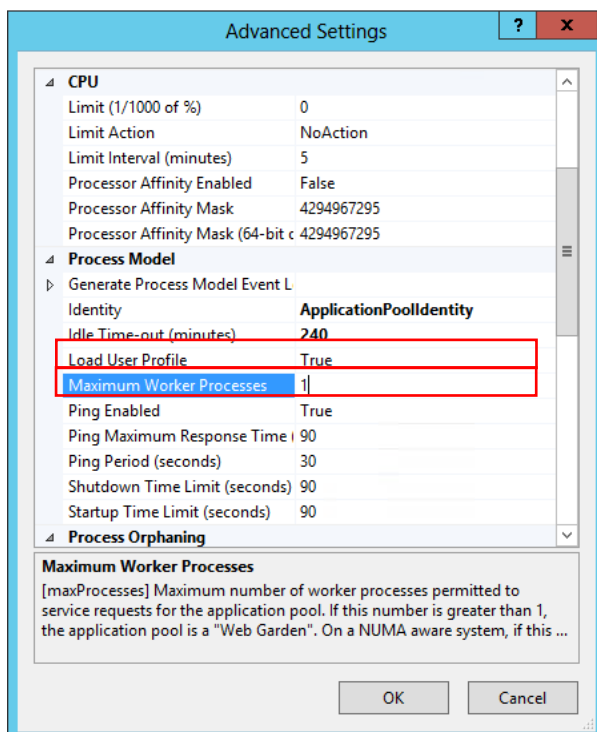
We highly recommended that you always run the commands to register asp.net in IIS even if the proper .NET version (.NET CLR version) is already listed here.

Right-click the "SMA" application pool and check the default values of the application pool:



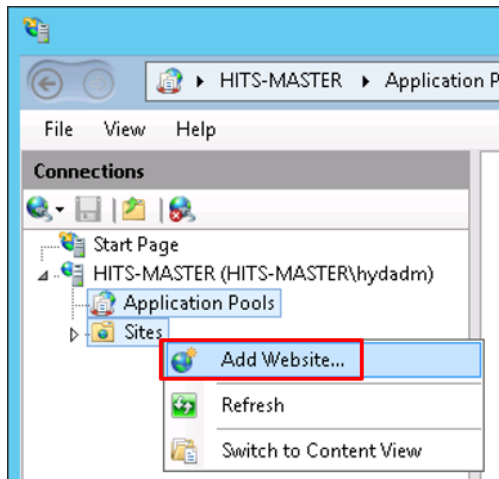
Set application pool defaults...

Make sure the option **"Maximum worker processes"** is set to **1** and set the option **"load user profile"** to **"True"**!



OK

Right-click "Sites":



Add website...

Site name: **SMA**

Application pool: **SMA**

Physical path: **c:\inetpub\wwwroot\SMA**

Port: you can choose any port, e.g.: **8082**

OK

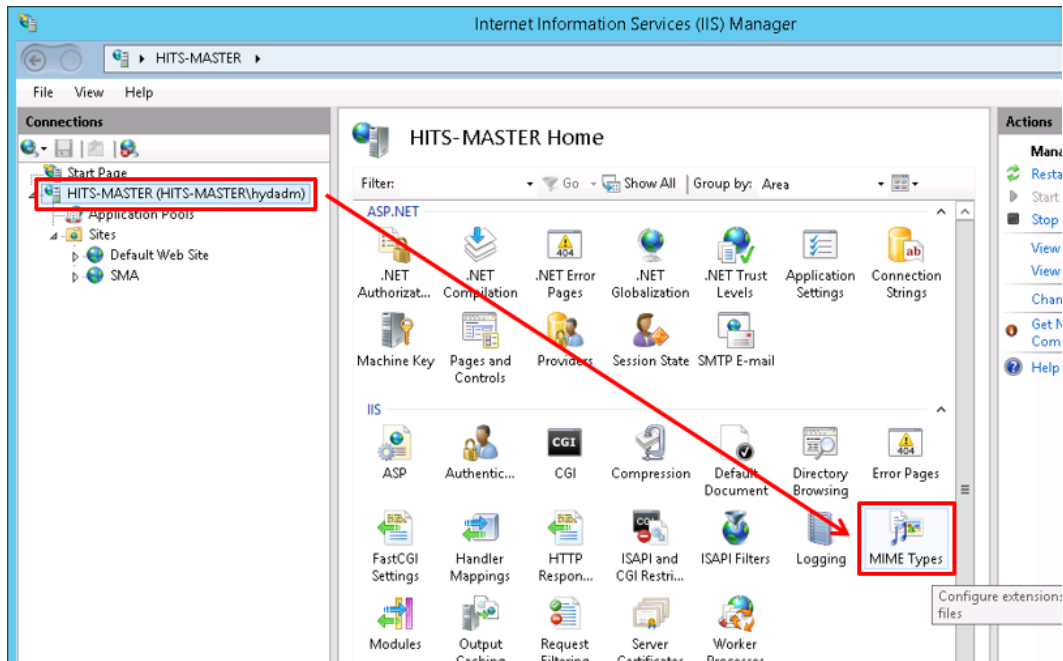


The ports you can select depend on the ports that are available on your server.

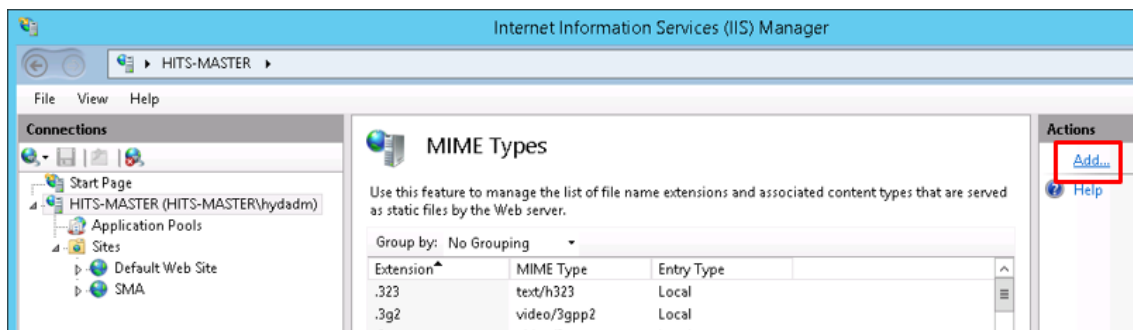
Check in any case if the ports 8080, 8081, 8082, etc. are already occupied by WSP installations

in case of a HYDRA multi-system installation.

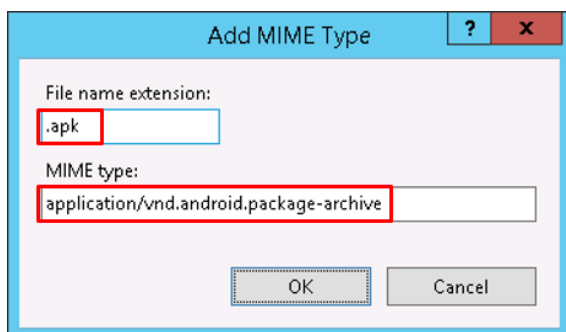
Permit the direct download of client installation packages by Android devices (APK download):



Double-click "MIME type"



Add



File extension: **.apk** (do not forget the dot ".!")

MIME_Type: **application/vnd.android.package-archive**

OK

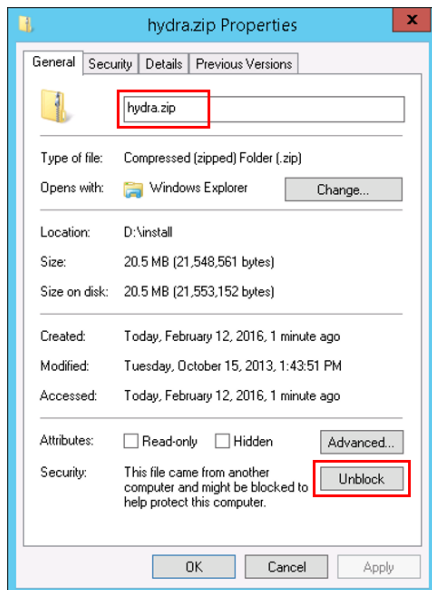
Restart your server.

2.4.3.3 Windows Server 2016 (IIS 10.0)

Create a new directory "SMA" in the root directory of the IIS:

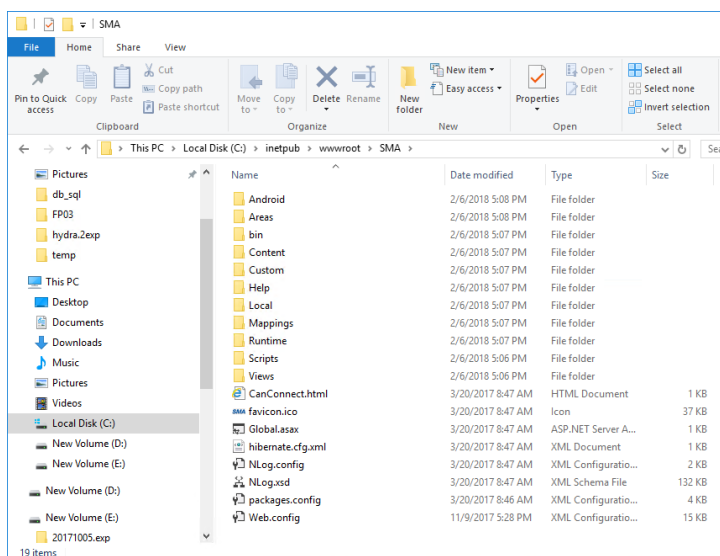
e.g.: `c:\inetpub\wwwroot\SMA`

Check the file properties of the SMA installation package **SMA_x64.zip** and make sure it is not blocked for security reasons, e.g.:

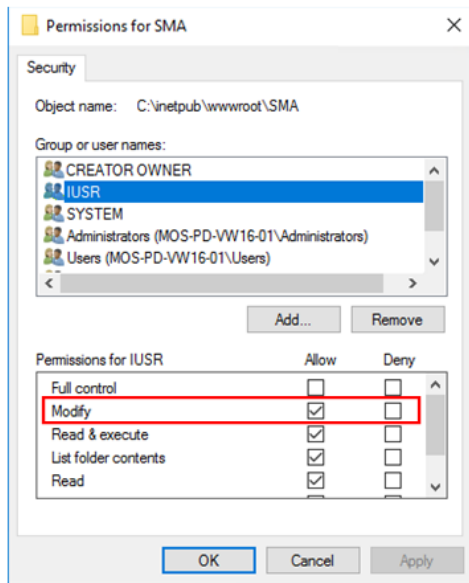


If it is blocked, then unblock it.

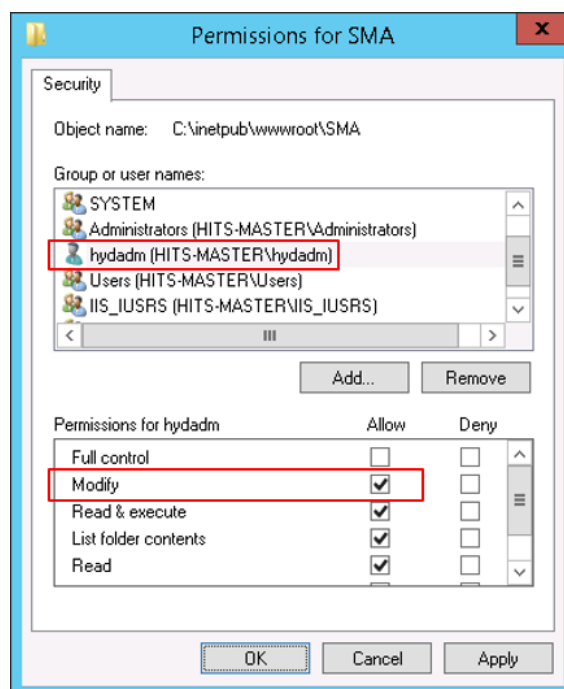
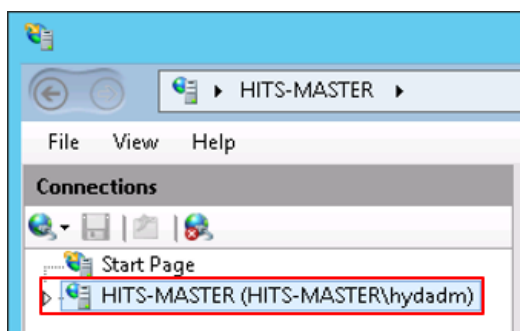
Unpack the SMA installation package **SMA_x64.zip** and copy the content to `c:\inetpub\wwwroot\SMA`



You have to assign the user “IUSR” to the SMA directory, e.g. **c:\inetpub\wwwroot\SMA**



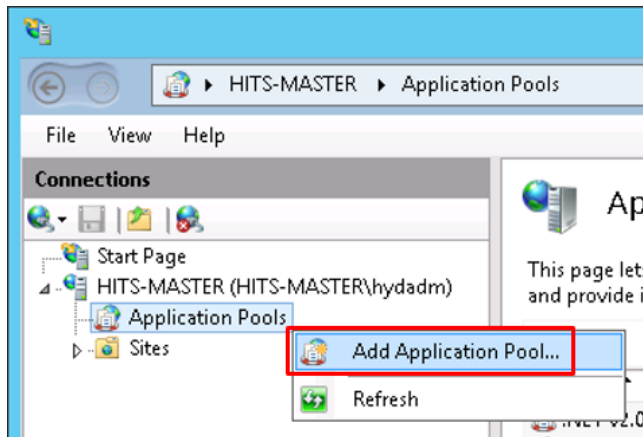
Assign the user displayed in the “Internet Information Services (IIS) Manager”, e.g. “hydadm”, to the SMA directory **c:\inetpub\wwwroot\SMA**.



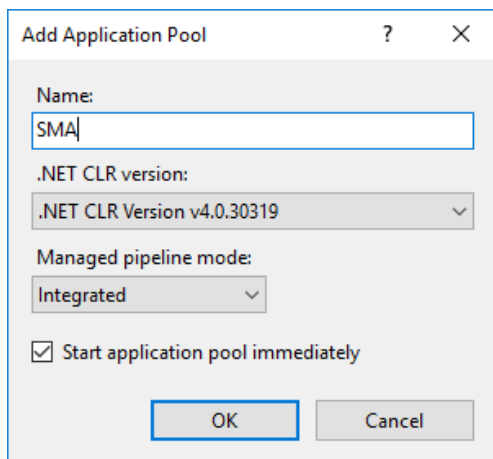
Start the "Internet Information Services (IIS) Manager":

Start → Administration → Internet Information Services (IIS) Manager

Right-click "Application pools"



Add application pool...

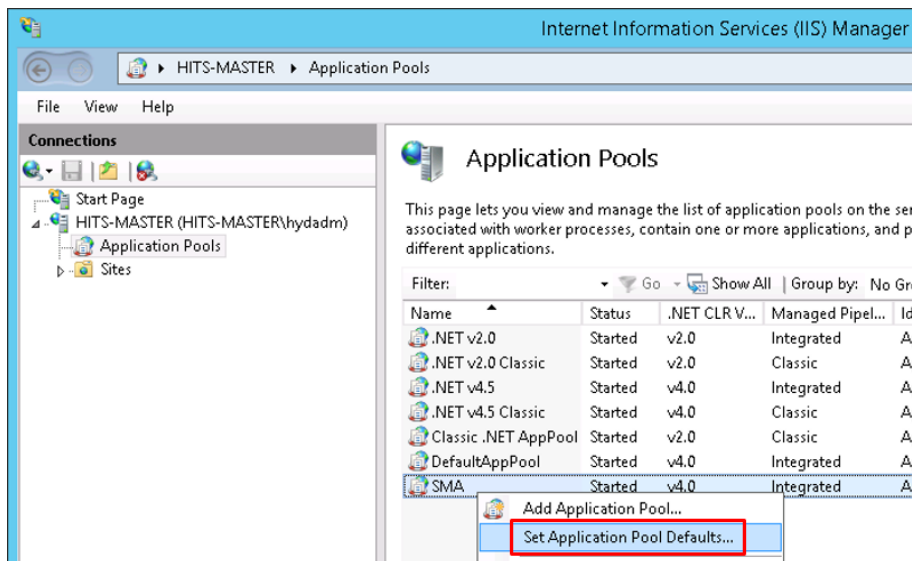


Name: **SMA**
.NET CLR version: **.NET CLR version v4.0.30319**
Managed pipeline mode: **Integrated**
OK

If the required .NET CLR version v4.0.30319 is not included in the selection list, you first have to register asp.net in IIS (see section [1.4.4. asp.net](#)).

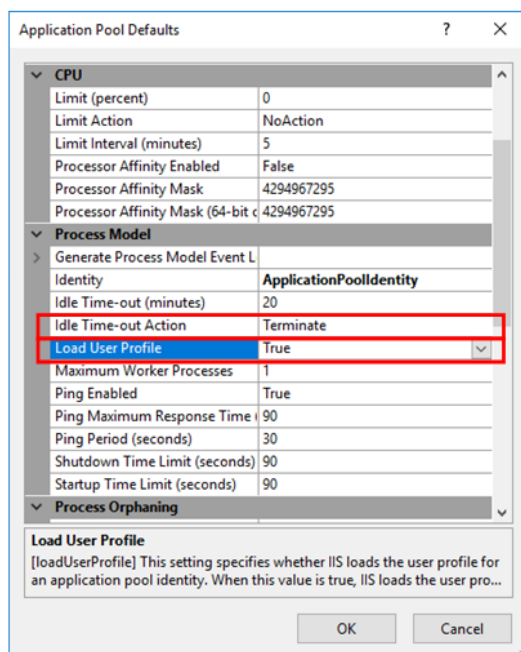
We highly recommended that you always run the commands to register asp.net in IIS even if the proper .NET version (.NET CLR version) is already listed here.

Right-click the "SMA" application pool and check the default values of the application pool:



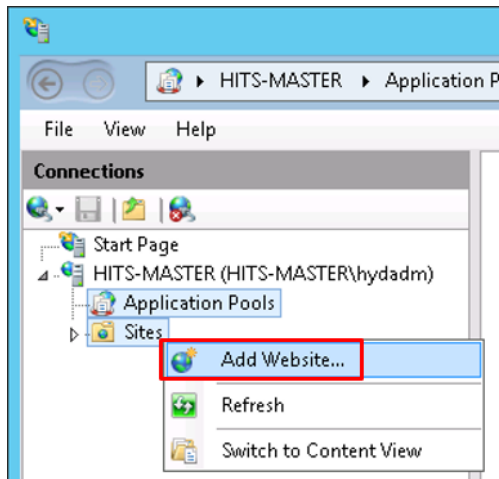
Set application pool defaults...

Make sure the option "Maximum worker processes" is set to **1** and set the option "load user profile" to "True"!



OK

Right-click "Sites":



Add website...

Site name: **SMA**
Application pool: **SMA**
Physical path: **c:\inetpub\wwwroot\SMA**
Port: you can choose any port, e.g.: **8082**
OK

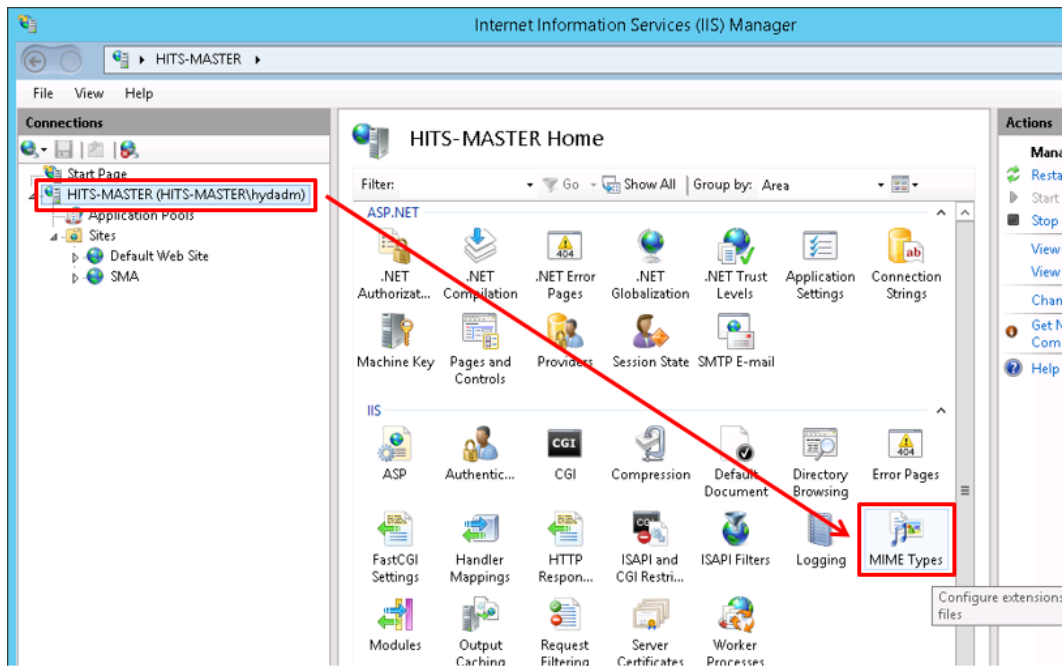


The ports you can select depend on the ports that are available on your server.

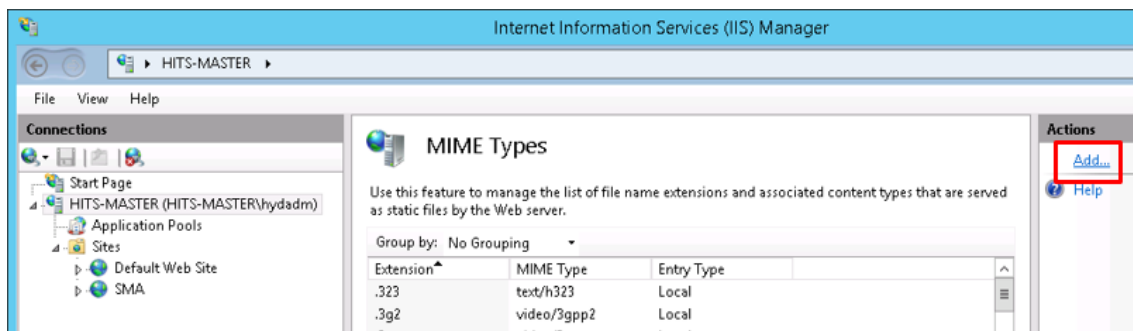
Check in any case if the ports 8080, 8081, 8082, etc. are already occupied by WSP installations

in case of a HYDRA multi-system installation.

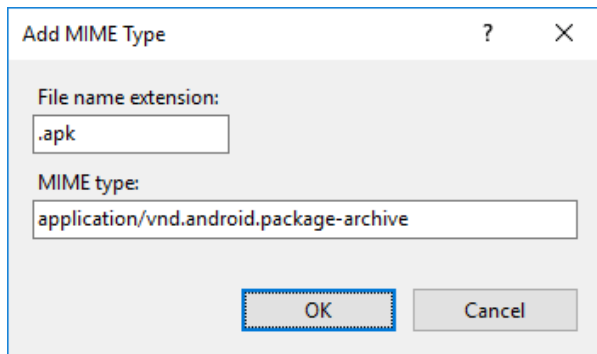
Permit the direct download of client installation packages by Android devices (APK download):



Double-click "MIME type"



Add



File extension: **.apk** (do not forget the dot ".!")

MIME_Typ: **application/vnd.android.package-archive**

OK

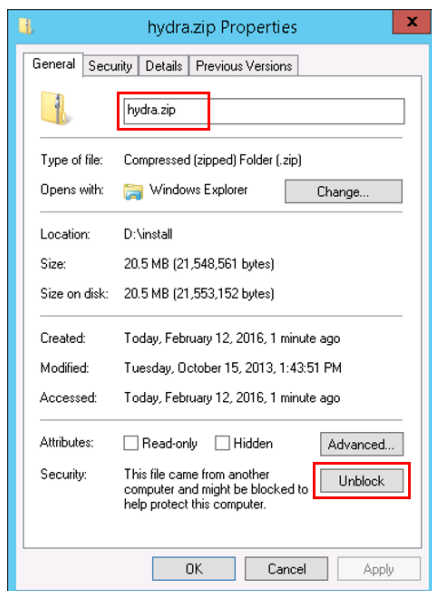
Restart your server.

2.4.3.4 Windows Server 2019 (IIS 10.0)

Create a new directory "SMA" in the root directory of the IIS:

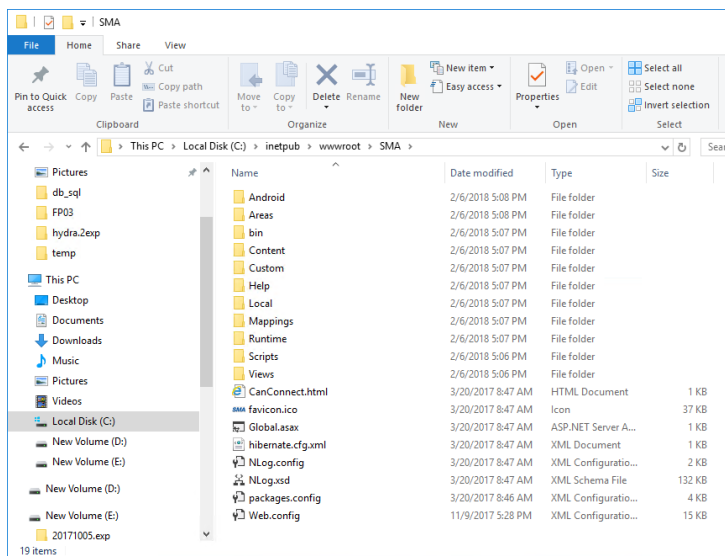
e.g.: **c:\inetpub\wwwroot\SMA**

Check the file properties of the SMA installation package **SMA_x64.zip** and make sure it is not blocked for security reasons, e.g.:

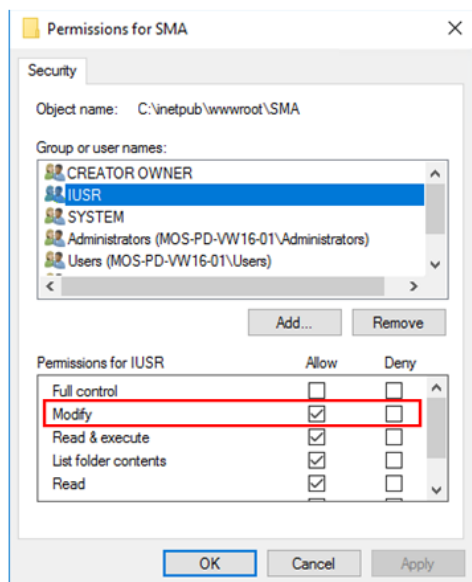


If it is blocked, then unblock it.

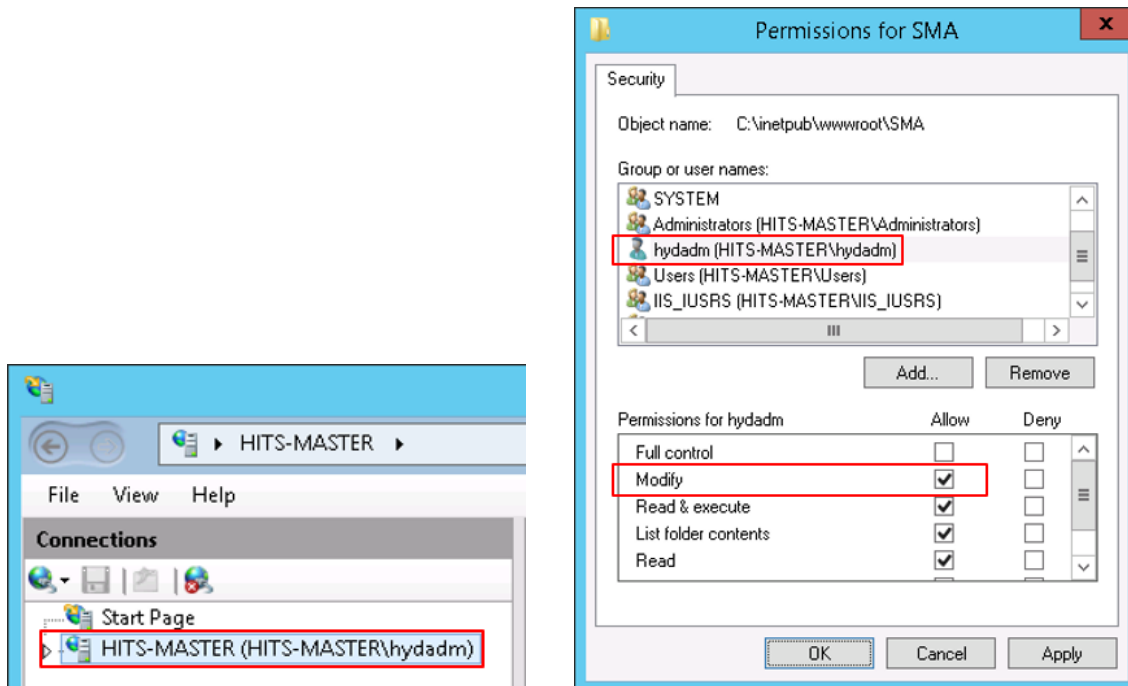
Unpack the SMA installation package **SMA_x64.zip** and copy the content to **c:\inetpub\wwwroot\SMA**



You have to assign the user **"IUSR"** to the SMA directory, e.g. `c:\inetpub\wwwroot\SMA`



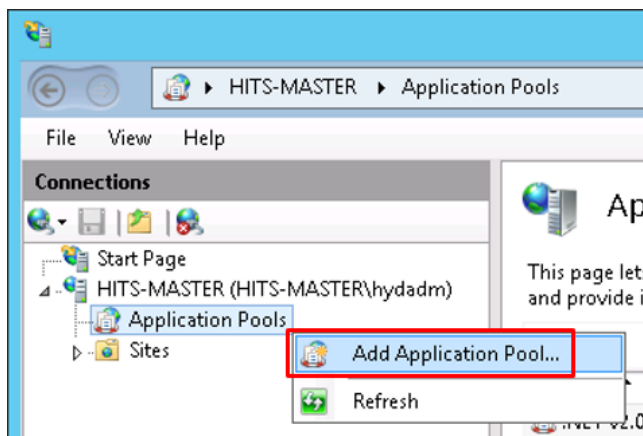
Assign the user displayed in the "Internet Information Services (IIS) Manager", e.g. "hydadm", to the SMA directory `c:\inetpub\wwwroot\SMA`.



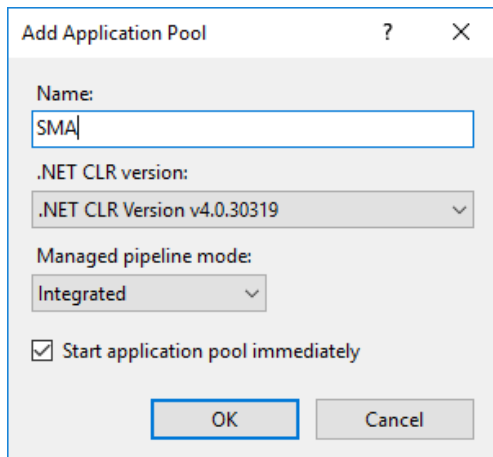
Start the "Internet Information Services (IIS) Manager":

❏ Start → Administration → Internet Information Services (IIS) Manager

Right-click "Application pools"



Add application pool...

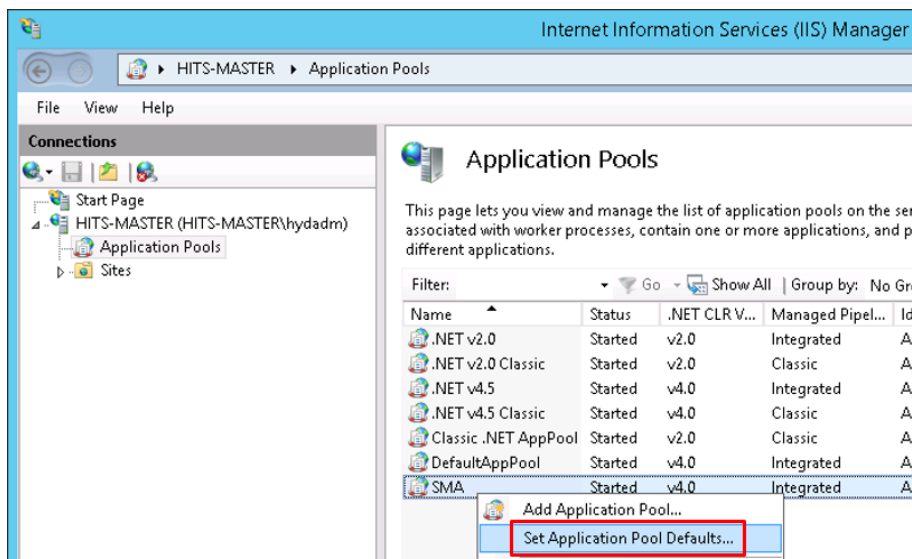


Name: **SMA**
.NET CLR version: **.NET CLR version v4.0.30319**
Managed pipeline mode: **Integrated**
OK

If the required .NET CLR version v4.0.30319 is not included in the selection list, you first have to register asp.net in IIS (see section [1.4.4. asp.net](#)).

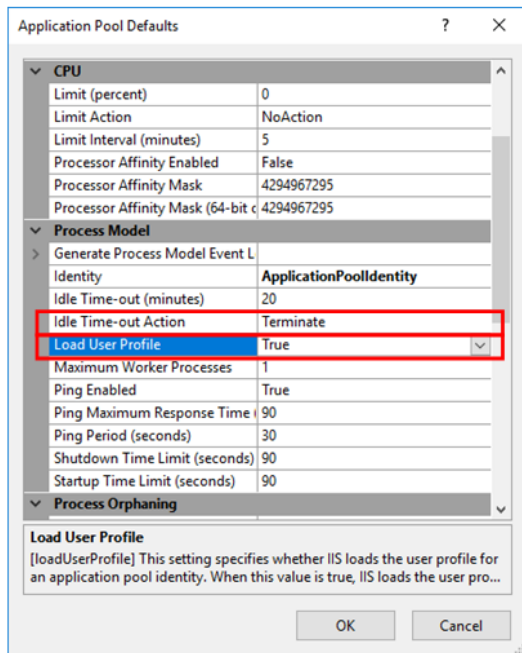
We highly recommended that you always run the commands to register asp.net in IIS even if the proper .NET version (.NET CLR version) is already listed here.

Right-click the "SMA" application pool and check the default values of the application pool:



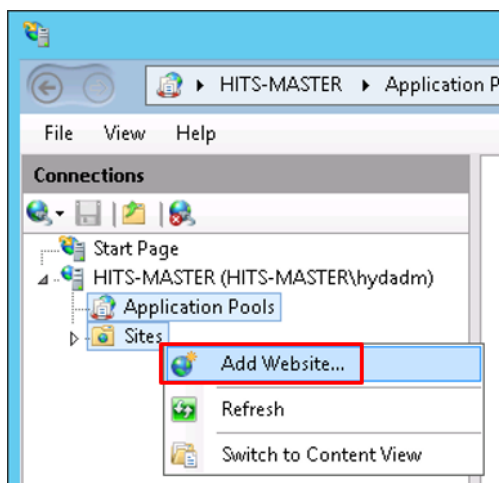
Set application pool defaults...

Make sure the option "Maximum worker processes" is set to 1 and set the option "load user profile" to "True"!



OK

Right-click "Sites":



Add website...

Site name: SMA Application pool: SMA Select...

Content Directory

Physical path: C:\inetpub\wwwroot\SMA ...

Pass-through authentication

Connect as... Test Settings...

Binding

Type: http IP address: All Unassigned Port: 8082

Host name: Example: www.contoso.com or marketing.contoso.com

☒ Start Website immediately

OK Cancel

Site name: SMA

Application pool: SMA

Physical path: c:\inetpub\wwwroot\SMA

Port: you can choose any port, e.g.: 8082

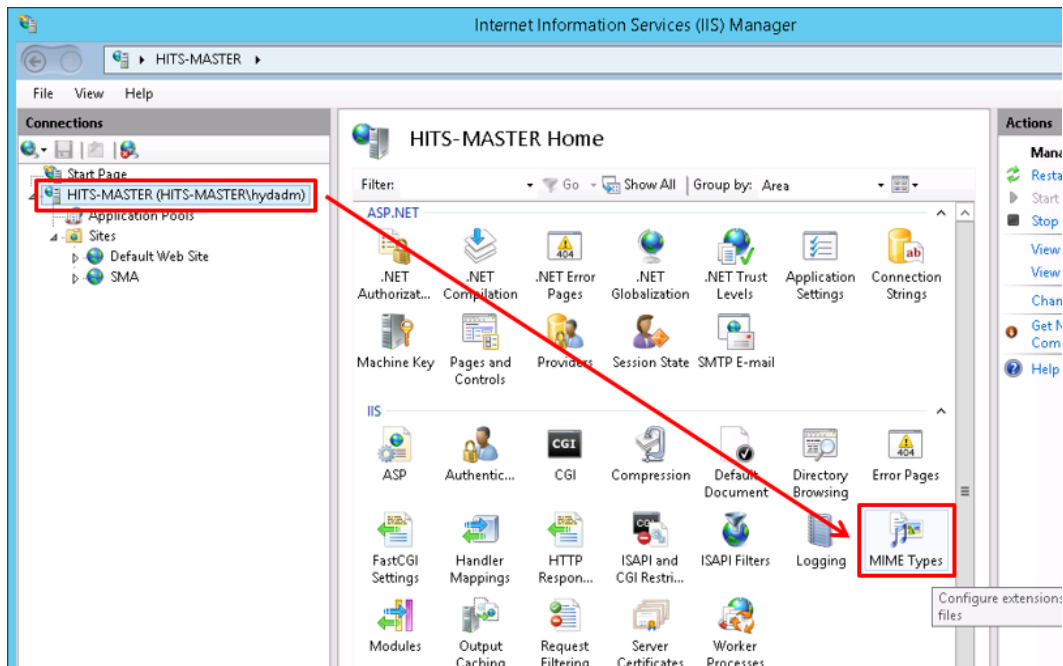
OK



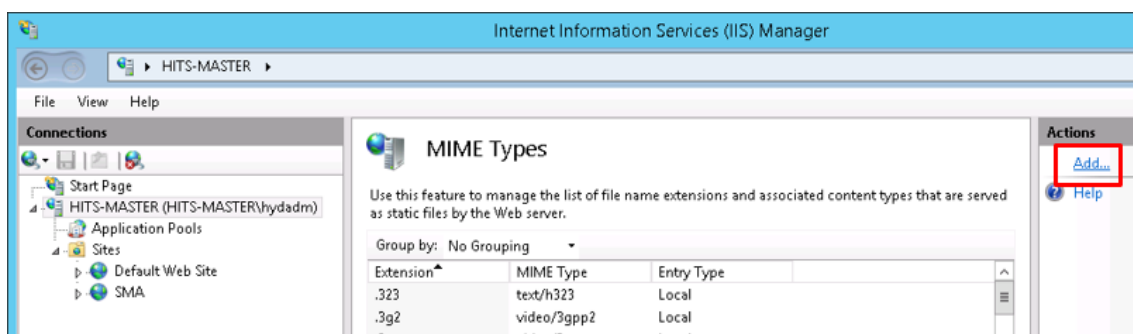
The ports you can select depend on the ports that are available on your server.

Check in any case if the ports 8080, 8081, 8082, etc. are already occupied by WSP installations in case of a HYDRA multi-system installation.

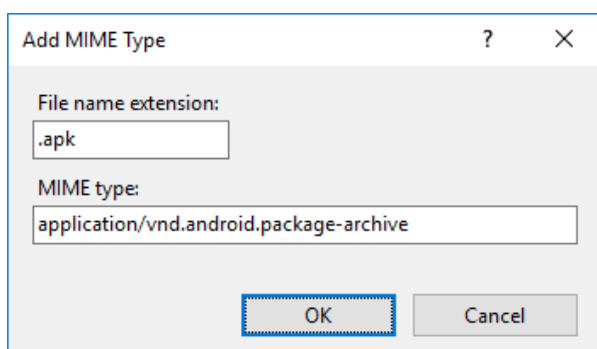
Permit the direct download of client installation packages by Android devices (APK download):



Double-click "MIME type"



Add



File extension: **.apk** (do not forget the dot ".!")

MIME_Typ: **application/vnd.android.package-archive**

OK

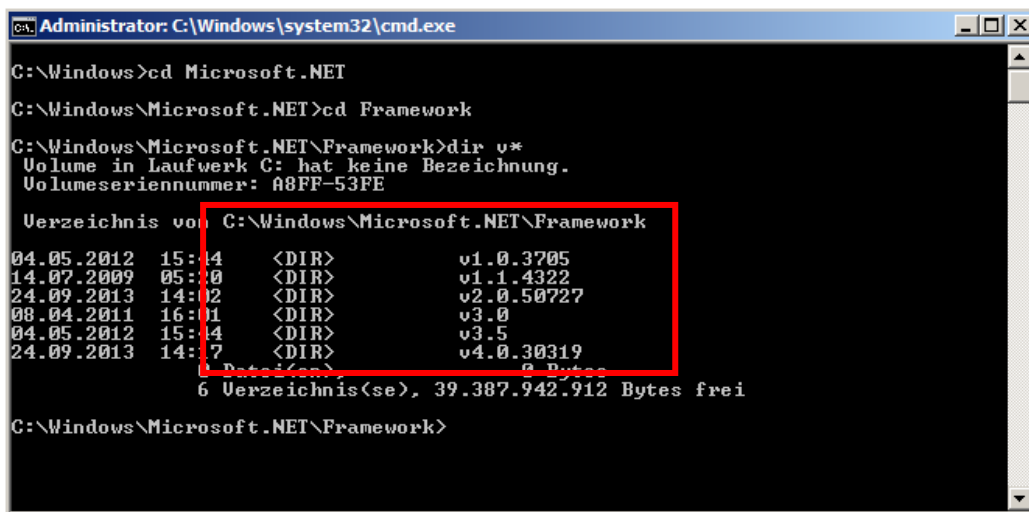
Restart your server.

2.4.4 asp.net

We highly recommend that you always run the commands to register asp.net in IIS , even if the proper .NET version (.NET CLR version) is already listed when adding a new application pool (see above).

2.4.4.1 Check installation

Use "dir v*" in the directory C:\Windows\Microsoft.NET\Framework to check which .NET versions are available on the computer:



```
Administrator: C:\Windows\system32\cmd.exe
C:\Windows>cd Microsoft.NET
C:\Windows\Microsoft.NET>cd Framework
C:\Windows\Microsoft.NET\Framework>dir v*
Volume in Laufwerk C: hat keine Bezeichnung.
Volumeseriennummer: A8FF-53FE

Verzeichnis von C:\Windows\Microsoft.NET\Framework

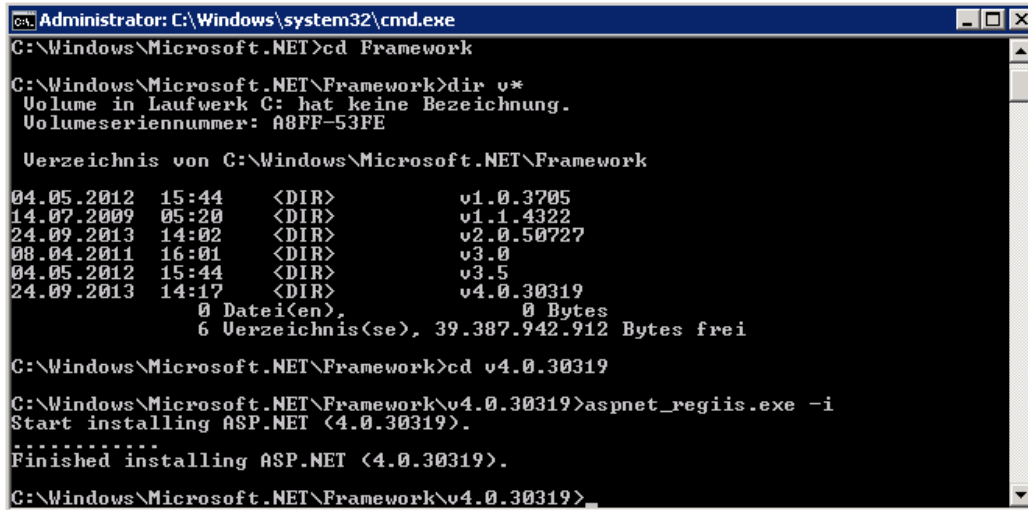
04.05.2012  15:14    <DIR>          v1.0.3705
14.07.2009  05:00    <DIR>          v1.1.4322
24.09.2013  14:02    <DIR>          v2.0.50727
08.04.2011  16:01    <DIR>          v3.0
04.05.2012  15:14    <DIR>          v3.5
24.09.2013  14:07    <DIR>          v4.0.30319
               6 Verzeichnis(se), 39.387.942.912 Bytes frei

C:\Windows\Microsoft.NET\Framework>
```

2.4.4.2 Register asp.net in IIS

To register asp.net in IIS, go to the required directory, e.g. **C:\Windows\Microsoft.NET\Framework\v4.0.30319** and execute the following command:

aspnet_regiis.exe -i



```
Administrator: C:\Windows\system32\cmd.exe
C:\Windows\Microsoft.NET>cd Framework
C:\Windows\Microsoft.NET\Framework>dir v*
Volume in Laufwerk C: hat keine Bezeichnung.
Volumeserienummer: A8FF-53FE

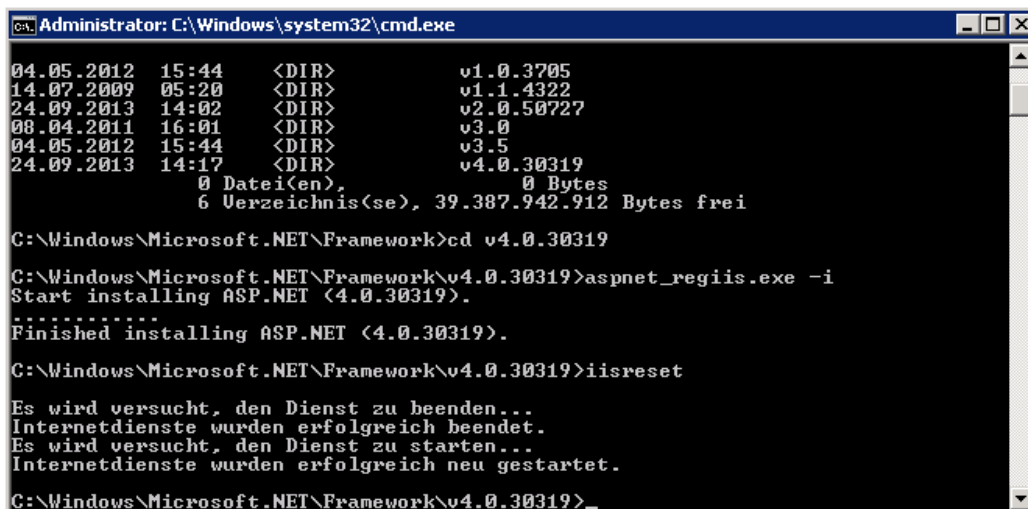
Verzeichnis von C:\Windows\Microsoft.NET\Framework

04.05.2012  15:44    <DIR>          v1.0.3705
14.07.2009  05:20    <DIR>          v1.1.4322
24.09.2013  14:02    <DIR>          v2.0.50727
08.04.2011  16:01    <DIR>          v3.0
04.05.2012  15:44    <DIR>          v3.5
24.09.2013  14:17    <DIR>          v4.0.30319
               0 Datei(en),               0 Bytes
               6 Verzeichnis(se), 39.387.942.912 Bytes frei

C:\Windows\Microsoft.NET\Framework>cd v4.0.30319
C:\Windows\Microsoft.NET\Framework\v4.0.30319>aspnet_regiis.exe -i
Start installing ASP.NET (4.0.30319).
.....
Finished installing ASP.NET (4.0.30319).
C:\Windows\Microsoft.NET\Framework\v4.0.30319>
```

Restart IIS with the following command:

iisreset



```
Administrator: C:\Windows\system32\cmd.exe

04.05.2012  15:44    <DIR>          v1.0.3705
14.07.2009  05:20    <DIR>          v1.1.4322
24.09.2013  14:02    <DIR>          v2.0.50727
08.04.2011  16:01    <DIR>          v3.0
04.05.2012  15:44    <DIR>          v3.5
24.09.2013  14:17    <DIR>          v4.0.30319
               0 Datei(en),               0 Bytes
               6 Verzeichnis(se), 39.387.942.912 Bytes frei

C:\Windows\Microsoft.NET\Framework>cd v4.0.30319
C:\Windows\Microsoft.NET\Framework\v4.0.30319>aspnet_regiis.exe -i
Start installing ASP.NET (4.0.30319).
.....
Finished installing ASP.NET (4.0.30319).
C:\Windows\Microsoft.NET\Framework\v4.0.30319>iisreset

Es wird versucht, den Dienst zu beenden...
Internetdienste wurden erfolgreich beendet.
Es wird versucht, den Dienst zu starten...
Internetdienste wurden erfolgreich neu gestartet.
C:\Windows\Microsoft.NET\Framework\v4.0.30319>
```

Restart your server.

2.4.5 SMA configuration

2.4.5.1 Basic configuration

Check and adjust the following rows according to your requirements in the SMA configuration file (by default: `c:\inetpub\wwwroot\SMA\Web.config`):

Directory for SMA log files:

```
<add key="LogFolder" value="C:\temp\SMA-Log" />
```

The default path is: `c:\temp\SMA-Log`.

The cache for the SMA system is also stored in "LogFolder".

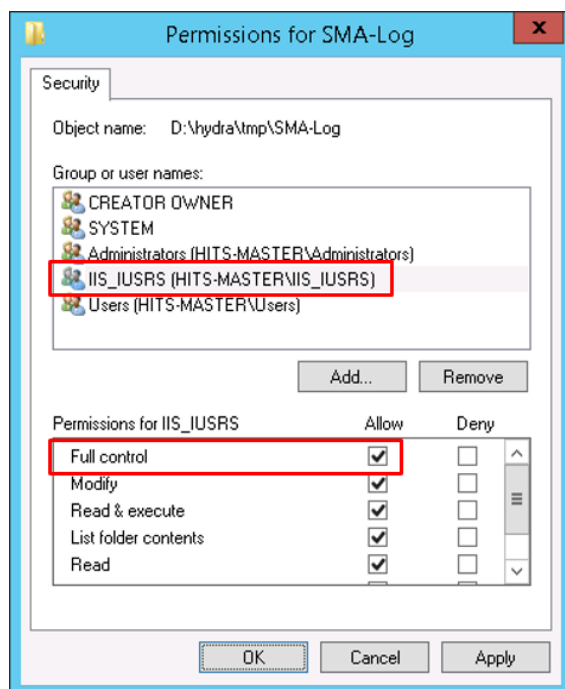
Important: Do not log data in the SMA directory!

We recommend using a directory within HYDRADIR, e.g.:

```
<add key="LogFolder" value="D:\hydra\tmp\SMA-Log" />
```

Generate this directory, if it does not yet exist!

The group "IIS_IUSRS" must have full access to the SMA log directory, e. g. `d:\hydra\tmp\SMA-Log`



Further settings:

Host name or IP address for the HYDRA MasterServer:

```
<add key="MasterServer" value="moc-tomcat-01" />
```

Port for the HYDRA MasterServer:

```
<add key="MasterServerPort" value="8080" />
```

Name of the service (the default value should be correct):

```
<add key="MasterServerWebApplication" value="MocServices/MesInstanceService" />
```

For example:

```
<add key="LogFolder" value="D:\hydra\tmp\SMA-Log" />

<!-- The name or IP address of the Master Server, which provides a list of available
HYDRA systems.
    Allowed values: A valid server name or IP address, e.g "HYDRASERVER"
    Default value: "" -->
<add key="MasterServer" value="moc-tomcat-01" />
<!-- The master server's port where the instance service is available.
    Allowed values:
    Default value: "8080" -->
<add key="MasterServerPort" value="8080" />
<!-- Name of the web application of the master service
    Default value: "MocServices" -->
<add key="MasterServerWebApplication" value="MocServices/MesInstanceService" />
```

2.4.5.2 Edit clockings

With HYDRA 8 you can edit clocking records.

Set the entry "PersonEditClockings" to "true" in the Web.config of the SMA installation (default storage location: "C:\inetpub\wwwroot\SMA\Web.config") in order to enable this function.

```
<!-- Person edit clockings -->
<add key="PersonEditClockings" value="true"/>
```

Set the corresponding entry to "false", if you do not want the person to edit the clocking records.

```
<!-- Person edit clockings -->
<add key="PersonEditClockings" value="false"/>
```

Possible as of SMA version 42420.

2.4.5.3 Hide unlicensed applications

Applications which are not licensed are grayed out by default on the home screen after a user logs in.

If you want to hide these applications completely, set the entry "ShowNotLicensedApps" to "false" in the web.config of the SMA installation (default storage location: C:\inetpub\wwwroot\SMA\Web.config).

```
<!-- Licensing -->
<add key="ShowNotLicensedApps" value="false" />
```

Use the following entry if you want to gray out unlicensed applications (default setting):

```
<!-- Licensing -->
<add key="ShowNotLicensedApps" value="true" />
```

2.4.5.4 Further configuration options

Find below a list of possible configurations that you can make for an SMA installation. The document describes the entries of the Web.config of the SMA installation (default storage location: "C:\inetpub\wwwroot\SMA\Web.config") that can be changed.

- **LogLevel**
 - o Valid values are "Off" (default) and "Trace".
 - o Off: Logging is not active.
 - o Trace: Logging is performed using the configured directory (LogFolder).
- **CheckboxTerminalMode**
 - o Valid values are "true" (default) and "false".
 - o true: The checkbox *Terminal mode* is initially activated (if no information is stored in the browser cache) on the person login page in tab *Settings*.

- false: The checkbox *Terminal mode* is initially deactivated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
- The help of the login page describes how the field works (SMA_Login.pdf).
- **CheckboxShowFieldPerson**
 - Valid values are "true" (default) and "false".
 - true: The checkbox *Show 'person' field* is initially activated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - false: The checkbox *Show 'person' field* is initially deactivated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - The help of the login page describes how the field works (SMA_Login.pdf).
- **CheckboxShowFieldCardId**
 - Valid values are "true" (default) and "false".
 - true: The checkbox *Show 'badge' field* is initially activated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - false: The checkbox *Show 'badge' field* is initially deactivated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - The help of the login page describes how the field works (SMA_Login.pdf).
- **DefaultTerminal**
 - Default assignment is "254"
 - The field *Terminal* is initially assigned this value (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - The help of the login page describes how the field works (SMA_Login.pdf).
- **CheckboxCostCenterBooking**
 - Valid values are "true" and "false" (default).
 - true: The checkbox *Cost center posting* is initially activated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - false: The checkbox *Cost center posting* is initially deactivated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - The help of the login page describes how the field works (SMA_Login.pdf).
- **DefaultCompanyCostCenter**
 - Default assignment is " " (empty)
 - The field *Company for cost center posting* is initially assigned this value (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - The help of the login page describes how the field works (SMA_Login.pdf).
- **DefaultCompanyLackBasicList**
 - Default assignment is " " (empty)
 - The field *Company for absence reason list* is initially assigned this value (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - The help of the login page describes how the field works (SMA_Login.pdf).
- **CheckboxRememberLanguageHide**

- Valid values are "true" and "false" (default).
- true: The checkbox *Save selection* is not displayed on the login pages in tab *Language*.
- false: The checkbox *Save selection* is displayed on the login pages in tab *Language*.
- If you save the selected language, the language selected last is preassigned on the login pages after logout.
- **ListItemSettingsHide**
 - Valid values are "true" and "false" (default).
 - true: the tab *Settings* is completely hidden on the login pages. The default values are still assigned to the fields of the setting page. Important: If some information is stored in the browser cache, this information overwrites the default configuration settings of the Web.config. Solution: When the Web.config is changed, the clients must empty their cache.
 - false: The setting does not have any effect. The tab *Settings* is displayed on the login pages.
- **ForcePzeLoginOnHomeScreen**
 - Valid values are "true" and "false" (default).
 - true: If a logout is performed on the SMA start screen (home) or if the login page is called using the button *Settings* (screwdriver), the person login page is loaded.
 - false: If a logout is performed on the SMA start screen (home) or if the login page is called using the button *Settings* (screwdriver), the user login page is loaded.
- **LanguageChoiceBlacklist**
 - Can include a pipe-separated list of languages that must not be available in the selection of the login.
 - Example:
 - The following files are available in the file system (SMA\Runtime\resources\standard\languages)
 - mpdvTexte.de-AT.resx
 - mpdvTexte.de-CH.resx
 - mpdvTexte.de-DE.resx
 - Entry: `<add key="LanguageChoiceBlacklist" value="de-AT|de-CH"/>`
 - Result:
 - German (Austria) is not available
 - German (Switzerland) is not available
 - German (Germany) is available
- **PersonLogOnHideInputField**
 - Default assignment is " " (empty)
 - " " (empty): On the person login page, you can enter the personnel number (person) or the badge number (bagde) to log in.
 - userPerson: To log in a person, you can only enter the badge number (badge).

- userCardId: To log in a person, you can only enter the personnel number (person).
- **PersonLogOnAllowEmptyPincode**
 - Valid values are "true" (default) and "false".
 - true: On the person login page, a person can also log in with empty field *Pin code*.
 - false: On the person login page, a person can only log in if the field *Pin code* is filled. Otherwise, an error message *Pincode required* is displayed.
- **PersonLogOnAutoSelectField**
 - Default assignment is " " (empty)
 - " " (empty): When you open the person login page, no input field is focused.
 - userPerson: When you open the person login page, the input field *Person* is focused.
 - userCardId: When you open the person login page, no input field *Badge* is focused.
 - userPincode: When you open the person login page, the input field *Pin code* is focused.
- **PersonLogOnPincodelsReadonlyIfIdentityIsMissing**
 - Valid values are "true" and "false" (default).
 - true: On the person login page, you can only fill the field *Pin code* when you have filled the field *Person* or *Badge*.
 - false: On the person login page, you can always fill the field *Pin code*.
- **NavigationBarHideActualLoggedInUser**
 - Valid values are "true" and "false" (default).
 - true: The currently logged on user or the currently logged on person is not displayed in the navigation bar.
 - false: The currently logged on user or the currently logged on person is displayed in the navigation bar.
- **PersonSessionTimeout**
 - Valid values are "true" and "false" (default).
 - The effect of this setting also depends on the configuration of setting "PersonSessionTimeoutEdit".
 - "PersonSessionTimeoutEdit" is filled with value "false":
 - true: When a person has logged on, a timeout is displayed in the navigation bar. When the timeout has expired, the person is automatically logged off.
 - false: There is no timeout.
 - "PersonSessionTimeoutEdit" is filled with value "true":
 - true: The checkbox *Timeout* is initially activated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
 - false: The checkbox *Timeout* is initially deactivated (if no information is stored in the browser cache) on the person login page in tab *Settings*.
- **PersonSessionTimeoutSeconds**
 - Can include a positive integer value. Default is "60"
 - The effect of this setting also depends on the configuration of setting "PersonSessionTimeoutEdit".

- "PersonSessionTimeoutEdit" is filled with value "false":
 - If the setting PersonSessionTimeout has the value true, this setting specifies the timeout in seconds when a logged in person is logged out.
- "PersonSessionTimeoutEdit" is filled with value "true":
 - The field *Timeout seconds* is initially assigned this value (if no information is stored in the browser cache) on the person login page in tab *Settings*.
- **PersonSessionTimeoutEdit**
 - Valid values are "true" and "false" (default).
 - true: When a person logs in, the person can specify the values for the configuration to activate the *Timeout* and the *Timeout seconds* on the person login page in tab *Settings*.
 - false: Only the configuration of the Web.config for the settings "PersonSessionTimeout" and "PersonSessionTimeoutSeconds" are valid for the "timeout" function. No changes are possible when a person logs in.
- **PersonSessionLkTimeout**
 - You can store a language key. Default is " " (empty)
 - If the configuration includes a language key, this key is translated and displayed before timeout (if configured and active).
 - If no relevant language key is found, the text is directly displayed before timeout (if configured and active).
- **LogOutRemoveUser**
 - Valid values are "true" and "false" (default).
 - true: When a user logs out, the login information is removed. The field *User* is not prepopulated on the user login page.
 - false: When a user logs out, the login information is still used to prepopulate the field *User* on the user login page.
- **LogOutRemovePerson**
 - Valid values are "true" and "false" (default).
 - true: When a person logs out, the login information is removed. The fields *Person* and *Badge* are not prepopulated on the person login page.
 - false: When a person logs out, the login information is still used to prepopulate the fields *Person* and *Badge* on the person login page.
- **LogOutLinkToHome**
 - Valid values are "true" and "false" (default).
 - true: After logout and a successful re-login, the login page directs to the start page (home).
 - false: After logout and a successful re-login, the login page directs to the page that was last opened when logged out.
- **NHibernateDbName**
 - Can include a name. Default is " " (empty)
 - Only the letters a-z, A-Z and _ are allowed

- The name must end with ".db"
 - Specifies the name of the SMA database. This database is used by some applications to store global SMA settings. Independent of the browser cache.
 - The default " " (empty) creates a file of name SMA.db
 - Do not change this setting.
- **NHibernateDbPath**
- Can include a Windows path. Default is " " (empty)
 - The default " " (empty) refers to the directory "C:\temp\SMA-DB"
 - The SMA database is stored in this directory. This database is used by some applications to store global SMA settings. Independent of the browser cache.
 - The default path or the configured path must have the same rights as the log directory (LogFolder)

2.4.6 How to configure a PZE Terminal on the MOC

Create a PZE terminal with terminal number **254 and type 254**.

System administration → Terminals → Terminal configuration

You must be careful to use the **terminal type 254** in the terminal configuration. This terminal type specifies the terminal as SMA terminal. This terminal type is not used when the licenses (e.g. AIP-HRF) are calculated and when system parameters for terminals are identified. This specification ensures that the number of required licenses is correctly calculated.

Define a PIN code for all persons who should use this function.

Master data → Staff → HR master data

The screenshot shows a web-based 'Edit' form for HR master data. The 'Personal data' tab is selected. The form contains several fields: 'Nationality' (dropdown with 'D'), 'Date of birth' (dropdown with '4/1/1953'), 'Standard console' (text input), 'Pin code' (text input with '****'), 'Confirmation' (text input with '****'), 'Gender' (radio buttons for 'Male' and 'Female'), and 'Family status' (radio buttons for 'Single', 'Married', 'Widowed', and 'Divorced'). The 'Pin code' and 'Confirmation' fields are highlighted with a red rectangular box.

2.4.7 MW3.0 services with MW2.1 system

Install the MW3.0 services for each MW2.1 instance you want to connect. For this purpose, Tomcat/WSP must be installed on the relevant system and connected with the MW2.1 system. Install the current version of the MW3.0 Java package.

The following instructions apply:

- MW30 document: **Inst_hydra_mw30.pdf**
- Document to install the MOC with MW 3.0 on an MW 2.0 / 2.1 system: **inst_moc30-mw20-21.pdf**

How to proceed:

- Disable the environment variable JHYDRADIR
- Install Tomcat/WSP (memory: 1024 MB)
- Configure JHYDRADIR according to the specifications included in the MW30 document.
- Import the Java package according to the MW30 document
- Configure the path Spoolwsc

2.4.7.1 Tomcat/WSP is already installed

Install a current MaintenanceManager (2), if Tomcat/WSP is already installed. Then you can import the current JavaPackage.

Important: HYDRA's Java interface must run (can be checked/started via the HYDRA manager)

Important: With MW 2.1, every SMA client occupies one HYD-KE license (console license). If you use SMA as a time and attendance client, a corresponding number of licenses should be available.

2.4.7.2 Applications for MW2.1

The following applications have been tested for MW2.1 and/or note the following exceptions:

General:

Service pack, week 49/2013 must be available in order to enable the **installation with MW2.1**

Time recording:

The program hyt_stmparc (version 7.2.1.22) must be available in the HYDRA server directory in order to use all time & attendance functions with MW2.1. **Important: Required version at least 7.2.1.22**

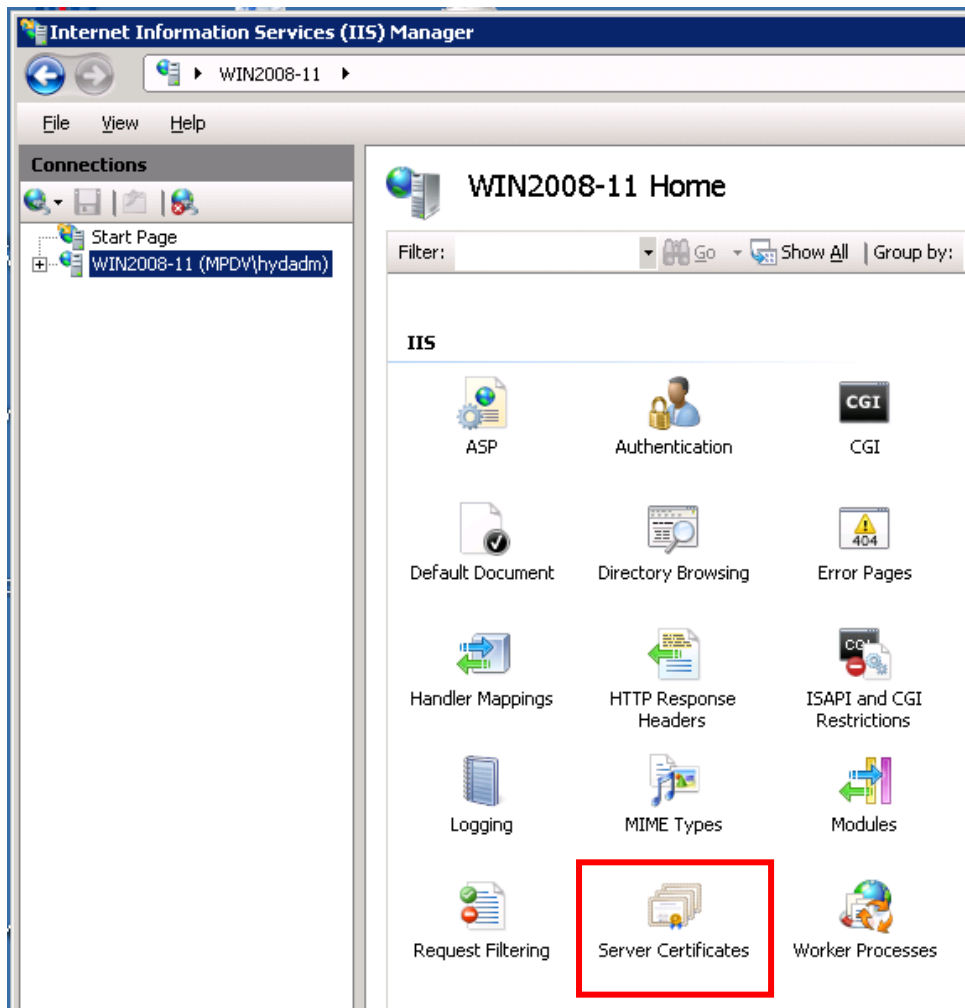
Windows: hyt_stmparc.exe

Linux: hyt_stmparc.out

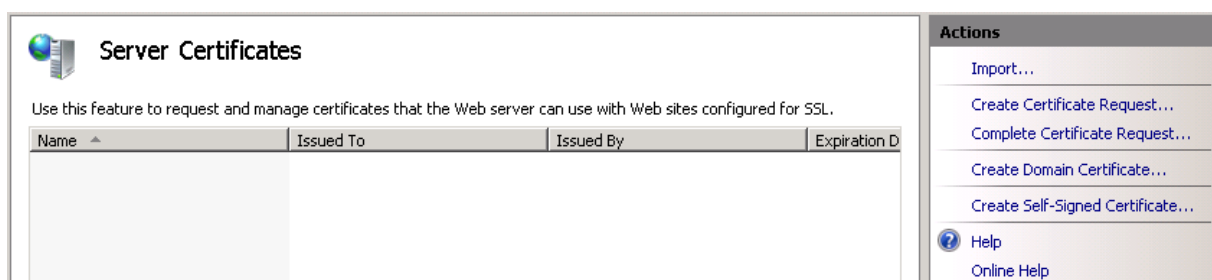
2.4.8 HTTPS activation – client to SMA server

If you want to secure the connection to the SMA server via HTTPS, then activate HTTPS as follows:

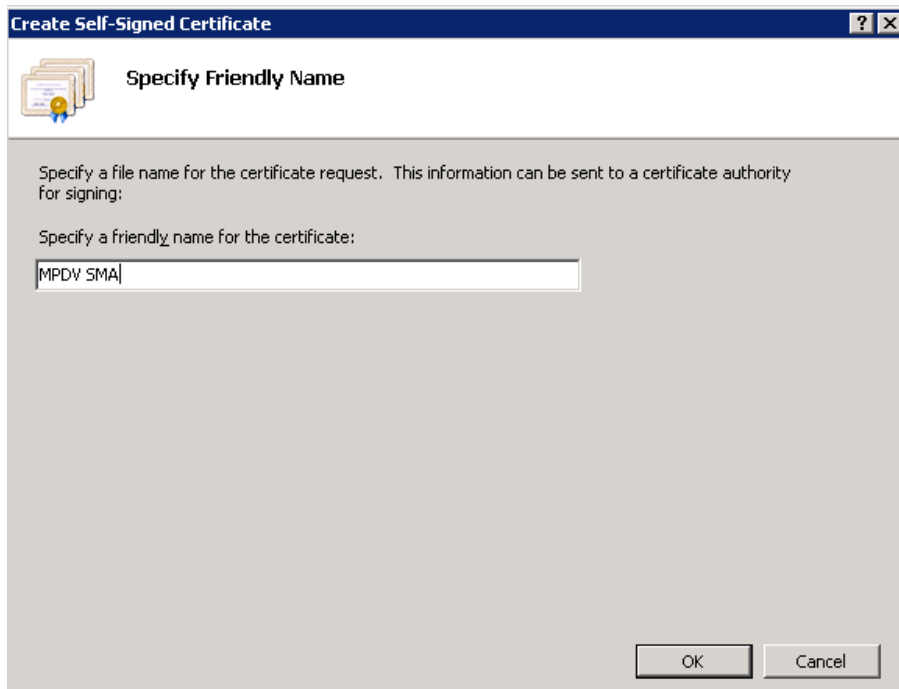
First of all, provide a certificate for the secured connection. To do so, click the IIS menu item "server certificates":



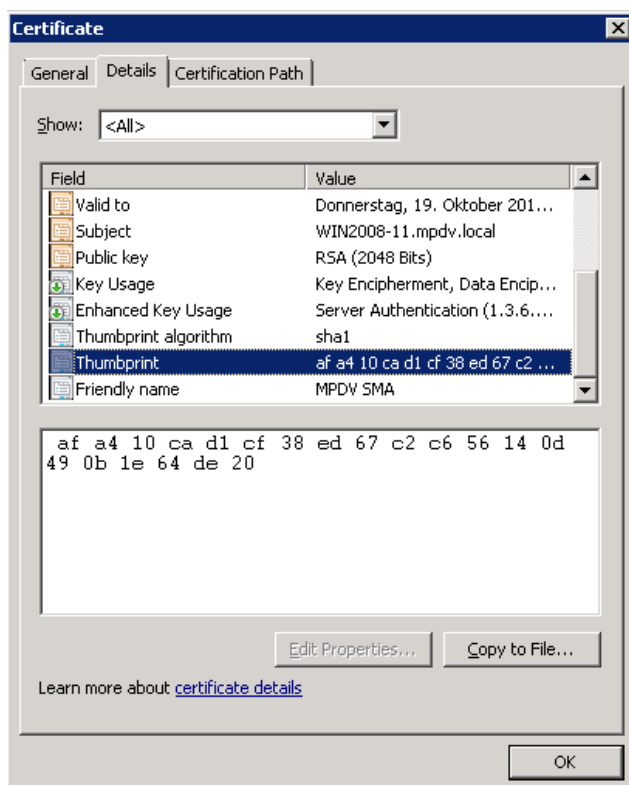
The dialog that opens shows the available certificates and the different methods to create new certificates or to import existing certificates:



This example recommends creating a self-signed certificate ("Create self-signed certificate").
Use "MPDV SMA" as the certificate name:



Enter the name and confirm by clicking OK. Then double-click the created certificate to specify its thumbprint.

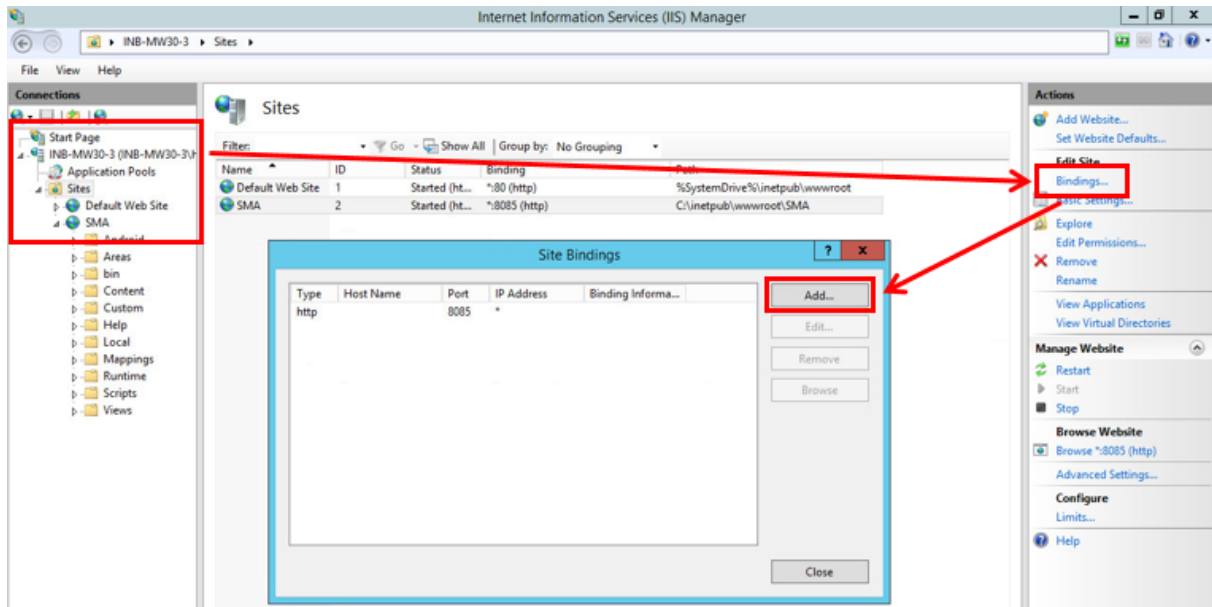


Copy the thumbprint and keep it for later use.

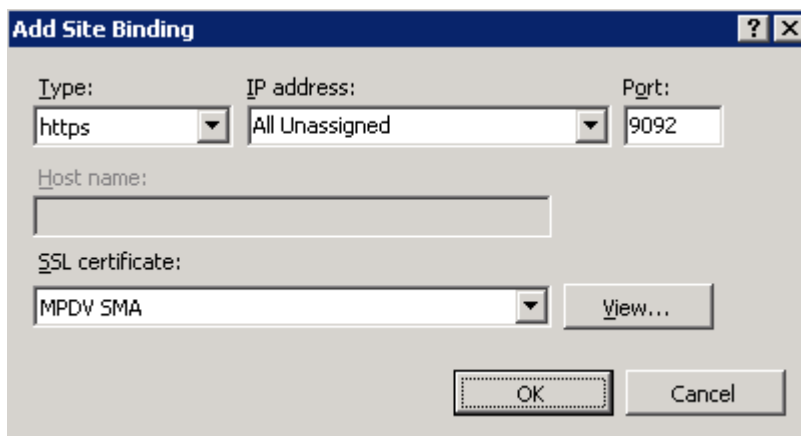
Then add an https site binding to the IIS.

To do so, select the option "Bindings..." in the SMA folder of IIS.

Select "Add..." in the "Site bindings" dialog that opens.



The following dialog is displayed:



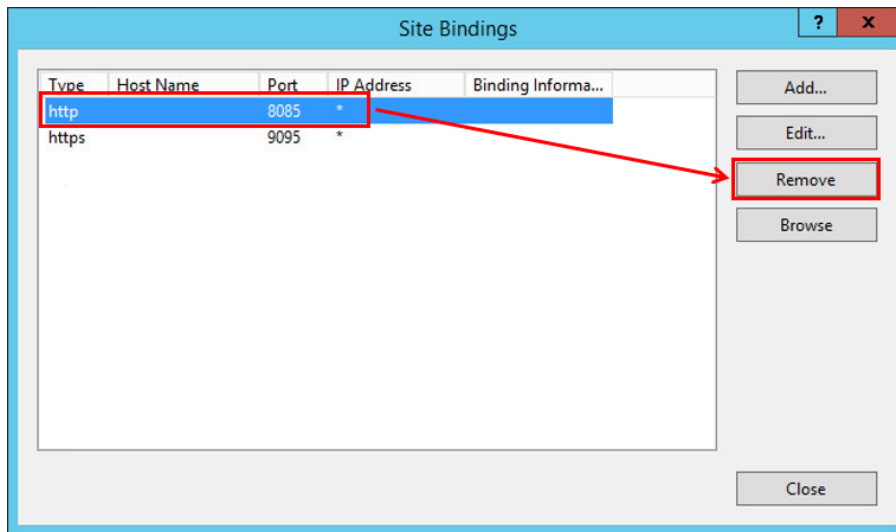
Select "https" as the site binding type. We recommend using port 9092.

You can also choose any other free ports.

For the "SSL certificate" option, select the certificate "MPDV SMA" to secure the https connection.

If your IT department provides a specific certificate, you can select it alternatively.

If required, remove the "insecure" connection via "http":



Finally, restart the IIS server service.

You can now reach the SMA server via the address <https://<SMA-Server>:9092/> (see example).

2.4.9 HTTPS activation – SMA server to HYDRA

If the HYDRA Instance Service provides an address via the https protocol, the SMA server connects to HYDRA via https protocol.

2.4.9.1 Configuration in SMA

If the HYDRA Instance Service expects a connection with the https protocol, you must specify the value for the entry "MasterServer" with protocol in the SMA configuration file (default storage location: `c:\inetpub\wwwroot\SMA\Web.config`).

Important: You must use the "fully qualified domain name"

Example: `https://server.company.de`

2.4.9.2 Configuration in HYDRA

Specify the entry for the system with the https protocol in the HYDRA Instance Service configuration (`Instancerepo.properties`) as follows:

CONFIG: HYDRA\HyInstMgrDir\Instancerepo.properties

HYDRA.host.1=https://<HYDRA-server>.<domain>

IMPORTANT!

This entry also affects the system selection during MOC login! If the SMA server communicates with HYDRA via HTTPS, then the MOC must do the same!

Important: You must use the "fully qualified domain name"

The WSP configuration (application.properties) must be extended by an SSL certificate so that the WSP communicates via https. Example:

CONFIG: HYDRA\WSP\config\application.properties

server.ssl.key-store=<HYDRA>\certs\<ssl-key-store>.jks

server.ssl.key-store-password=<key-store-password>

IMPORTANT!

If the SMA server communicates with HYDRA via HTTPS, then the MOC must do the same!

2.4.10 MQTT

Some SMA applications can update data via push notifications. You use MQTT for this feature. You use the WebSocket protocol for the connection to the MQTT broker.

Important: The browser directly connects to the MQTT broker. It might be required to change the IT infrastructure here. Consider this for the system architecture.

The connection is made according to the configuration stored in HYDRA for the MQTT broker.

2.4.11 Label printing

If you use the data collection functions, you can also trigger a label print if a relevant configuration is available for the data collection functions in HYDRA.

In SMA, you must only specify the label print service in the SMA configuration.

SMA configuration file (default storage location: **c:\inetpub\wwwroot\SMA\Web.config**): Set the value for the setting "LabelPrinterService".

Example: `http://<label-printer-service-server>:<port>/`

2.4.12 Optional: Import SMA function profile

The database patch `dbp_sma_function_profiles.hsc` creates an SMA function profile that includes all currently available SMA functions authorizations.

2.5 Test SMA

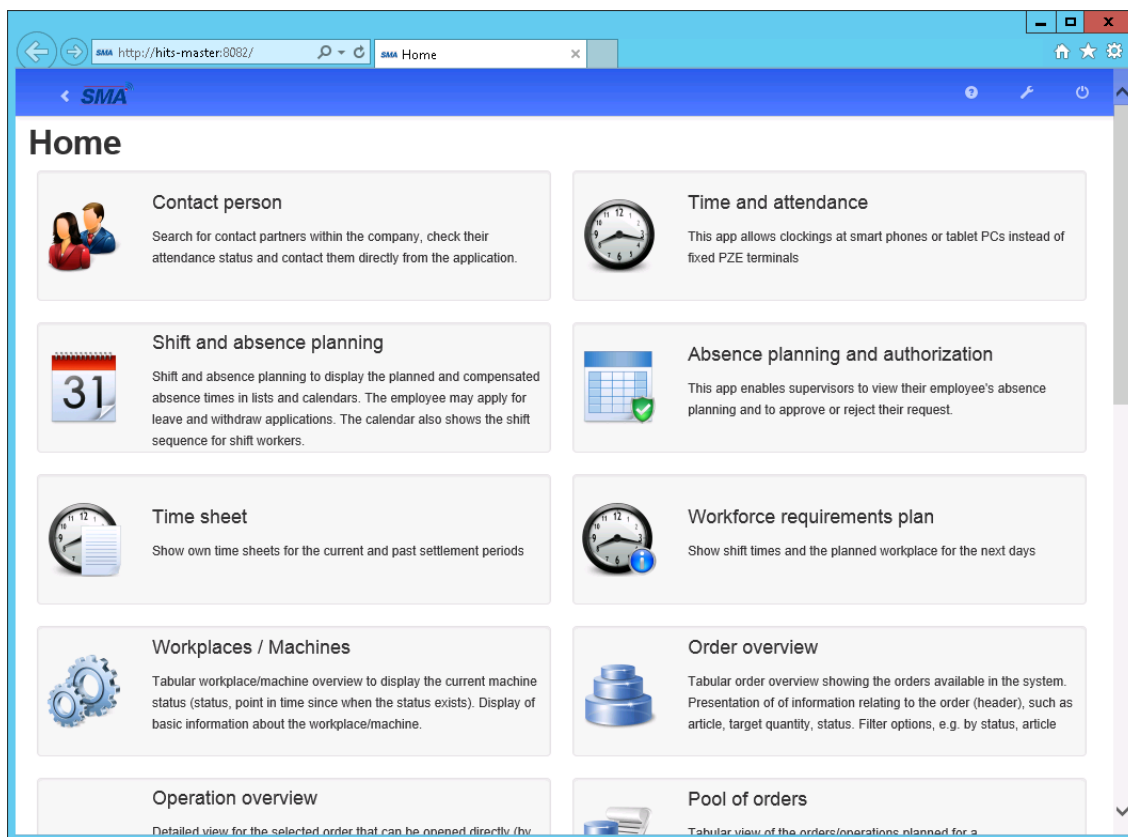
In order to test SMA applications successfully, create a user with appropriate SMA function authorizations (see section 2.4.12).

Start the **application** in the web browser (**important:** as stated in the specifications, you need an HTML5 capable browser. See also the HW_SW_GUIDE.pdf; section 3 HYDRA Smart MES Applications (SMA).)

[http://%hostname%:\[port\]](http://%hostname%:[port])

Example:

<http://mos-srv-01:8082>



Click *Workplaces / machines*

User: 12345

HYDRA password (by default: mpdv)

Workplaces / Machines

Machine ID	Machine Name	Status
50610	Engel	REPAIR
50611	Arburg Selogica 320S	OK
50612	Arburg Selogica 320S	Warning
50614	Arburg C/Dialogica 270C	Warning
50623	Krauss Maffei	OK
60610	Engel speed 380 / 55	OK
60611	Arburg Selogica 320S	Warning
60612	Arburg Selogica 320S	Warning
60614	Arburg C/Dialogica 270C	OK

50610 - Engel

Status

Status	REPAIR
Status since	09.03.2015 21:05:02
Duration so far	37:53:20
Expected end	-

Order

Article	SAR-FKA-9005
Article designation	Mirror right, black
MES order number	S30000810200
Operation designation	Quality check

Workplace master data

Comp.	01
Cost center	4010
Group	SPRGUSS1

Workplaces / Machines

Machine ID	Machine Name	Status
50610	- Engel	REPAIR
50611	- Arburg Selogica 320S	
50612	- Arburg Selogica 320S	
50614	- Arburg C/Dialogica 270C	
50623	- Krauss Maffei	
60610	- Engel speed 380 / 55	
60611	- Arburg Selogica 320S	
60612	- Arburg Selogica 320S	
60614	- Arburg C/Dialogica 270C	

50610 - Engel

Status

Status	REPAIR
Status since	09.03.2015 21:05:02
Duration so far	37:53:20
Expected end	-

Order

Article	SAR-FKA-9005
Article designation	Mirror right, black
MES order number	S30000810200
Operation designation	Quality check

Workplace master data

Comp.	01
Cost center	4010
Group	SPRGUSS1

Click the *Home* button

Home

Contact person
Search for contact partners within the company, check their attendance status and contact them directly from the application.

Time and attendance
This app allows clockings at smart phones or tablet PCs instead of fixed PZE terminals

Shift and absence planning
Shift and absence planning to display the planned and compensated absence times in lists and calendars. The employee may apply for leave and withdraw applications. The calendar also shows the shift sequence for shift workers.

Absence planning and authorization
This app enables supervisors to view their employee's absence planning and to approve or reject their request.

Time sheet
Show own time sheets for the current and past settlement periods

Workforce requirements plan
Show shift times and the planned workplace for the next days

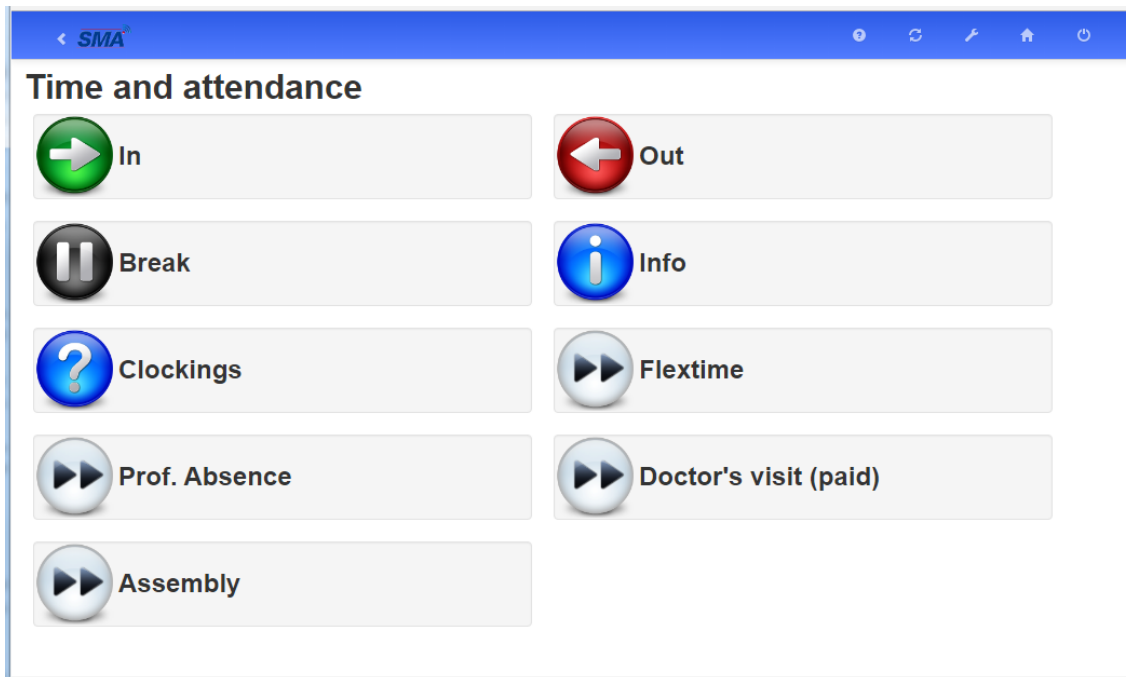
Workplaces / Machines
Tabular workplace/machine overview to display the current machine status (status, point in time since when the status exists). Display of basic information about the workplace/machine.

Order overview
Tabular order overview showing the orders available in the system. Presentation of information relating to the order (header), such as article, target quantity, status. Filter options, e.g. by status, article

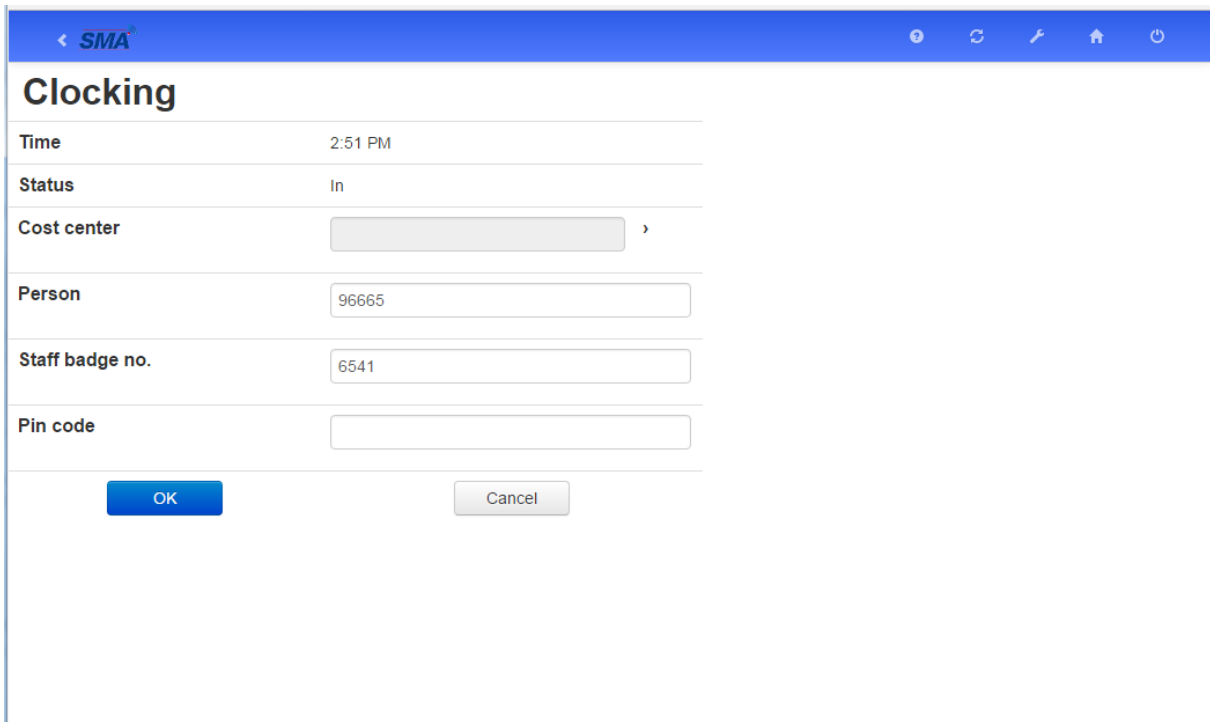
Operation overview
Detailed view for the selected order that can be opened directly (by clicking on the order number)

Pool of orders
Tabular view of the orders/operations planned for a

Click *Time and attendance*:



Click *In*

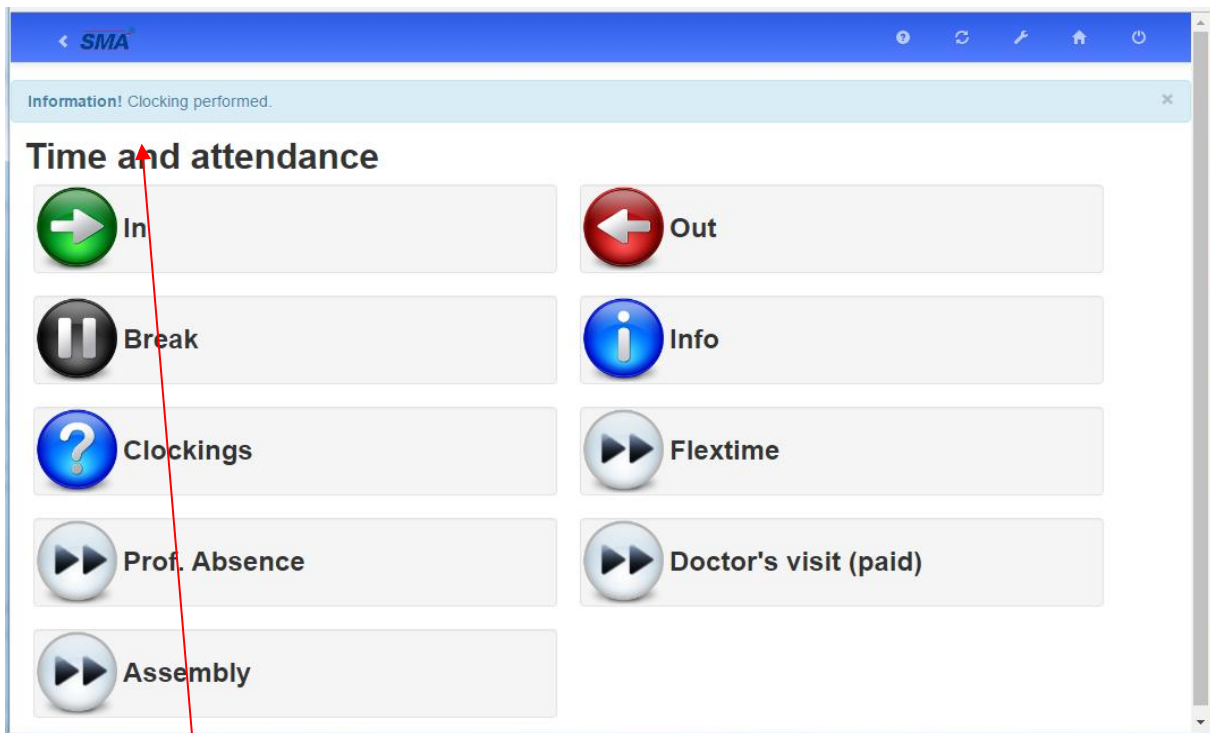


The screenshot shows the 'Clocking' dialog box in the SMA-CMS interface. The dialog box has a title bar with the SMA logo and several icons. The main content area contains the following fields and labels:

- Time:** 2:51 PM
- Status:** In
- Cost center:** [Empty field with a dropdown arrow]
- Person:** 96665
- Staff badge no.:** 6541
- Pin code:** [Empty field]

At the bottom of the dialog box, there are two buttons: 'OK' (blue) and 'Cancel' (gray).

Click *OK*



The clock-in has been carried out:

2.6 Install the SMA app

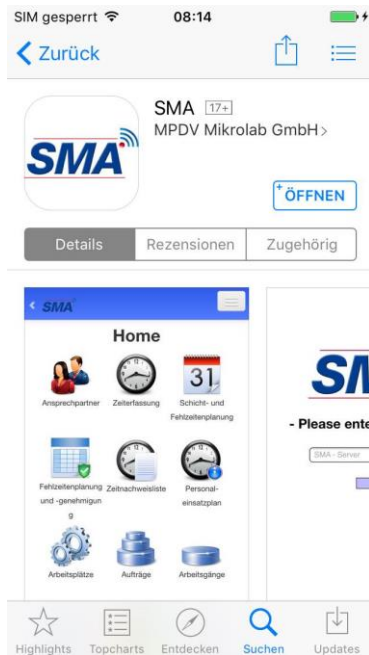
2.6.1 iOS – SMA App

2.6.1.1 Check requirements

- 1.) SMA is already installed.
- 2.) Apple Mobile Devices with operating system iOS 7.1 (or higher)
- 3.) Internet connection available on the device

2.6.1.2 Install SMA on the Apple Mobile Device

The Apple Store delivers SMA for iOS. Go to the Apple Store of the device and search for the SMA app:



You can install the app as usual.

2.6.1.3 Start the application

The overview of all installed applications shows an icon called "SMA". Use this icon to start the application. Clicking the icon starts the application.

An input field appears on the screen. Enter the IP address of the SMA server and its port in this field.

Structure: <SMA server>:<Port>

Example: **10.11.12.13:8082**

This entry is stored in the app cache, once a successful connection to the SMA server has been set up. When you start the SMA app the next time, the system automatically attempts to connect to the last successfully connected SMA system.

2.6.2 Android - SMA app

2.6.2.1 Check requirements

- 1.) SMA is already installed.
- 2.) The SMA installation includes the required files.
 - a. SMA server installation (mostly: C:\inetpub\wwwroot\SMA)

- b. SMA_armv7.apk available (mostly: C:\inetpub\wwwroot\SMA\Android)
- c. SMA_x86.apk available (mostly in: C:\inetpub\wwwroot\SMA\Android)

2.6.2.2 Transfer APKs to Android device

- 1.) According to the process architecture, transfer the package *SMA_armv7.apk* or *SMA_x86.apk* to the device.
- 2.) You can choose from the following options to transfer APKs to the Android device:
 - a. via USB
 - i. Copy the relevant APK to a computer that includes the required drivers to receive data from an Android device.
 - ii. Connect the device with the computer.
 - iii. Transfer the corresponding APK to the device.
 - b. via download
 - i. Open the following website in the browser of the device:
 1. SMA_armv7.apk:
http://<SMA server>:<Port>/Android/SMA_armv7.apk
 2. SMA_x86.apk:
http://<SMA server>:<Port>/Android/SMA_x86.apk
 - ii. Files are stored in the download folder.

2.6.2.3 APK installation

If you download APKs to the device, the APK files are included in the download history of the device.

If you transfer APKs to the device via USB, the APK files are included in the folder you selected during the transfer.

If the device's security policies prevent APKs from being installed, go to the device's properties and allow installation of "unknown sources".

Depending on the Android version and manufacturer, you can find the settings for installing "unknown sources" elsewhere. If you cannot find the settings, ask your IT department how to enable the installation of "unknown sources" for APK files with your device and Android version.

2.6.2.4 Start the application

The overview of all installed applications shows an icon called "SMA". Use this icon to start the application. Clicking the icon starts the application.

An input field appears on the screen. Enter the IP address of the SMA server and its port in this field.

Structure: <SMA server>:<Port>

Example: **10.11.12.13:8082**

This entry is stored in the app cache, once a successful connection to the SMA server has been set up. When you start the SMA app the next time, the system automatically attempts to connect to the last successfully connected SMA system.

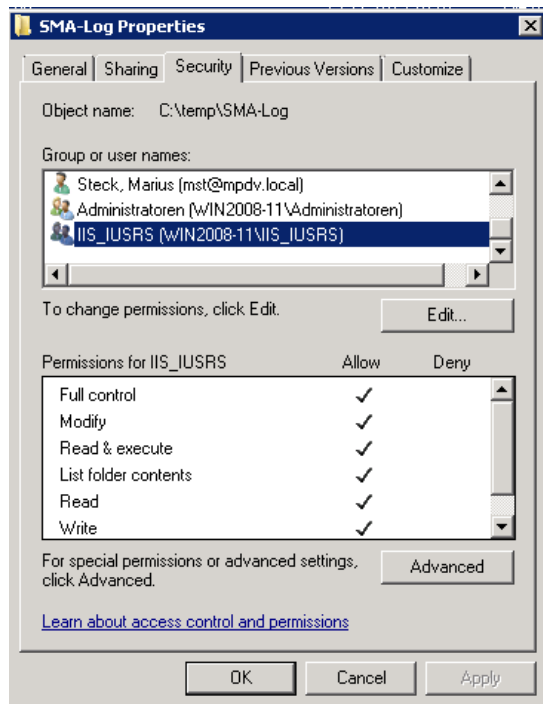
2.7 Tips + Tricks

2.7.1 Write permissions to the log directory

You need SMA write permissions to the log directory. You can configure this log directory in the WEB.config file.

In order for the SMA to function properly, the user executing the IIS must have full access to the log folder, as configured in c:\inetpub\wwwroot\SMAWEB.config.

Right-click the *log* folder → Properties, here: SMA

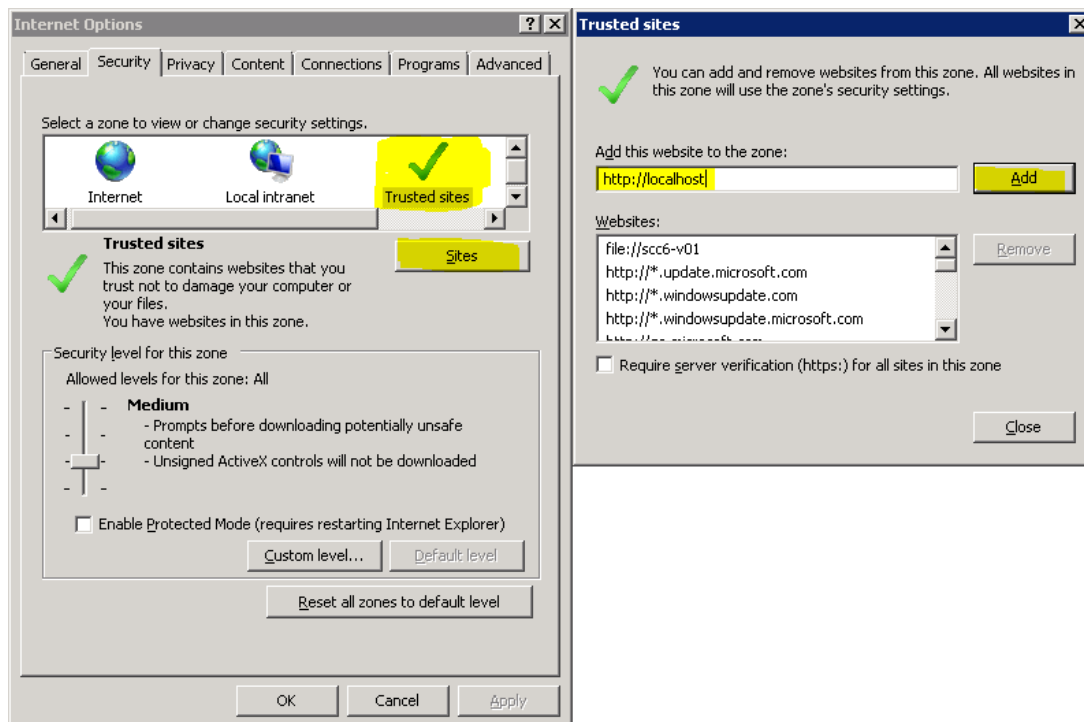
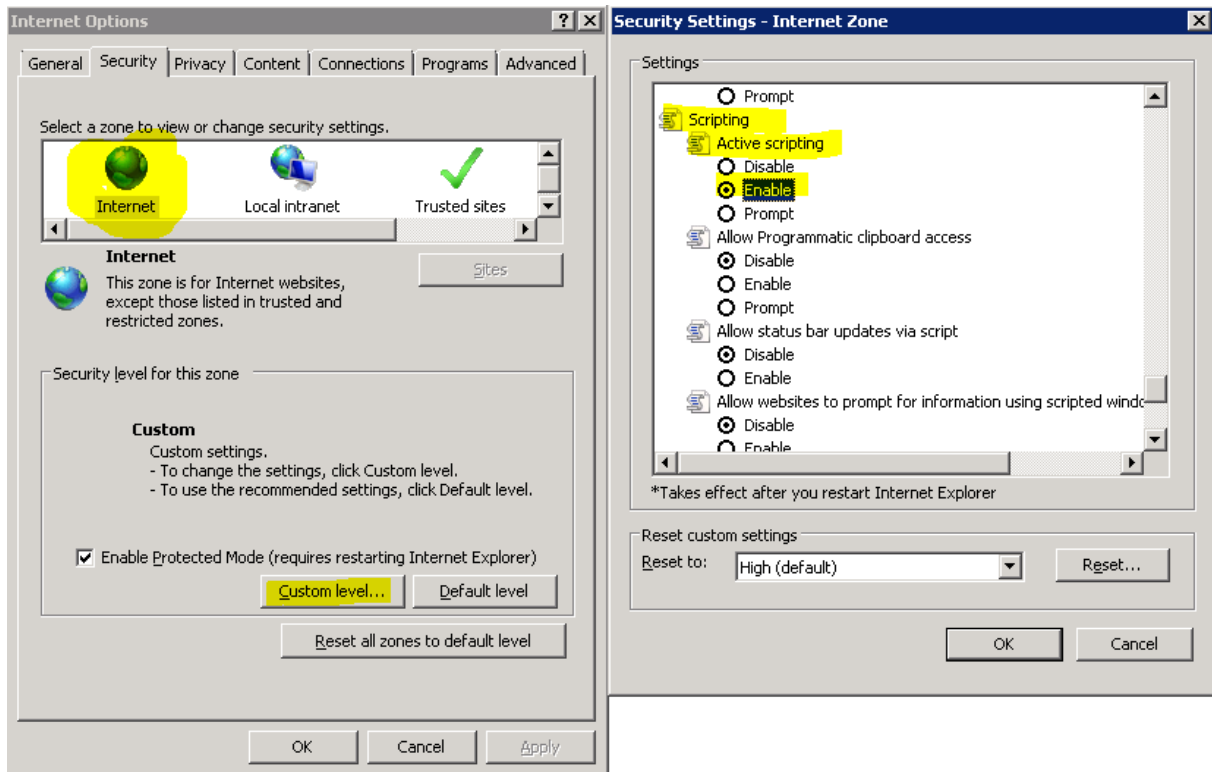


Go to the tab *Security* and assign full access to the user whose name starts with IIS.

2.7.2 The IE cannot identify mouse clicks

It is a security problem if the IE does not identify mouse clicks. The security settings for the website are too high and scripting is disabled.

In order to change this, enable *scripting* for the zone *Internet* and/or for *trusted sites*.



2.7.3 HTTP Error 403.14 - Forbidden

If the Windows Server 2008 R2 shows an error message 403.14 instead of the home screen, the following patch might not have been installed on the system: "Update for Windows Server 2008 R2 x64 Edition (KB980368)".

You can download this patch from the following address:

<http://www.microsoft.com/de-DE/download/details.aspx?id=5272>

2.7.4 McAfee VirusScan Enterprise (VSE) 8.8 Patch 4

If this version of the McAfee VirusScan is installed on the SMA server, then SMA cannot work without errors as this version changes the meta-information of files and therefore considerably affects the SMA system behavior.

For details on the topic, refer to: <https://community.mcafee.com/thread/74874?start=0&tstart=0>

Or to: <https://kc.mcafee.com/corporate/index?page=content&id=KB81595>

You can solve this issue by installing patch 5 for McAfee VirusScan Enterprise (VSE) 8.8. The problem has been sorted for this version of the VirusScan.

2.7.5 Request the SMA installation version

From revision "42972" of the SMA installation (dated 17 December 2015), you can use the following command to get the version of the SMA installation via a browser:

<http://<SMA-Server>:<SMA-Port>/?fct=GetVersion>

A site with the following content is displayed:

SMA Revision 42972

2.7.6 window.localStorage – path cannot be found

Problem:

Problem: The local storage is unknown to the Internet Explorer.

Cause:

A "cleaning" software was executed with administrator rights. This software changed the rights of files/folders. The XML files storing the local storage entries were also affected.

Effect:

That means, the user has no longer access rights to these files and/or cannot create files.

Solution:

Rename the following folder:

C:\Users\<user>\AppData\Local\Microsoft\Internet Explorer\DOMStore in DOMStore_save

Restart the IE.

Challenge:

The storage location varies from operating system to operating system and from IE version to IE version.

3 SMA Scripting

3.1 Introduction

This document describes the processing of SMA scripts.

In SMA, two types of scripts are available:

Changes to standard applications

Customer-specific applications

The following links can be helpful for the development:

jQuery: <http://jquery.com/>

jqPlot: <http://www.jqplot.com/>

moment.js: <http://momentjs.com/>

Bootstrap: <https://getbootstrap.com/docs/3.3/css/>

3.2 General

3.2.1 Reading the current SMA version

To read your current SMA version, open a browser and enter the following:

```
<server>:<port>/?fct=getversion
```

The output is as follows:

```
"SMA Revision 57709"
```

3.2.2 Activating SMA logging

The file "SMA.log" contains the logging of the SMA server. You can configure the file directory and the log level in the file "WEB.config" in the installation directory of the SMA server.

To activate the logging, make the following changes:

SMAWeb.config:

```
<appSettings>
  <!-- as default set to: C:\temp\SMA-Log -->
  <add key="LogFolder" value="C:\temp\SMA-Log" />
</appSettings>
```

```
<!-- LogLevel: Off|Trace|Debug|Info|Warn|Error|Fatal default = Off -->
<add key="LogLevel" value="Trace"/>
```

3.2.3 SMA structure

Create the following directory structure below the installation directory of the SMA server:

```
[Android]
[Areas]
[bin]
[Content]
[Custom]
    [Areas]
    [Content]
    [Scripts]
    [Views]
    [www]
[Help]
[Local]
    [Areas]
    [Content]
    [Scripts]
    [Views]
    [www]
[Mappings]
[Runtime]
[Scripts]
[Views]
Web.config
favicon.ico
CanConnect.html
NLog.xsd
packages.config
```

3.2.3.1 Areas

The folder *Areas* includes all standard applications.

Example for "Workplaces/machines"

\\SMA\\Areas\\Resource\\Views\\Resource\\index.cshtml

3.2.3.2 Custom or Local

The folder "Custom" includes all customer-specific applications, which were created by MPDV.

The folder "Local" includes all customizations, which were created by the customer.

The substructure is the same in both folders.

3.2.3.2.1 Areas

- The folder "Areas" includes all standard applications, which have been changed by MPDV/the customer.

Example: \SMA\Custom\Areas\Resource\Views\Resource\Index.cshtml for the application "Workplaces/machines".

- Data collection applications: \SMA\Custom\Areas\DataCapturing\DataCapturing.config.xml or \SMA\local\Areas\DataCapturing\DataCapturing.config.xml
- Web service configuration: \SMA\Custom\Areas\CustomService\Configurations\ or \SMA\local\Areas\CustomService\Configurations\

3.2.3.2.2 Views

In the folder "Views", you can change the Home screen.

\SMA\Views\Home\menu.xml

or \SMA\Views\Home\menu_add.xml

For more details, see section "Integrating an application in the Home screen".

3.2.3.2.3 www

Storage location for customer-specific applications:

css

js

html

3.2.3.3 Runtime

3.2.3.3.1 Language key

You use the directory "Runtime" to store the respective language keys/translations.

Standard: \SMA\Runtime\resources\standard\languages\

Custom: \SMA\Runtime\resources\custom\languages\

Local: \SMA\Runtime\resources\local\languages\

3.2.3.3.2 *Storage location for web services:*

Standard:

Properties: \SMA\Runtime\resources\standard\data\properties\

Services: \SMA\Runtime\resources\standard\data\services\

The subdirectory "Standard" includes the Configuration.xml and GuiConfiguration.xml of the standard.

Custom:

Properties: \SMA\Runtime\resources\custom\data\properties\

Services: \SMA\Runtime\resources\custom\data\services\

In the subdirectory "Custom", the Configuration.xml and GuiConfiguration.xml are stored by MPDV.

Local:

Properties: \SMA\Runtime\resources\local\data\properties\

Services: \SMA\Runtime\resources\local\data\services\

In the subdirectory "Local", the Configuration.xml and GuiConfiguration.xml are stored by the customer.

3.2.4 Custom web services

If a customer-specific web service has been created, the files configuration.xml and do.xml must be stored in the SMA directory and additionally in the following directories:

<server>\jhydradir\MOC\1\listInterpreter\custom\. for changes made by MPDV or

<server>\jhydradir\MOC\1\listInterpreter\local\. for changes made by the customer.

3.2.5 Changing the language



Note

GUI texts are texts included in *.cshtml files. Texts integrated into the GUI using a script/data are **not** included.

If you have created/customized mapping files, you must restart the SMA server. Changes only become effective after the restart.

Changing the language

All GUI texts in SMA applications use language keys.

Assignment of language keys to the currently used language:

Mapping files of the standard: SMA\Runtime\resources\standard\languages

Mapping files of MPDV: SMA\Runtime\resources\custom\languages

Mapping files created by the customer: SMA\Runtime\resources\local\languages

Language keys existing in the "local" scope overwrite standard language keys.

Standard	MPDV Customizing	Customer	Result
lkHome = Home			Home
lkHome = Home	lkHome = Your Home		Your Home
	lkHome = Home	lkHome = My Home	My Home
		lkHome = My Home	My Home

Finding the language keys used / customizing pages:

Special case Home screen (menu.xml)

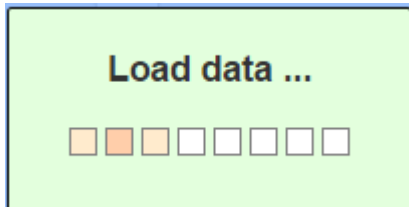
Heading of a page (example "SMA\Views\Home\Index.cshtml")

➔ ViewBag.Title = **Html.Translate("lkHome")**;

3.2.6 Show progress bar

To show a progress/loading bar, the following functions are available:

3.2.6.1 Infinite loading bar



```
$mpdv.progress.showDefaultProgress();
```

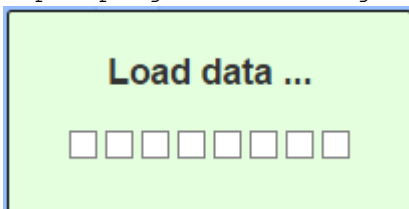
3.2.6.2 Loading bar showing progress

The value can be specified from 0 to 100:

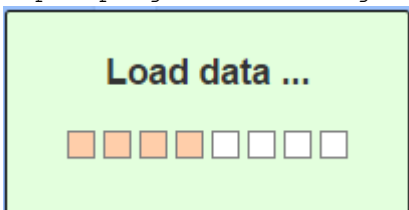
```
$mpdv.progress.showProgress(value);
```

Example:

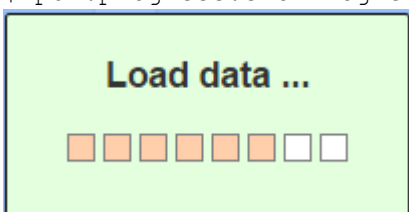
- ```
$mpdv.progress.showProgress(0);
```



- ```
$mpdv.progress.showProgress(50);
```



- ```
$mpdv.progress.showProgress(80);
```



### 3.2.6.3 Update

To implement an update for a progress bar, you must implement the following:

```
$mpdv.progress.updateProgress(value);
```

### 3.2.6.4 Reset

To reset a loading bar, use the following function:

```
$mpdv.progress.hideAndClearProgress();
```

## 3.3 Changes to Standard Applications

### 3.3.1 Customize applications

#### 3.3.1.1 General

#### Customizing a page using the example "Workplace/machine overview"

Copy the original application "SMA\Areas\Resource\Views\Resource\Index.cshtml" to "SMA\Local\Areas\Resource\Views\Resource\Index.cshtml".

Copy the file Web.config "SMA\Areas\Resource\Views\Web.config" to "SMA\Local\Areas\Resource\Views\Web.config".

Now you can change the file Index.cshtml in "SMA\Local\Areas\Resource\Views\Resource\".



If you change MPDV standard applications, the system always loads the customized pages.

That means:

- You cannot make use of MPDV's further developments for the application, as the system always uses the customized application by default.
- Therefore, no bug fixes can be issued for customized applications.

The customer is responsible for customized applications.

### 3.3.2 Web service calls

To integrate a web service in JavaScript, two functions are available.

The question if a filter parameter is required or not, identifies the correct web service.

### 3.3.2.1 Call without filter

```
$.post("/CustomService/CustomService/GetData", {
 configName: "training_mdunits_list"
})
.done(function (data) {
 console.log(data);
})
.fail(function (data) {
 showError(data);
});
```

### 3.3.2.2 Call with filter parameter

```
$.post("/CustomService/CustomService/GetData", {
 configName: "training_mdunits_list",
 filterJson: '{"Parameters":{"units.unit": {"0":"ST"}}}'
})
.done(function (data) {
 console.log(data);
})
.fail(function (data) {
 showError(data);
});
```

## 3.3.3 MPDV data collection - Version 1

You require the license "SMA-AMF" for the data collection applications of MPDV in SMA.

Configure the data collection in the following file:

SMA\Custom\Areas\DataCapturing\DataCapturing.config.xml

or

SMA\Local\Areas\DataCapturing\DataCapturing.config.xml

If the file "DataCapturing.config.xml" is not available:

Copy the file "DataCapturing.config.bsp.xml" and rename it to "DataCapturing.config.xml". Then change the file "DataCapturing.config.xml".

The "type" specifies which data collection function is started. If you customize a data collection function that is available by default (e.g. A\_AN), this standard function is overwritten.



### 3.3.3.1 Calling a data collection application

Calling a data collection application

```
http://<SMA-Server>:<Port>/DataCapturing/DataCapturing?captureType=<Type>
```

The parameter <type> includes the name of a configured data collection function.

Example: "Capturing" with attribute "Type" and value "A\_AN" exists:

```
http://<SMA server>:<Port>/DataCapturing/DataCapturing?captureType=A_AN
```

Calling a data collection application with transfer of value:

Example:

```
http://<SMA-Server>:<Port>/DataCapturing/DataCapturing?captureType=A_AN&MNR=1168&ANR=000040450010&MST=1
```

### 3.3.3.2 Configuration of a <Snippet>

**Attribute:**

Type: specifies the value type (ANR, MNR, ...)

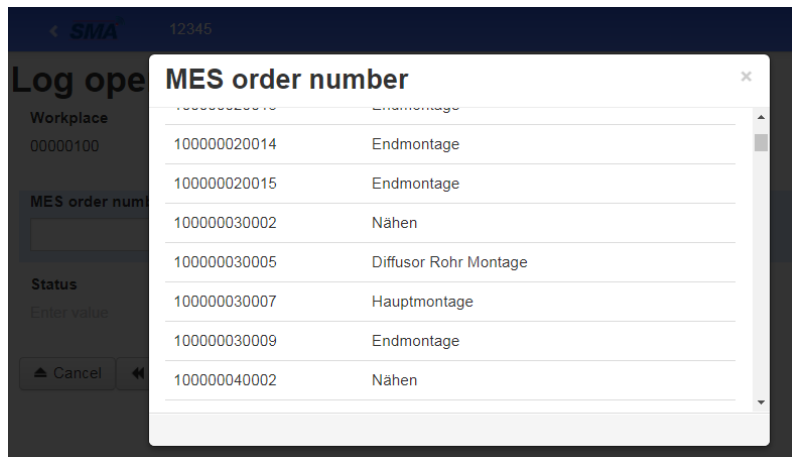
Lk: display name (language keys can be used)

InputType: "text", "number" or "password"; "text" is used by default

#### 3.3.3.2.1 *Selection list*

**SelectionList**

If the value is set to "true", a web service must be configured in the child nodes.



### 3.3.3.2.2 Displaying units

#### UnitList

If the values of both attributes, "UnitList" and "SelectionList", are set to "true", a web service must be configured in the child nodes.

### 3.3.3.2.3 Dependencies

#### IsDependent

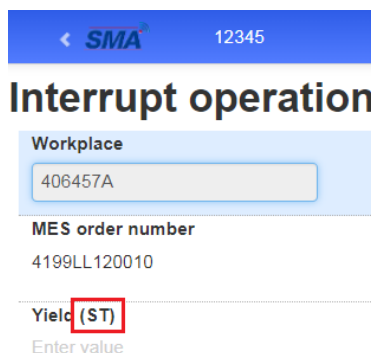
If the value is set to "true", the data is only loaded if the following conditions are fulfilled:

- A selection dialog is opened.
- An element responsible for the display of the unit has been changed.

#### DependingOn

Here, you define the dependent snippet.

For example, the display of a unit can depend on the order.



#### DependingFilterAcronym

If the attributes "SelectionList", "IsDependent" and/or the snippet "UnitList" is set to "true" and a dependency to a snippet is defined in the attribute "DependingOn", then you can dynamically set a configured filter in the web service configuration of the "snippet".

The attribute "DependingFilterAcronym" refers to the configured filter that you want to set.

Example: A filter by the operation number is set when the unit for yield is requested (depending on the selected operation).

DependingFilterAcronym (illustrated with an example):

```

MES order number
4199LL120010

Yield (ST)
Enter value

<Snippet Type="EGR:GUT" Lk="lkYieldQuantity" InputType="number" SelectionList="true" UnitList="true"
IsDependent="true" DependingOn="ANR" DependingFilterAcronym="operation.id">
 <WebService Name="BOOperationOverview" />
 <ResultParameters Primary="operation.plan.unit.base"/>
 <FilterParameters>
 <Parameters>
 <Parameter Acronym="operation.id" WebServiceType="string" InputAsArray="Y" Operator="IN">
 <Values>
 <string></string>
 </Values>
 </Parameter>
 </Parameters>
 </FilterParameters>
</Snippet>

```

"The value of the "snippet" ANR is used as value for the filter operation.id to request the unit.

### 3.3.3.2.4 Return parameter of the service

#### DefaultViaService

If the value is set to "true", you can identify the default value via the web service configuration that is included in the child nodes of the "snippet".

```

<Snippet Type="TLG" Lk="lkoperation.partitioning.target" InputType="text" DefaultViaService="true"
DefaultValue="operation.partitioning.target" IsDependent="true" DependingOn="ANR" DependingFilterAcronym="operation.id">
 <WebService Name="BOOperationOverview" />
 <ResultParameters Primary="operation.partitioning.target"/>
 <FilterParameters>
 <Parameters>
 <Parameter Acronym="operation.id" WebServiceType="string" InputAsArray="Y" Operator="IN"/>
 </Parameters>
 </FilterParameters>
</Snippet>

```

### 3.3.3.2.5 Setting the default of a field

#### DefaultValue

If no value is passed in the request, this value is set as default in the "snippet".

Example, preassignment of the badge number.

```
<Snippet Type="KNR" Lk="lkperson.card_id" InputType="number" SelectionList="false"
DefaultValue="GET_USER" Invisible="false"/>
```

GET\_USER provides the value of the user logged on.

#### 3.3.3.2.6 *Further attributes of a snippet*

**Invisible:** "Snippet" is not displayed.

Passes default value/transferred value (URL) in the background.

```
<Snippet Type="KNR" Lk="lkperson.card_id" InputType="number"
SelectionList="false" DefaultValue="GET_USER" Invisible="true"/>
```

**Readonly:** "Snippet" cannot be edited.

Protects a transferred value:

```
<Snippet Type="MNR" Lk="lkWorkplace" InputType="text" SelectionList="false"
Readonly="true"/>
```

**AdditionalClasses:** Permits to set style sheet classes for the "snippet".

Recommended class: "dottedDataCaptureBottomBorder".

#### 3.3.3.2.7 *Forwarding*

Forwarding after completion of data collection:

### Forward to customized page

Set the transfer parameters:

returnArea → for customized pages = "CustomService"

returnController → for customized pages = "CustomService"

returnAction → for customized pages = "Index"

returnCustomPageId → <application> (example: schulung\_ws)

**Note:**

Do not configure the attributes "SubmitButtonDestinationArea", "SubmitButtonDestinationController" and "SubmitButtonDestinationAction". Otherwise the parameters transferred are ignored.

Forwarding to a customized page in SMA. Examples:

**For example, using a configuration:**

```
<Capturing Type="A_AN" Lk="lkAAn" SubmitButtonLk="lkLog_on"
SubmitButtonDestinationArea="CustomService"
SubmitButtonDestinationController="CustomService"
SubmitButtonDestinationAction="Index"
SubmitButtonDestinationPageId="schulung_ws">
 <Snippets>
```

**For example, using a transfer parameter in the URL:**

```
http://<SMA server>:<Port>/DataCapturing/DataCapturing?captureType=A_AN
&returnArea=CustomService&returnController=CustomService
&returnAction=Index&returnCustomPageId=schulung_ws
```

### 3.3.4 MPDV data collection - Version 2

Version 2 is used as of service pack 13.

You can find the standard configurations of data collection functions in:

SMA\Areas\DataCapturing2\Configurations\

The configuration is described here:

SMA\Areas\DataCapturing2\Configurations\DataCapturing2.config.bsp.xml

Storage location for data collection functions you customized:

SMA\Custom\Areas\DataCapturing2\Configurations\

or

SMA\Local\Areas\DataCapturing2\Configurations\

This directory imports all files (also files stored in subfolders) matching the following pattern:

DataCapturing2.\*config.xml

Example: DataCapturing2.cut\_sma.config.xml

You can configure multiple data collection functions in one configuration file for data collection functions.

For a better overview, we recommend to use one data collection function per configuration file.

### 3.3.4.1 Request a data collection application

Request a data collection application

`http://<SMA-Server>:<Port>/DataCapturing2/DataCapturing2?captureId=<Id>`

The parameter <Id> includes the ID of a configured data collection function.

Example: "Capturing" with attribute "Type" and value "A\_AN\_2":

`http://<SMA server>:<Port>/DataCapturing2/DataCapturing2?captureId=A_AN_2`

Request a data collection application and transfer values:

Example:

`http://<SMA-Server>:<Port>/DataCapturing2/DataCapturing2?captureId=A_AN_2&MNR=1168&ANR=000040450010&MST=1&cancelUrl=`

You can use the parameters CancelUrl or SubmitUrl to declare the target URLs when you have interrupted or completed the data collection function.

The URL /DataCollection/DataCollection is set by default.

### 3.3.4.2 Configuration of a <Capturing>

`<Capturing Id="" Type="" Lk="" SubmitButtonLk="" CancelUrl="" SubmitUrl="">`

#### Attributes:

ID: configuration ID, unique ID

Type: key/name/web service of the data collection function (example of a dialog: A\_AN, of a web service: MDUnits.list)

Lk: heading/title (language keys can be used)

SubmitButtonLk: text of the button to complete a data collection function

CancelUrl: defines a target URL to cancel the collection function

SubmitUrl: defines a target URL after completing the collection function

### 3.3.4.3 Configuration of a <Snippet>

```
<Capturing>
 <Snippets>
 <Snippet Type="" Lk="" Inputtype="" SelectionType="" Readonly="" Invisible=""
 DefaultUnit="" DefaultValue="" TargetId="" Barcode="" SubmitEmptyValues="">
 <DataWebService />
 <UnitWebService />
 <DataMapping />
 </Snippet>
 </Snippets>
</Capturing>
```

#### Attributes:

Type: specifies the value type (ANR, MNR, ...)

Lk: display name (language keys can be used)

InputType: "text", "number" or "password"; "text" is used by default

#### 3.3.4.3.1 Setting the default of a field

##### DefaultValue

If no value is passed in the request, this value is set as default in the "snippet".

Example, preassignment of the badge number.

```
<Snippet Type="KNR" Lk="lkperson.card_id" InputType="number" SelectionList="false"
DefaultValue="GET_USER" Invisible="false"/>
```

GET\_USER provides the value of the user logged on.

GET\_BATCH provides a batch number that is specified using the web service BOBatch.createBatchId.

##### DefaultUnit

If no value is transferred via URL, the system sets the configured value.

(You can use language keys.)

### 3.3.4.3.2 Further attributes of a snippet

**Invisible:** "Snippet" is not displayed.

Passes default value/transferred value (URL) in the background.

```
<Snippet Type="KNR" Lk="lkperson.card_id" InputType="number"
SelectionList="false" DefaultValue="GET_USER" Invisible="true"/>
```

**Readonly:** "Snippet" cannot be edited.

Protects a transferred value:

```
<Snippet Type="MNR" Lk="lkWorkplace" InputType="text" SelectionList="false"
Readonly="true"/>
```

**Barcode** The "snippet" can be scanned via bar code and camera.

**SelectionType:** The "snippet" can be declared as list, table or input field, for example.

The following values are available:

<empty>: If you do not enter a configuration, the system sets the configuration value "simple" by default.

simple: You can enter data in a simple input field.

list: You can enter data using a modal selection list.

table: You can choose data from a table selection below the input field.

view: You cannot enter values directly. The value is not passed to the server.

viewList: You cannot enter values directly. The value is not passed to the server.

viewTable: You cannot enter values directly. The value is not passed to the server.

button: Shows a button that opens a configured data collection function. Once this function is completed, the current data collection function is opened again. Configure the attribute "TargetId" with the ID of the data collection function you want to open with the button.

label: You cannot enter values directly. The value is passed to the server.

**TargetId:** Only required for the SelectionType "button". To configure this Id, use the Id of the data collection application that you call using SelectionType "button".



## SubmitEmptyValues:

If the value is "true", the system also passes "empty" values when the input dialog is sent.

If the value is "false", the system does not pass "empty" values when the input dialog is sent.

### 3.3.4.4 Data configuration of a snippet

**DataWebService** includes the data configuration for selection lists/tables.

```
<DataWebService Name="" ServiceType="">
 <ResultParameters>
 <Parameters>
 <Parameter Acronym="" IsKey="" IsFor="" Lk="" FormatType="" IsDescription=""/>
 </Parameters>
 </ResultParameters>
 <FilterParameters>
 <Parameters>
 <Parameter Acronym="" WebServiceType="" InputAsArray="" Operator="" DependingOnType="">
 <Values>
 <string></string>
 </Values>
 </Parameter>
 </Parameters>
 </FilterParameters>
</DataWebService>
```

## Configuration of attributes

**Name:** Name of the HYDRA web service in dot notation, i.e. MDUnits.list becomes MDUnitsList.

**ServiceType:** Includes the web service type.

The following types are available:

<empty> or "webservice": A web service requests the data.

"dlg": The data is identified using a dialog (DLG) that returns result parameters.

"dlgList": The data is identified using a dialog (DLG) that returns a result table.

**UnitWebService:** includes the configuration for the display of the unit.

**DataMappings:** includes the value assignment for the transfer parameters when you call a data collection function via SelectionType and value "button".

### 3.3.4.5 Customization of the Home screen

You can add further applications in the file SMA\Local\Views\Home\menue\_add.xml. This will add the application to the end of the Home screen.

```
<?xml version="1.0"?>
<ArrayOfButtonView xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xmlns:xsd="http://www.w3.org/2001/XMLSchema">
```

```

<ButtonView>
 <id>QuantityCalculator</id>
 <Label>lkQuantityCalculator</Label>
 <LabelMini>lkQuantityCalculator</LabelMini>
 <Description>lkQuantityCalculatorDesc</Description>
 <Icon>Content/img/ShippingBoxesClosed_70x70.png</Icon>
 <Action>
 <Target>location='../CustomService/CustomService?pageId=quantity_calculator'</Target>
 </Action>
 <LicensedIn>SMA-CMS</LicensedIn>
</ButtonView>
</ArrayOfButtonView>

```

The file menu\_add.xml includes a list of ButtonView elements. Each ButtonView element stands for a new menu entry.

Structure of the ButtonView element:

ID: unique reference of the element

Label: title of the menu item

LabelMini: title of the menu item shown on small displays

Description: descriptive text (the description is not shown on small displays)

Icon: image of the application

Action: includes a "target" tag that is executed when you click the element.

LicensedIn: authorization key that controls the visibility of the application on the Home screen

To delete an application in the Home screen, you just configure the application with the unique ID in the file SMA\Local\Views\Home\menu\_rm.xml.

Use the file SMA\Views\Home\menu.xml to identify the IDs of the standard applications.

```

<?xml version="1.0"?>
<ArrayOfButtonView xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xmlns:xsd="http://www.w3.org/2001/XMLSchema">
 <ButtonView>
 <id>WorkplaceOverview</id>
 </ButtonView>
 <ButtonView>
 <id>OrderOverview</id>
 </ButtonView>
</ArrayOfButtonView>

```

The file SMA\Local\Views\Home\menu\_sorting.xml creates the sorting of the Home screen.

The file menu\_sorting.xml includes a list of ButtonView elements. You only have to configure the ID (tag: "Id") to specify the sort order of applications.

If an application is not included in the sort order, the application is added according to the standard configuration (menu.xml).

```

<?xml version="1.0"?>

```

```
<ArrayOfButtonView xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">

 <ButtonView>
 <id>WorkplaceOverview</id>
 </ButtonView>
 <ButtonView>
 <id>OrderOverview</id>
 </ButtonView>
</ArrayOfButtonView>
```

For further configurations of the SMA Home screen, refer to the following section 3.4.2 Title of a customer-specific application.

### 3.3.5 Modal dialog

The HTML of a modal dialog is based on twitter bootstrap 2.3.2 and is made up of three main parts:

Header

Body

Footer

Example (creating a modal dialog with the name "exampleModal"):

```
<div id="exampleModal" class="modal hide" tabindex="-1" role="dialog" aria-labelledby="detailActionLabel" aria-
hidden="true">

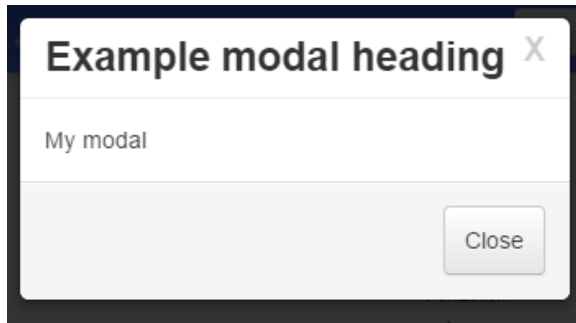
 <div class="modal-header">
 <button type="button" class="close" data-dismiss="modal" aria-hidden="true" id="exampleModal-
closeDialogX">X</button>
 <h3 id="exampleModal-group_header">Example modal heading</h3>
 </div>

 <div class="modal-body" style="overflow-x: hidden;" id="exampleModal-body">
 My modal
 </div>

 <div class="modal-footer">
 <div style="float: left;">
 <!-- maybe an other button -->
 </div>
 <div style="float:right;">
 <button class="btn button-big-auto" data-dismiss="modal" aria-hidden="true" id="exampleModal-
closeDialog">Close</button>
 </div>
 </div>
</div>
```

You use the following function to call the modal dialog created:

```
$mpdv.components.modal.openModal("exampleModal");
```



## 3.4 Customer-Specific Applications

### 3.4.1 Calling the application via URL in the browser:

Structure of the URL:

http://<SMA server>:<Port>/CustomService/CustomService?pagelId=<application>

SMA server: Name/IP of the server where SMA is installed.

Port: Port of the SMA server where SMA runs.

Application: Name of the created HTML file without file extension.

### 3.4.2 Title of a customer-specific application

You can specify a title for an application in HTML.

Example for the title "Test":

```
<!--
#mpdv_title: Translate(lkTest)
-->
```

### 3.4.3 Integrating files in HTML

#### 3.4.3.1 Integrating CSS files

If you want to use style sheets in an HTML application, you must integrate the following in the HTML page.

Example for the integration of the file test.css, which is included in the directory SMA\local\www\css\:

```
<link href="../../local/www/css/test.css" rel="stylesheet">
```

### 3.4.3.2 Integrating JS files

If you want to use Javascript files in an HTML application, you must integrate the following in the HTML page.

Example for the integration of the file test.js, which is included in the directory SMA\local\www\js\:

```
<script src="../../../local/www/js/test.js"></script>
```

### 3.4.4 Integrating an application in the Home screen

Copy the Home screen and its configuration and store both in the folder for customizations.

Copy from SMA\Views\Home\Index.cshtml to SMA\Local\Views\Home\Index.cshtml

Copy from SMA\Views\Home\menu.xml to SMA\Local\Views\Home\menu.xml

If you already use a customized Home screen, you may skip this step.

### Integrating the application into SMA (customizing Home screen):

Extend the following line of code in the file "SMA\Local\Views\Home\Index.cshtml":

```
var menuPath = Path.Combine(AppDomain.CurrentDomain.BaseDirectory, "Views", "Home", "menu.xml");
new
var menuPath = Path.Combine(AppDomain.CurrentDomain.BaseDirectory, "Local", "Views", "Home",
"menu.xml");
```

With this extension, you can customize the file menu.xml to your requirements.

You can also add a new menu item as follows:

\SMA\Views\Home\menu\_add.xml

This file must only include the new entry.

Example:

```
<?xml version="1.0"?>
<ArrayOfButtonView xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
 <ButtonView>
 <Label>lkTest</Label>
 <LabelMini>lkTest</LabelMini>
 <Description>lkTestDesc</Description>
 <Icon>/Content/img/Generators_70x70.png</Icon>
 <Action>
 <Target>location='../CustomService/CustomService?pageId=test'</Target>
 </Action>
```

```

 <DoNotDissMissModal>false</DoNotDissMissModal>
 <LicensedIn>SMA-CMS</LicensedIn>
 </ButtonView>
</ArrayOfButtonView>

```

The file is merged with the menu.xml and attached to it.

**Note:** <LicensedIn> includes the respective license required.

Customizations made by MPDV: SMA-CMR

Customizations made by the customer: SMA-CMS

## 3.4.5 Web service calls

### 3.4.5.1 Web service configuration

Stored in: SMA\Local\Areas\CustomService\Configurations

Structure of the file:

```

<?xml version="1.0"?>
<CustomServiceConfig xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
 <!-- Name of the web service -->
 <WebService Name="" />
 <!--
 what are you asking for
 -->
 <ResultParameters>
 <Parameters>
 <Parameter Acronym="" />
 </Parameters>
 </ResultParameters>

 <!-- if you want to filter data (performance, efficiency, usability) define filterParameters -->
 <FilterParameters>
 <Parameters>
 <!--
 acronym: e.g.: operation.act.production_indicator
 webServiceType: if empty, default = string (string, integer, decimal, datetime, boolean)
 inputAsArray: if empty, default = N (N, Y)
 operator: if empty, default = "EQUAL" (LIKE, EQUAL, BETWEEN, IN, NOT_EQUAL, LIKE_OR_NULL, EQUAL_OR_NULL,
 BETWEEN_OR_NULL,
 IN_OR_NULL, NOT_EQUAL_OR_NULL, GT, LT, GTE, LTE, GT_OR_NULL, LT_OR_NULL, GTE_OR_NULL, LTE_OR_NULL)
 -->
 <Parameter Acronym="" WebServiceType="" InputAsArray="" Operator="" />
 </Parameters>
 </FilterParameters>
</CustomServiceConfig>

```

## 3.4.5.2 Test call

### 3.4.5.2.1 Call without filter parameters

```
http://<SMA-Server>:<Port>/CustomService/CustomService/GetData?configName=<Konfiguration>
```

Configuration: Name of the created XML file without file extension.

Example: Result as JSON

```
{
 "Error": false,
 "Msg": "no errors",
 "DataSet": null,
 "DataTable": [
 {
 "units.designation": "Milimeter",
 "units.unit": "mm"
 },
 {
 "units.designation": "Millimeter",
 "units.unit": "MM"
 },
 {
 "units.designation": "Stück",
 "units.unit": "ST"
 }
]
}
```

### 3.4.5.2.2 Call with filter parameters

```
http://<SMA-Server>:<Port>/CustomService/CustomService/GetData?configName=<Configuration>&filterJson=<filterJson>
```

Configuration: Name of the created XML file without file extension.

#### **filterJson:**

Using the acronym filterJson, you can add parameters in JSON format. These parameters are used to filter the configured web service. You can only use parameters that have been configured.

Example:

Configuration InPutAsArray= "Y":

```
{"Parameters":{"units.unit":{"0":"ST"}}
```

Configuration InputAsArray="N":

```
{"Parameters":{"units.unit":"ST"}}
```

### 3.4.5.3 Call without filter

```
$mpdv.communication.post("/CustomService/CustomService/GetData", {
 configName: "training_mdunits_list",
 }, fillList, showError);}
```

### 3.4.5.4 Call with filter parameters

```
var parameters = '"units.unit": {"0":"ST"}';
$mpdv.communication.post("/CustomService/CustomService/GetData", {
 configName: "training_mdunits_list",
 filterJson: '{"Parameters":{"' + parameters + '}}'
}, fillList, showError); }
```

### 3.4.5.5 Processing of the result

#### 3.4.5.5.1 Successful processing

Example:

```
function fillList(data){
 save_materials = null;

 if (data) {
 var error = data.Error;
 if (!error) {
 // use data
 if (data.DataTable && data.DataTable.length > 0) {
 //do something
 }
 }else {
 showError(data);
 }
 }
}
```

#### 3.4.5.5.2 Error

This function describes the output in case of an error:

```
function showError(data) {
 $mpdv.progress.hideAndClearProgress();

 if (data) {
 var msg = data.Msg;
```



```

 if (msg) {
 $mpdv.messages.showErrorMessage(msg);
 } else {
 $mpdv.messages.showErrorMessage(lkServerReturnedError);
 }
} else {
 $mpdv.messages.showErrorMessage(lkServerReturnedError);
}
}

```

### 3.4.5.5.3 Sorting of web service results

You can use the following to sort the data of a web service:

```
data.DataTable.sort(dynamicSortMultiple("value"));
```

```

function dynamicSortMultiple() {
 var props = arguments;
 return function (obj1, obj2) {
 var i = 0, result = 0, numberOfProperties = props.length;

 while (result === 0 && i < numberOfProperties) {
 result = dynamicSort(props[i])(obj1, obj2);
 i++;
 }
 return result;
 };
}

function dynamicSort(property) {
 var sortOrder = 1;
 if (property[0] === "-") {
 sortOrder = -1;
 property = property.substr(1);
 }
 return function (a, b) {
 var result = (a[property] < b[property]) ? -1 : (a[property] > b[property]) ? 1 : 0;
 return result * sortOrder;
 };
}

```

## 3.4.6 BAPI call

```

function loadData() {
 var aunr = $("#getOrderNumber").val();

 if (aunr) {
 var dlg = "DLG=CNR.INSERT|CNR="+cnr+"|";

 $mpdv.communication.post("/CustomService/CustomService/SendDlg", {
 keyValueDlg: dlg
 }, handleInfo, showError);
 }
}

```

BAPI call including forwarding to the page **InspectionPoints**:

```
var dlg = "DLG=CNR.INSERT|CNR="+cnr+"|";

$mpdv.communication.post("/CustomService/CustomService/SendDlg", {
 keyValueDlg: dlg
}, function (data) { window.location.href = "../InspectionPoints"; },
showError);
```

### 3.4.7 Forwarding

Example:

Change to page "Resource/ResourceDnc" with Cancel:

```
$("#CancelButton").on("click", function () {
 window.location.href = "/Resource/ResourceDnc";
});
```

### 3.4.8 Barcode processing

```
$('#body').keyup(function (e) {
 //alert(e.keyCode);
 if (e.keyCode === 13) {
 //do something
 }
});
```

### 3.4.9 Read UserPerson from session

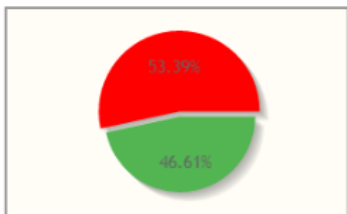
```
var personID = window.sessionStorage.getItem('UserPerson');

getPersonName(personID);
```

### 3.4.10 Integrating charts

Use jqplot to integrate charts. <http://www.jqplot.com/>

Example:



You must integrate the following in the HTML page:

```

<script type="text/javascript" src="../../Scripts/jquery.jqplot.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.canvasTextRenderer.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.canvasAxisLabelRenderer.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.canvasAxisTickRenderer.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.barRenderer.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.categoryAxisRenderer.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.pointLabels.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.pieRenderer.min.js"></script>
<script type="text/javascript" src="../../Scripts/plugins/jqplot.dateAxisRenderer.js"></script>

```

The JavaScript processing is as follows:

```

var dataSeriesPie = null;
function fillKpi(data){
 if (data) {
 var error = data.Error;
 if (!error) {
 var table = data.DataTable;
 if (table) {
 $.each(table, function (i, r) {
 if (i !== 0)
 return;
 plotTarget = 'pie';
 $('#'+plotTarget).html(null);
 dataSeriesPie = [
 ['QUA', r["efficiencyreport.quality"]],
 ['TQUA', 1- r["efficiencyreport.quality"]],
];
 paintKpiPlot(plotTarget, dataSeriesPie);
 });
 }
 } else {
 showError(data);
 }
 }
}

```

### 3.4.11 Lists with fixed values

To create a list with fixed values, you can implement the following in JavaScript:

```

function callBatchStatus(mode) {
 $(".bufferTableInfoBox").css("display", "none");
 $(".chooseDialogTable").css("display", "inline-table");

 var data = {
 DataTable: [
 {
 "status.id": "F",
 "status.designation": "Frei"
 },
 {
 "status.id": "S",
 "status.designation": "Gesperrt"
 }
]
 };
}

```

```

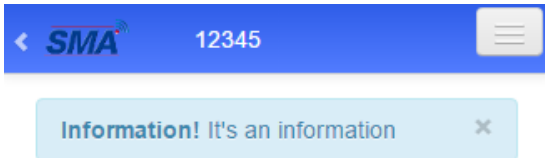
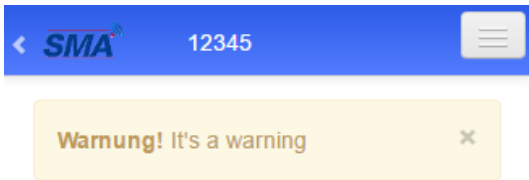
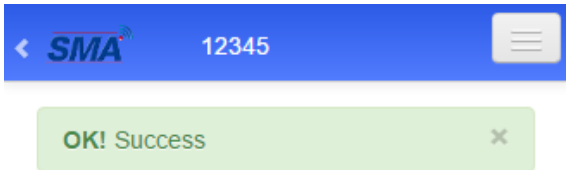
 },
 {
 "status.id": "P",
 "status.designation": "Prüfung"
 },
 {
 "status.id": "V",
 "status.designation": "Verfallen"
 },
 {
 "status.id": "T",
 "status.designation": "Transport"
 }
]
};

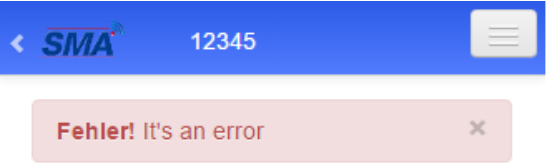
showDataTable(data, mode);
}

```

### 3.5 Script Functions/Variables

Code	Description
<code>\$mpdv.config.getApplication()</code>	Reads the application name from URL.   Example:   <code>http://localhost:8081/CustomService/CustomService?pa geld=training_calculator</code>   Return value: "CustomService"
<code>\$mpdv.utils.convertSecondsToHhmmss(value)</code>	Conversion of seconds to format: hh:mm:ss   Example:   <code>\$mpdv.utils.convertSecondsToHhmmss(3666)</code>   Return value: "01:01:06"   <code>\$mpdv.utils.convertSecondsToHhmmss(-3666)</code>   Return value: "-01:01:06"
<code>\$mpdv.utils.convertSecondsToHhmm(value)</code>	Conversion of seconds to format: hh:mm   Example:   <code>\$mpdv.utils.convertSecondsToHhmm(3666)</code>   Returns: "01:01"   <code>\$mpdv.utils.convertSecondsToHhmm(-3666)</code>   Returns: "-01:01"
<code>\$mpdv.components.modal.openModal(v</code>	Opens a modal dialog   Example:

alue)	\$mpdv.components.modal.openModal("exampleModal")
\$mpdv.communication.getUrlParam(value)	Reads the value from specified URL.   Example: URL = http://localhost:8081/?myParam=super   \$mpdv.communication.getUrlParam("myParam") Return value: "super"
\$mpdv.messages.showInfoMessage(value)	Output of an information box   Example: \$mpdv.messages.showInfoMessage("It's an information")  
\$mpdv.messages.showWarningMessage(value)	Output of a warning   Example: \$mpdv.messages.showWarningMessage("It's a warning")  
\$mpdv.messages.showSuccessMessage(value)	Output of a success message   Example: \$mpdv.messages.showSuccessMessage("Success")  

<code>\$mpdv.messages.showErrorMessage(value)</code>	Output of an error message   Example:  <code>\$mpdv.messages.showErrorMessage("It's an error")</code> 
<code>\$mpdv.components.list.matchWildcardString(wildcardString, string)</code>	Functionality to query wildcards   ? for one character   * for several characters   Example:  <pre>&gt; \$mpdv.components.list.matchWildcardString("e", "Hello") &lt; false &gt; \$mpdv.components.list.matchWildcardString("*e*", "Hello") &lt; true &gt; \$mpdv.components.list.matchWildcardString("e???", "Hello") &lt; true &gt; \$mpdv.components.list.matchWildcardString("e??", "Hello") &lt; false &gt; \$mpdv.components.list.matchWildcardString("*e*1*", "Hello") &lt; true &gt; \$mpdv.components.list.matchWildcardString("e*b*", "Hello") &lt; false</pre>
<code>Translate(&lt;language key&gt;)</code>	Function to integrate language keys   The language key is edited in the file "mpdvDictionaryCustomer.xlsm".   Store the files created here in SMA\Runtime\resources\local\languages.
<code>"GET_USER"</code>	GET_USER provides the value of the user logged on.
<code>\$("#&lt;VALUE&gt;").focus();</code>	Set focus on a field   Example:  <code>\$("#KNR").focus();</code>

## 4 Repository Client

You use the MPDV Repository Client MRC to display and edit repository data. It provides a user-friendly access.

### 4.1 Quick start

This section provides a quick overview of how to work with the Repository Client. The individual steps are only briefly described. For further information on the individual steps, refer to the sections in the following, if required.

#### Installation

##### Requirements

To use the Repository Client, you must have installed the Microsoft DotNet framework (at least version 4.5.2).

##### Program installation

To install the program, just copy the folder including the binary files into your system. An installation program is not required.

##### Installation of developer license

If the developer license is not available, you can only read the data. You cannot save or export the data.

The developer license is handed out once you have attended a respective Customizing Training. The developer license is provided as \*.lic file. This file is included in the data medium that you have received during the training: Folder "Repository Client", subfolder "tools/licence", e.g.

`x:\CUT-MOC_81_files\Tools\MPDVRepositoryClient\tools\licence\mpdvWrite.lic.`

Copy the folder "licence" with its content into the folder of the Repository Client in the roaming directory of the Windows user, e.g.:

`C:\Users\%User%\AppData\Roaming\MPDV\RepositoryClient\licence\mpdvWrite.lic`

This folder is automatically created on the first start of the Repository Client.

#### Before you start

Before you start working with the Repository Client, you must make sure that the Repository Client has been installed according to the installation instructions and that the required license files are stored as described there.

In general, the repository is empty when you start work. However, if you do not want to start with an empty repository, it is recommended to make sure that the data you want to work with is available.

## First steps

Start the Repository Client.

➔ Start `.\bin\mrc.exe`

Now load or create a [Workset](#).

A workset specifies the sources of the repository that you want to edit.

➔ Click the button *Load work set* in the file-based repository  
-> select `workingset.work`

Click the button [Repository](#) ➔ [Load Repository](#) to load the data from the sources specified in the workset.

How to proceed further depends on what you want to do via the Repository Client.

## Tip: Working with perspectives

The Repository Client provides the possibility to use different perspectives for different tasks. Do not hesitate to close views you do not need or drag them to another open view in order to tab them and hence provide for more space and clarity. You can also use more than one view of the same type. Simply adapt your perspective to the requirements of your tasks.

### Example:

If you create a service and you would like to check with an existing service how to populate the fields, you can simply open another service view. Thus, you do not need to destroy your current view and re-orient yourself later.

### Default perspectives

The installation of the Repository Client provides default perspectives:

#### default

This is a good perspective to start with. It provides a combined view for server and client-related contents.

Select a domain on the top left. Via the included relations, the top right area shows the services, servicesGUI and properties of this domain. The bottom right area shows the ServiceParameters of the service selected above, the ServiceParameterGui of the ServiceGui selected above and the ControlDataSources, ReferenceData and Authorizations of the selected domain.



**default client**

Similar to the "default" perspective but limited to the contents concerning client development.

**default server**

Similar to the "default" perspective but limited to the contents concerning server development.

**DB schema**

Shows documentation of the database structure.

**Validation**

Use this perspective to validate contents.

Proceed as follows:

- Open perspective and load data from workset.
- Perform validation: tab *Repository* --> button *Validate*.
- A CSV file listing the identified irregularities is generated in the sub-directory "validation\_logs" of the Repository Client's installation directory. The application that is linked to this file type in the operating system opens the CSV file.
- In the views for Domains, Properties, ServiceParameters, etc. the Repository Client only shows the entries with detected irregularities.

You should analyze and, if necessary, correct the detected irregularities. Not every irregularity leads to an error.

## 4.2 Start and exit Repository Client

Start the Repository Client via the Windows start menu, a link on the desktop or the command line. As soon as all required components have been loaded, the application window is shown.

You can start and run the Repository Client multiple times in parallel on a PC. You can access different repository data in each of the started instances of the Repository Client. For example, you can view several versions of the repository at the same time.

You can also start the client by opening one of the workset files. On start of the client, the system attempts to load the contents of the workset defined in this file. This option is available after the first start of the Repository Client.

**Command line parameters**

You may transfer parameters to the Repository Client upon the start. The following parameters are supported:

`-perspective/-p <perspectivename>` Use this parameter to start the Repository Client with a specific perspective. If you do not use this parameter, the last active perspective is started by default.

`-workset/-w <worksetfile>` Use this parameter to specify the workset to be loaded. If you do not specify the workset on start of the application, the last loaded workset is loaded.

`--autoload/--a` If you use this parameter, the Repository Client loads the repository defined in the last active workset or in the workset transferred via parameters. This repository is loaded directly on start of the application.

`--trim/--t` If you use this parameter, the Repository Client removes so-called leading and trailing "whitespaces" when loading the repository. Only select this option if needed, because the load time increases extremely.

### Exit

To exit the application, click  in the title bar.



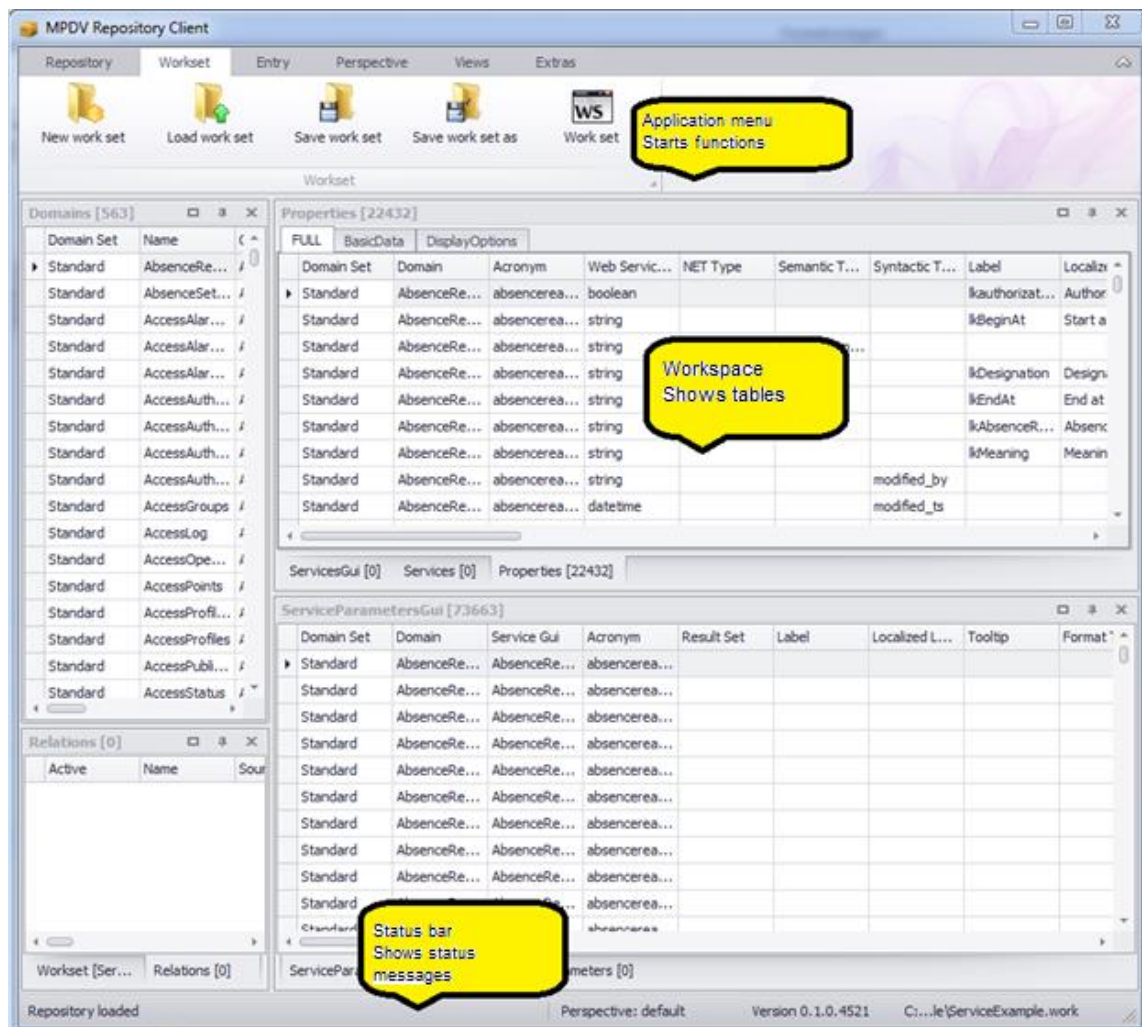
If the loaded repository includes active changes when you exit the application, a respective message is issued asking you to save the changes. If you do not want to save the changes, you can discard the changes or stop exiting the application.



Note: Changes to the workset and perspective are discarded when you exit the application, if you have not saved the changes.

## 4.3 The Application Window

The application window forms a framework for the display of different tables. It includes the application menu with control elements to call and control different functionalities. The menu is on top of the window. A status bar is at the bottom of the window. The status bar shows progress and event messages.



You can individually dock the grids/table views. To do so, click the title bar of a table view/grid and drag it out of the docking position. For orientation purposes, the system shows the docking positions where you can drop the table view. You can also drop a table view without docking it.

## 4.4 Grids/table views

Grids/table views are components to present data records in a table. You can change the tables in the Repository Client according to your requirements. For each grid/table view, the functions described below are available.



The settings, that you make in a table, are saved with the perspective. To undo changes, you can switch to the standard perspective (in the application menu: *Perspective* → *Change perspective*).

### Sort table data

Click the table header to sort table data in descending order. If you click the table header once more, data is sorted in ascending order. The selected sorting option is shown.

You can sort data by several columns: Press the *Shift* key of your keyboard after sorting the first column. Then click the other column headings by which you want to sort.

You can also use the context menu of the table header to sort data.

### **Group data in the table**

You can group table data if the *group by* area is shown. If the *group by* area is not shown, you can show the area via the context menu of the table header (*Show/Hide group by box*). To group by a column, click the column header and drag it to the grouping pane. Multiple grouping is also supported.

### **Optimum column width (best fit)**

Select the option "*Best fit*" in the context menu of the table header to adjust the column width of the selected column to the optimum width. In this case, "optimum" means that the column is as wide as the largest entry in the selected column.

### **Optimum column width (all columns) / Best fit (all columns)**

Click this function to adjust all columns to the optimum width.

### **Change column width**

You can also change the column width using the mouse, i.e. move the space between two cells to the left or right.

### **Show and/or hide columns and entire categories**

Use the context menu function *Select columns* to show and/or hide individual columns and entire categories. For this purpose, select the function in the context menu and then drag the required columns and/or categories from the table to the pool or from the pool to the table.

### **Change the sequence of columns and categories**

Also use the mouse to change the display order of columns and categories. To do so, drag the column or category you want to move and drop it at the required location. The system will indicate the location to which the column and/or category will be allocated when you release the left mouse button.

### **Freezing columns to prevent horizontal scrolling**

You can freeze columns at the left and right-hand side to keep the columns in view while scrolling. These column settings are included in the perspective and can be saved with the perspective. Right-click the column header and press one of the below-mentioned shortcuts to freeze columns:

- CTRL + right click: freeze at the left-hand side.
- ALT + right click: freeze at the right-hand side.
- SHIFT + right click: Unfreeze.

### Filter table data

Click the filter icon  in the column where you intend to set the filter and select the required filter option from the list; i.e. you select one of the values available in the table or compose a combination of values in the *user-defined* filter.

You can also set several filters in different columns. The table footer indicates that the table has been filtered and also shows the filter criteria. Select the function *Edit filter* on the right of the footer to open the filter editor. Use the filter editor to create complex filter criteria across all columns. You may also open the filter editor via the context menu of the table header.

### Search box

Use the context menu of the table header to access the option *Show search box*. This option provides a search box within the table. Use this box to quickly search and/or filter the requested data. Simply start typing in this box and the system will only show those rows matching the data you typed. The more characters you enter, the more you narrow down the result.

### Filter row

Use the context menu of the table header to open the option *Show filter row*. This option provides an additional row shown below the table header. You can enter a search term in any column, and the system will narrow down the displayed rows appropriately. The system supports wildcards. You can also combine search terms in various columns to restrict the search result.

### Edit rows

Double-click a row or press the "Enter" button to switch to the editing mode and edit the respective row. When you have finished editing and leave the row, the editing mode is terminated. Edited rows are highlighted in color.

## 4.5 The application menu

You can use the application ribbon menu of the Repository Client to control various functions of the tool. It includes several tabs that are described in the following.

### Workset

Includes functions to administer worksets. A workset specifies the sources included in the repository that you want to edit. To display a workset, use the workset panel which consists of a grid/table view.



- **New workset:** This function creates a new workset. If a workset is loaded that has been modified, a dialog pops up asking you to save the changes.  
Click "Yes" to save the changes, "No" will discard them. In both cases, a new workset is created. Click "Cancel" to cancel the process of creating a new workset.
- **Load workset:** This functions loads a workset from an existing file. You can select the workset to be loaded via a file dialog. If the currently loaded workset has been modified, you can save the changes as described above.
- **Save workset:** This function saves the current workset in the file from which it was loaded. If the current workset is new, a file dialog pops up asking you to select the file in which you want to save the workset.
- **Save workset in:** Use this function to save the currently loaded workset in a file. You can select the files using a file dialog.
- **Workset:** Use this button to show and/or hide the grid/table view presenting the currently loaded workset. For details on the workset table, please refer to section Workset.

## Repository

Includes functions to load and save data in the repository. In addition, you may display repository-specific data here.



- **Load repository:** Use this function to load the repository. The currently loaded workset specifies the data sources that are used to load the repository. If a repository is loaded that has been modified, a dialog pops up asking you to save the changes. Click "Yes" to save the changes, "No" will discard them. In both cases, the repository will be reloaded subsequently. Click "Cancel" to cancel the process of loading a repository.
- **Save repository:** **\*only available if used in development mode**  
Use this function to save changes in the repository. If no changes have been made, an appropriate note will be displayed.
- **Export repository:** **\*only available if used in development mode**  
In contrast to saving the repository, you can use this function to export parts of the repository. For details on this function, please refer to section **Error! Reference source not found.**
- **Validate:** **\*only available if used in development mode**  
You can use this function to validate your data records manually. (See section "Validation").
- **Value list:** Use this menu entry to show and/or hide the value list. The list includes permissible entries for specific fields of the repository.
- **References:** Use this entry to show and/or hide the table with repository references. For details on references, please refer to section References.

- **Changes: \*only available if used in development mode**

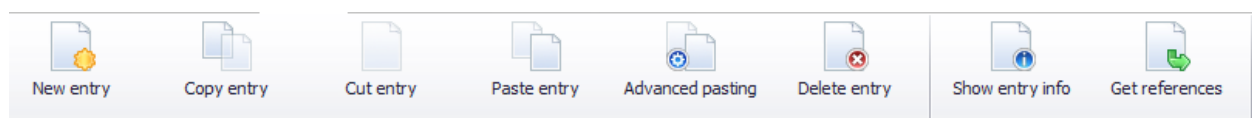
Use this button to show and/or hide the change view. This view shows the current modifications in the loaded repository.

- **Service documentation**

Use this button to show the extended documentation of selected standard services. For further information on the service documentation, refer to section "4.9 Service documentation".

## Data collection

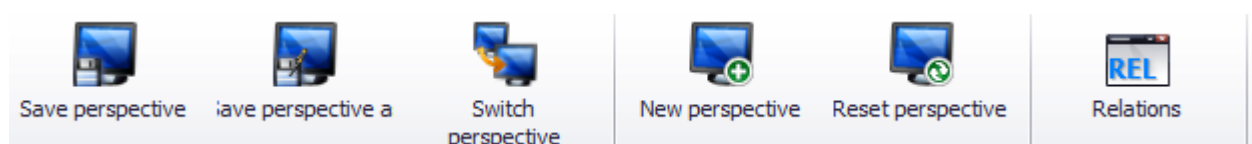
The *Entry* tab summarizes the functions that you can use to edit the loaded repository. The entries refer to the currently focused table view/grid.



- **New entry:** Use this function to create a new entry. For details on this function, please refer to [Context menu → New](#).
- **Copy entry:** Use this function to copy selected table entries. For details on this function, please refer to [Context menu → Copy](#).
- **Cut entry:** Use this function to cut selected table entries. For details on this function, please refer to [Context menu → Cut](#).
- **Paste entry:** Use this function to insert (paste) entries from the cache/clipboard. For details on this function, please refer to [Context menu → Insert](#).
- **Advanced pasting:** Use this function to edit entries in the clipboard prior to inserting them. For details on this function, please refer to [Context menu → Advanced pasting](#).
- **Delete entry:** Use this function to delete the selected entries. For details on this function, please refer to [Context menu → Delete](#).
- **Show entry info:** Use this function to open a dialog showing information on the currently selected entry. For details on this function, please refer to [Context menu → Info](#).
- **Get references:** Use this function to open a new grid/table view showing the currently selected data record including referenced values. For details on this function, please refer to [Context menu → Get references](#).

## Perspective

These entries of the menu refer to the administration of perspectives. A perspective is a layout of table views/grids and includes also the associated relations between table views/grids.





- **Save perspective:** Use this function to save the currently shown arrangement of tables.
- **Save perspective as:** Use this function to save the currently shown perspective under a different name. You can enter the new name in the displayed dialog.
- **Switch perspective:** Use this function to change the perspective. A dialog with all available perspectives is shown.
- **New perspective:** Use this function to create a new perspective. Similar to the "Save perspective as" function, you can select the name of the perspective in a dialog. After entering the name, you can immediately switch to the new perspective.
- **Reset perspective:** Use this function to reset the current perspective to the status saved last.
- **Relations:** Use this menu entry to show/hide the grid/table view showing the relations between the grids/table views. For details on relations, please refer to section Relations.



Note: Changes to the perspective are discarded when you exit the application, if you have not saved the changes explicitly.

## Views

Use these entries to open table views/grids. The entries will open a new grid/table view each showing the relevant data records of the repository.



For clear identification of the data records shown in the tables, the *Parent* column is included in each of the tables. The *Parent* column includes the identifier for the father node in the repository tree. The other columns of these tables are defined by the repository documentation.

The *View* area additionally includes a group with entries for the remaining grids/table views of the application.

## 4.6 Workset

Worksets define a set of data sources that make up the repository that you want to edit. Use the workset management function to organize your work on different projects and create an appropriate workset for each of your projects. You can show/hide the workset table via the application menu ([Workset](#) → [Workset](#)).



Note: The workset loaded last will be loaded on start of the Repository Client.



Workset [example.work]							
Name	Server Source	Client Source	Prio...	Is Writeable	Active	Overrides	M
ZIP	D:\DevRepo\ServerDomains.zip	D:\DevRepo\ClientDomains.zip	1	<input type="checkbox"/>	<input type="checkbox"/>		
Runtime	\\servername\hydra1\jhydradir\MOC\1	C:\Program Files (x86)\MPDV\HYDRA 8\MOC	2	<input type="checkbox"/>	<input checked="" type="checkbox"/>		
▶ Dev	d:\DevSrc\Repository\Data\server	d:\DevSrc\Repository\Data\client	3	<input checked="" type="checkbox"/>	<input checked="" type="checkbox"/>		

The workset table includes the following columns:

### Name

You can specify the domain set that you want to use to load the data source. The repository data include the information on the domain set that was used to load the data. You can copy data from a domain set into another domain set. Example: Copy existing applications from the domain set "Runtime" (read only) into your development directory, e.g. the domain set "Dev" (writable) where you can make changes.

### Client Source

You can specify the data source that you want to use to load client data. The following options are available:

- Load data from the runtime structure of the client reference in the server  
In the server, the client configurations are stored as client reference with runtime structure. You can load the repository data from this structure.

Example:

HYDRA: x:\jhydradir\MaintenanceManager\rt\client\MOC

MIP: x:\wsp\_config\MaintenanceManager\rt\client\MOC

The access is read-only.

- Load data from the runtime structure of a local MOC client  
If you enter a path to an MOC installation directory, the respective client data are directly loaded from the MOC runtime installation.

Example:

C:\Program Files (x86)\MPDV\HYDRA 8\MOC

The access is read-only.

- Load data from a local directory with domain structure  
You use local directories with domain structure for the administration of your own developments. Using the Repository Client, you can read data in this directory and you can also save data into this directory.

Example: d:\DevSrc\Repository\Data\client

- Load data from a ZIP archive  
You can enter a ZIP file (including path) that includes the data in domain structure. MPDV provides the ZIP archives as part of the trainings or on the support portal.  
The access is read-only.

## Server Source

You can specify the server source that you want to use to load the server-specific data. The following options are available:

- Load data from the runtime structure in the server

You can read the configurations for the server in a server installation. To this end, the configuration directory of the web service provider (WSP) up to the instance number is specified.

Example:

HYDRA:        \\<servername>\<install\_dir>\jhydradir\MOC\1

MIP:        \\<servername>\<install\_dir>\jdir\MOC\1

The configuration is loaded from the standard scope by default. As an alternative, you can also load the configuration from the *custom* scope, if you enter the value "custom" in the field "name".

The access is read-only.

- Load data from a local directory with domain structure

You use local directories with domain structure for the administration of your own developments. Using the Repository Client, you can read data in this directory and you can also save data into this directory.

Example:        d:\DevSrc\Repository\Data\server

- Load data from a ZIP archive

You can enter a ZIP file (including path) that includes the data in domain structure. MPDV provides the ZIP archives as part of the trainings or on the support portal.

The access is read-only.

## Priority

The priority of a data record specifies the loading sequence of sources that are allocated to the same data record. In the above example, data is first read from the local development directory "d:\DevSrc\Repository" and then from the runtime installations "Server" and "Client". Data records with low priority are overridden and consequently not loaded.

## Is Writeable

In this column you can specify if a data source grants write access. You only require write access in workset entries where you want to make local developments.



Please note that ZIP archives do not grant write access. For this reasons, you must not enable "is Writeable" in case of a ZIP data source.



Please note that it is not supported to save data in the runtime structure (client directory or server runtime directory). You must therefore not enable "Is Writable" for data sources with

runtime structure.

### Overrides

You can specify in this column which domain set is overridden by the current one. This affects the resolving of references (for details please refer to section References).



An entry in the "Overrides" column does not have any effect on the loading of the data sources. See column "Priority".

### Active

Use this option to enable or disable an entry.

## 4.7 Relations

Relations are a property of perspectives and define table filters. Tables that include relations are dynamically adapted to the selected values of another table by setting a filter. For example: Using relations, you can specify that only the service parameters of the service currently selected in another service view are displayed in a service parameter view. The *Relations* table lists the relations of the current perspective. You can call this table via the application menu (*Perspective* → *Relations*).

Relations [13]							
Active	Name	Source	Target	Filter	Var0	Var1	Var2
<input checked="" type="checkbox"/>	Authorizations for Domain	Domains	Authorizations	[Parent] = 'id->Authorizations'			
<input checked="" type="checkbox"/>	ControlDataSources for Domain	Domains	ControlDataSources	[Parent] = 'id->ControlDataSources'			
<input checked="" type="checkbox"/>	DataObject for Service	Services	Dataobjects	[DomainSet] = '\$0' and [Domain] = '\$1' AND [Name] = '\$2'	DomainSet	Domain	Name
<input checked="" type="checkbox"/>	Properties for Domain	Domains	Properties	[Parent] = 'id->Properties'			
<input checked="" type="checkbox"/>	ReferenceData for Domain	Domains	ReferenceData	[Parent] = 'id->ReferenceData'			
<input checked="" type="checkbox"/>	SchemaColumns for ServiceParameter	ServiceParameters	SchemaColumns	((TableName] = '\$0' AND [ColumnName] = '\$1' AND [TableNam...	DBTabelle	DBField	HydraAcronym
<input type="checkbox"/>	ServiceParameter for ServiceParameterGui	ServiceParametersGui	ServiceParameters	[Domain] = '\$0' AND [Service] = '\$1' AND [Acronym] = '\$2'	Domain	ServiceGui	Acronym
<input checked="" type="checkbox"/>	ServiceParameters for Service	Services	ServiceParameters	[Parent] = 'id'			
<input checked="" type="checkbox"/>	ServiceParametersGui for Service	Services	ServiceParametersGui	[DomainSet] = '\$0' and [Domain] = '\$1' AND [ServiceGui] = '\$2'	DomainSet	Domain	Name
<input type="checkbox"/>	ServiceParametersGui for ServiceGui	ServicesGui	ServiceParametersGui	[Parent] = 'id'			
<input checked="" type="checkbox"/>	Services for Domain	Domains	Services	[Parent] = 'id->Services'			
<input type="checkbox"/>	Service forServiceGui	ServicesGui	Services	[Name] = '\$0'	Name		
<input checked="" type="checkbox"/>	ServicesGui for Domain	Domains	ServicesGui	[Parent] = 'id->ServicesGui'			

The following columns are displayed:

### Active

This checkbox specifies if the relation is used.

### Name

The name of the relation – free choice.

### Source

The table and its selection that are used to set the filters. If you edit an entry in this column, the currently possible assignments, i.e. all currently existing views are presented in a selection box.

### Target

The table where the filter is applied. If you edit an entry in this column, the currently possible assignments, i.e. all currently existing views are presented in a selection box.

### Filter

You can store the filter expression here that you want to apply to the target table. Variables ranging between \$0 and \$9 are supported. These variables are dynamically filled with the values of the columns Var[0-9].

### Var[0-9]

In these columns, you can specify the columns of the source table that are used to adapt the filters dynamically.

As soon as a correct (and activated) relation is entered in this table, it is applied. If you close one of the referenced views of a relation, the relevant view is removed from the relation. If this results in a double entry in the *Relations* table, this entry is removed. You can therefore use this view to administer the relations between concrete table instances and to administer unbound relations that can serve as template for relations.

## 4.8 References

References show the inherent connections between data records of the repository. They are defined by the repository structure and cannot be edited in the Repository Client. For example: A value in the column "Syntactic Type" of the *Property* table references another data record in the *Property* table. The *References* table lists the defined references and may be activated via the application menu ([Repository](#) → [References](#)).

Name	Source	Source Column	Dependency	Condition	Target	Filter	Priority
ControlDataSource1	Property	ControlDataSource	ControlDataSourceMode	Lookup	ControlDataSource	[Name] = \$value and [Domainset] = \$domset	-1
ControlDataSource2	Property	ControlDataSource	ControlDataSourceMode	Reference	ReferenceData	[type] = \$value and [Domainset] = \$domset	-1
SyntacticType	Property	SyntacticType			Property	[Acronym] = \$value	0
SemanticType	Property	SemanticType			Property	[Acronym] = \$value	1
Acronym1	ServiceParameter	Acronym			Property	[Acronym] = \$value	3
Acronym2	GuiServiceParameter	Acronym			Property	[Acronym] = \$value	3

The following columns are displayed:

- **Name:** Name of the reference
- **Source:** The repository object type that can include this reference.
- **SourceColumn:** The source type property that can include the reference.
- **Dependency:** Source type property specifying the reference target.

- **Condition:** Value that the property specified under *Dependency* must have. Only then, the reference is pursued. For example: The value of *ControlDataSourceMode* (lookup, reference) specifies the target of the reference (*ControlDataSource* or *ReferenceData*) which is specified in the *ControlDataSource* property.
- **Target:** The repository object type that is referenced.
- **Filter:** Filter that selects the referenced data from the overall quantity of this type of data. Such a filter can include the variable *\$value* (value of column *Value* in the current row) and *\$parent* (value of column *Parent* in the current row).
- **Priority:** Specifies the priority of the reference. You can find further details in section "Get references".

References provide two general functions:

**Show reference:** You can use this function to display the referenced data in a new table. You can call this function via the context menu ([Context menu](#) → [Show reference](#)).



Note: This function is only available in the context menu of cells which can include references.

**Get references:** Use this function to complete missing values of a data record with values of referenced data records. For example: You can use this function to show the inherited values of a property of the *SemanticType* or the *SyntacticType*.

You can call this function via the context menu ([Context menu](#) → [Get references](#)) of the table views/grids. A new data record is generated and shown in a new panel. The generated data record is a copy of the currently selected data record. The values that are not filled are filled by those in the referenced data records. The reference priority specifies the filling sequence.

## 4.9 Service documentation

The Repository Client contains an extended documentation for selected standard services. The documentation mainly includes services released in the system and to be used with the Service Interface (SCS-SIF).

The documentation is available as of MRC version 1.8.STD.65500 (beginning of 2019).

There are two options to access the service documentation:

- In the toolbar "Repository", you can use the button *Service Documentation* to open the table of contents of the services included. Via hyperlinks, you can navigate to the different services.
- In the table views "Services" and "ServicesGui", you can use the context menu to open the documentation of the selected service if it is available.

**Printing a service documentation:**

You can print the service documentation using the shortcut Ctrl-P.

## 5 The Repository

### 5.1 Overview

The data of the repository is used in multiple ways:

- The repository defines and describes the **interface between client and server**. The input parameters and the result sets of service requests are described.
- In case of interpreted service types, the **processing and the business logic of a service** is specified via configuration in the repository. Only in exceptional cases, an actual programming in the server is required.
- For the client, the repository defines **how the data is displayed on the client** and which GUI elements are used to enter data. The repository also defines how the client checks the user input. You can generate most of the applications on the client using the configurations of the repository. Here, programming on the client is not required.

The repository data is grouped and structured using **domains**. A domain summarizes all data that logically belongs to an application.

The domain contains hierarchically structured and typed data. A domain includes *services* and *service parameters*, the respective *GUI settings*, *properties*, *authorizations*, *ReferenceData* and *ControlDataSources*.

Find below a detailed description of the repository elements.

### 5.2 Domain

Domains have properties and provide services within the domain context.

A domain is the smallest software unit. You can update the domain using an update package. Create a separate domain for each application. This domain then includes the services implemented for this application. You can also use the services and client attributes of a domain in applications of other domains. For example, a client application in its own domain can use a service of a different domain.

You can assign global contents to a global domain: for example, client menu configurations or separate global syntactic types.

#### Name

Each domain has a unique name. For the name, you use the notation "UpperCamelCase".

## 5.3 Service

Services have transfer parameters and return values, which are often identical to the properties of the domains.

### 5.3.1 Name

Name of a service. The service name usually consists of the domain name that includes the service and the **function**, separated by a dot.

### 5.3.2 Function

This field describes the requested service function. Typical functions are list, update, insert, delete, new, ...

### 5.3.3 ServiceType

There are several service types.

***InterpretedJavaService2***: Services of this type are used to display lists and evaluations. The services are interpreted at runtime using repository data. Contrary to the *InterpretedJavaService*, the services of type *InterpretedJavaService2* are prepared to stream data and provide more elegant options for Java user exits.

***InterpretedJavaService* (obsolete)**: Services of this type are interpreted at runtime using repository data. These services have been replaced with the service type *InterpretedJavaService2*.

***InterpretedBAPIService***: You use services of this type to edit data. The services are interpreted at runtime using repository data.

***ExternalJavaService***: Services of this type are completely implemented in Java. You can use these services to implement lists or editing functions. You use these services if the possibilities of the interpreting service types are not sufficient and the logic must be converted into Java programming.

***InterpretedWrapper***: Services of this type are interpreted at runtime using the repository data. The service is implemented as wrapper of an existing PDM dialog and is therefore subject to specific limitations, e.g. it does not support any dynamic Where.

***Wrapper* (obsolete)**: Services of this type are programmed and wrap an existing BAPI function. They are therefore subject to specific limitations, e.g. no dynamic Where.

***JavaService* (obsolete)**: Services of this type are completely implemented in Java.

Recommendation:



- The type *InterpretedJavaService2* is recommended for services that you use to read data.
- The type *InterpretedBAPIService* is recommended for services that you use to write data.
- If the interpreted service types cannot meet the requirements (or only with great effort) even if they include Java user exits, you should use the services implemented in Java of type *ExternalJavaService*.
- The other service types are older technologies and should not be used for new developments.

### 5.3.4 ListMode

For services of type *Wrapper* or *InterpretedWrapper*: This column must be populated for each service. The column specifies whether the requested PDM dialog returns a file as result or whether it is only a return string. "Y" => The result is a file, otherwise only a string.

### 5.3.5 DLG

For all services of **ServiceType** *InterpretedWrapper* or *Wrapper*, you must fill in either **DLG** or **SystemCall**. You fill in DLG, if **ServiceType** is *Wrapper* or *InterpretedWrapper* and if the service requests a PDM dialog with the structure "DLG=<content in this column>|..."

### 5.3.6 SystemCall

For all services of **ServiceType** *InterpretedWrapper* or *Wrapper*, you must fill in either **DLG** or **SystemCall**. Fill in SystemCall, if the you want to run a program in the server. In the column, the name of the external program is specified. The result is a PDM dialog with the structure: "DLG=SYSTEM.CALL|PROG=<content of this column>|..."

## 5.4 ServiceGui

The ServiceGui data define the use and the presentation of the services on a client. You can clearly allocate the ServiceGui to a service via their name.

### 5.4.1 Name

The name of the service for which this data record provides presentation information.

### 5.4.2 Package

This field is obsolete and must be left empty.

### 5.4.3 Extended

This field is obsolete and must be left empty.

### 5.4.4 AdditionalDataLogics

This field is obsolete and must be left empty.

### 5.4.5 ApplicationID

Application ID used for generating applications in the client. In case of editing applications, the ApplicationID is edited with the main data source of the application that you want to generate.

### 5.4.6 ApplicationTitle

Language key for the title of the generated application. In case of editing applications, the ApplicationTitle is edited with the main data source of the application that you want to generate.

### 5.4.7 ApplicationHelpFile

File name of help file (including file extension) of the generated applications. In case of editing applications, the ApplicationHelpFile is edited with the main data source of the application that you want to generate.

The name of the help file should be independent of the technology of a used client. The client should therefore put a prefix in front of the file name. You can then design the help file displayed according to the client's technology.

Example for the client MOC: In ApplicationHelpFile, you enter "Article.pdf". The client MOC then loads the document "MOC\_Article.pdf" as online help. The client automatically uses the prefix "MOC\_".

### 5.4.8 ApplicationHelpIndex

Bookmark that is activated when Help is opened. In the main application, it is usually "Overview". You must only edit this bookmark for the main data source of the application that you want to generate.

## 5.4.9 Description

### 5.4.9.1 General

Language key for short description of service.

You can show this description on the client when the selection of services is displayed.

### 5.4.9.2 Processing in the MOC client

The MOC shows the description if you add a data source while configuring an application.

## 5.5 ServiceParameter

ServiceParameters specify the parameters of a service. They provide information on the data source and value ranges.

The service parameters include selection criteria and the columns of the result set. A service parameter can be a selection criterion or be included in the result set. The attributes described below specify if a service parameter is used as selection criterion and/or is included in the result set.

### 5.5.1 Acronym

Name of the parameter. The combination of **Acronym** and **ResultSet** must be unique for each service.

### 5.5.2 ResultSet

If the associated service returns more than one **ResultSet**, a name must be indicated here. This way, you can return results in parallel that have been calculated at the same time but have a different structure. The combination of **Acronym** and **ResultSet** must be unique for each service.

### 5.5.3 WebServiceType

Data type of the parameter (*decimal, integer, string, boolean, binary, datetime*). This value must be identical to the configured value of the property configuration. **IMPORTANT:** *binary* parameters are not supported by default. You can only use these parameters in user exits.

### 5.5.4 DefaultValue

Specifies a service default value for a parameter.

### 5.5.5 IsResult

Specifies whether this service parameter is part of the ResultSet (return value). If you want to use the **DefaultValue**, do not set this field (IsResult).

In case of services of type InterpretedWrapper, you must only set the column IsResult to "Y" for UPDATE, LOCK, UNLOCK, DELETE, INSERT and COPY, if the BAPI actually returns a value, e.g. a new internal\_id when you create new data records.

### 5.5.6 IsDynamicResult

Required for the generation of the Java function (for dynamic ResultSets, the column number must automatically be extended to the fixed number). Missing columns are added as empty columns (i.e. these columns are not computed).

## 5.5.7 InputAsArray

The client must transfer values in form of an array. **InputAsArray** is only reasonable in case of a quantity input parameter, i.e. if at least one of the two columns, **IsSpecialParameter** and **IsFilterParameter**, is set and a quantity operator such as BETWEEN or IN is possible.

Specify if a field is an array or not (with filters always yes except for Boolean type).

If true and no array or empty, then exception. Is currently only verified in case of mandatory special parameters.

## 5.5.8 IsSpecialParameter

Specifies whether or not the parameter is a special type controlling the service functionality (i.e. is not a filter parameter). For the **ServiceType Wrapper**, this is the only possible parameter type. In case of the **ServiceType JavaService**, it represents a special parameter not directly included in the WHERE condition but with different "controlling" effects. If you want to use the *Default Value* on the server side, do not set this field. In addition to the defined special parameters of standard processing, you can also use other special parameters in user exits.

## 5.5.9 IsFilterParameter

Specifies whether it is a filter parameter. If you want to use the **DefaultValue** on the server side, do not set this field.

## 5.5.10 IsMandatory

Specifies whether it is a mandatory parameter for the service. If true and parameter is missing, an exception is thrown. Is currently only checked for special parameters.

## 5.5.11 Can\* (filter) operators

This option specifies whether the service supports the relevant filter operator for this parameter. Set the "Can\*" fields for filter parameters.

Available operators:

- **CanEqual**
- **CanLike**
- **CanBetween**
- **CanIn**
- **CanNotEqual**
- **CanLt (Can Less Than)**

- **CanLte (Can Less Than or Equal To)**
- **CanGt (Can Greater Than)**
- **CanGte (Can Greater Than or Equal To)**

For technical reasons, each operator has a second operator that you should select is a data record must be selected, if the operator is applicable or if the comparative value is NULL. The operator **CanEqual** will only return a data record in case of equal values, **CanEqualOrNull** in case of equal values or if the data record value is NULL. Accordingly, there are the following operators:

- **CanEqualOrNull**
- **CanLikeOrNull**
- **CanBetweenOrNull**
- **CanInOrNull**
- **CanNotEqualOrNull**
- **CanLtOrNull**
- **CanLteOrNull**
- **CanGtOrNull**
- **CanGteOrNull**

Especially with List Services you should make sure that generally all parameters support all operators in order to achieve the highest possible selectivity. In general, the framework supports this for Java services.

- You may only set **CanIn**, **CanBetween**, **CanBetweenOrNull** and **CanInOrNull**, if **InputAsArray** is also set.
- **CanLike** is only useful if the **WebServiceType** is *string*.
- With **WebServiceType** boolean, only **CanEqual** is useful.
- With **WebServiceType** string, all operators are possible.
- With all other types, all operators except for **CanLike** and **CanLikeOrNull** are useful.

Before you set wrappers, you must check which operators are actually supported by the PDM dialog or the system command.

## 5.5.12 HydraAcronym

With service type *InterpretedWrapper*, the HYDRA acronym is specified.

## 5.5.13 HydraResultAcronym

If the acronym of the selection criterion is different to the acronym in the result file, you can enter an acronym that is different to the HydraAcronym for the service type *InterpretedWrapper* and **ListMode=Y**.

### 5.5.14 TransferEmptyValuesToHydra

Specifies whether blank values, too, are to be transferred to the server, or whether the ID is simply omitted. "Y" => blank values are transferred, otherwise => ID is completely omitted.

Note: You must set this field for Insert and Update (editing screens). Only then, you can enter blank values and/or overwrite existing values with blank values.

### 5.5.15 HydraShiftPart

The following components are combined with the **Reference** field: Start of shift date, start of shift time, end of shift, end of shift time stamp, start of shift time stamp. These components are marked as belonging together. The column "HydraShiftPart" can include the following values:

- beginDate
- beginTime
- beginDatetime
- endTime
- endDatetime

Important: The column can only be populated if the parameter is part of a group that includes the following five components: Start of shift date, start of shift time, end of shift, end of shift time stamp, start of shift time stamp. The column must not be populated if it is only a group of three components including date, time and date + time field. In this case, ONLY populate the **Reference** column.

### 5.5.16 Reference

Is used to generate a DateTime data type from one field each for the date and the time (in seconds after midnight) and to identify the shift parameters.

### 5.5.17 TransformationType

Use this field to specify transformations for input and result parameters for List Services/wrappers (e.g. convert Bool to J/N and vice-versa or correct filtering for DateTime fields that consist of two fields in the database). For further details on this field, refer to section 5.10.

### 5.5.18 PlugName

Specifies whether the result parameter for this service is directly derived from the specified DataObject or whether it is added to the DataObject via plug.

Example:

Service A.List uses a plug of service B.List in the service parameter b. Consequently, the following configuration applies to service A.List:

ServiceParameter	DataObjectName	PlugName
a	A.List	
b	A.List	B.List

If the field *PlugName* includes a value, the Interpreter replaces the values of the *ServiceParameter* with those values of the plugged service when creating the SQL statement.

In the special case where an interpreted List Service does not use an own table but only plugs, and subsequently adds fields via user exit, these fields should state USEREXIT!



If you create new services, it is recommended to avoid plugs and to provide data directly via the DataObject via Join. Dependencies between several services are thus avoided.

### 5.5.19 DBField

Database field that you use to make a selection. Write the database field in lower case. You can either enter simply the field name or (for complex expressions) the expression with placeholders for the alias (e.g. `hydadm.get_datetime(%1$s.bearb_date,%1$s.bearb_time)` or `{fn substring(%1$s.field,2,1)}`).

Proceed as follows for joins to other tables:

Entry: <ALIAS>.<DBfield>

Example:

DB field: STA1.status\_bez

Acronym: gage.status.designation

Table: caq\_status (STA1)

Conditions: status\_typ = 'PMSTATUS', status\_nr = status

### 5.5.20 DBAlias

The alias for the table that is used to select the value for the acronym.

### 5.5.21 DBTabelle

The table that is used to select the value for the acronym.

### 5.5.22 DBFieldAlternative

If you cannot use the **DBField** because the **ConditionalFieldKey** is not applicable, you use the **DBFieldAlternative**.

You can enter a number, 'null', 'string', {fn ...} or another field / subselect. If it is another field or subselect, you MUST enter %1\$s for the alias of the table.

If **DBFieldAlternative** is empty, but you require an alternative field, NULL is selected.

### 5.5.23 DataObjectName

If a service uses several data sources to identify its data, you can store the data source (= DataObject = DO) that issues the result parameter in this field. For example: A service includes the parameters a, b and c:

- a is computed,
- b is identified using data object (DO) F and
- c is identified using data object (DO) G.

For a: the field is blank. For b: the field contains F. For c: the field contains G. Is used as reference for the ...do.xml configuration.

### 5.5.24 ConditionalFieldKey

This field specifies if a DB field is only conditionally available. The ConfigurationManager checks the condition for the existence of the field. Enter the feature key of the Configuration Manager (**feature set**) in this repository field to enable the check.



If a parameter is a conditional field and the condition is not fulfilled, the entries for the MOC acronym are removed from the ComplexSelectMap and the SpecialFilterMap.

As a result, the changes in the Special Filter Map via user exits and transformation type are also lost!



## 5.5.25 Constraints

Constraints are processing parameters that are used for **ServiceType** *InterpretedBAPIService*. Constraints are structured as keys with optional values. The separator between keys is the pipe character (|). You use a semicolon to separate various values. You use the equal sign (=) to separate key and value. The general structure is as follows:

Key1=Value;Value;Value|Key2|Key3=Value|

The following constraints are available:

Constraint Key	Constraint value(s)	Description
KEY	exactly one number between 1 and 5	Define field as key including key number for hyd_lock table
SERIAL	none	Field is a SERIAL (and/or auto-increment)
SEP_DATETIME	1st parameter refers to the date field 2nd parameter refers to the time field	Allows processing of separate date and time fields
BOOL	1st parameter is the value to be entered into the DB if true 2nd parameter is the value to be entered into the DB if false 3rd parameter is the value to be entered into the DB if null (null for null) 4th parameter is the type of DB field, e.g. BOOL=J;N;null string  BOOL=1;0;null;integer	Use this to write Boolean values into a string or Integer Field.
MODIFY_TS	None	
MODIFY_BY	None	
CREATE_TS	None	
CREATE_BY	None	

## 5.6 ServiceParameterGui

The ServiceParameterGui define how ServiceParameters are displayed on the client. Use **Acronym** and **ResultSet** to clearly allocate ServiceParameterGui to a service parameter.

A number of settings exist in the `ServiceParameterGui` and in the properties. You normally use the properties to define how data is displayed on the client. You only fill the respective field in the `ServiceParameterGui` if you want to display specific services on the client in a way that is different to the settings in the properties. The `ServiceParameterGui` fields overwrite the property fields of the same name.

### 5.6.1 Acronym

Name of the parameter for which this data record provides presentation information. There must be a corresponding property for each **acronym** of a parameter.

### 5.6.2 ResultSet

See **ResultSet** with `ServiceParameter`.

### 5.6.3 Label

#### 5.6.3.1 General

The label includes a language key. Using this language key, the parameter on the user interface is identified (by default), e.g. as label text of a column header. Overwrites the value from the property configuration.

#### 5.6.3.2 Processing in the MOC client

The label is displayed as label text of a field or a column title.

### 5.6.4 Tooltip

Specifies a specific tooltip for the parameter in the service context. Entry as language key.

### 5.6.5 FormatType

Use this field to overwrite specific values of a property in relation to the service (currently **Label**, **Length**, **ControlType**, **ControlTypeMode**, **ControlDataSource**, **ControlDataSourceMode**, **ControlDataSourceResult**).

For example: If you enter *workplace.id* as **FormatType** for the parameter *resource.id*, you can define for the parameter to be a *resource.id* in this service, however its length, label and control properties are taken from *workplace.id*.

In this case (other than in case of semantic and syntactic types), the value from **FormatType** takes priority. For this reason, we have a new hierarchy:

- Value from **FormatType**
- Value from ServiceParameterGUI
- Value from Property
- Value from **SemanticType**
- Value from **SyntacticType**

## 5.6.6 ClientDefaultValue

Input fields have a **ClientDefaultValue** property. The value entered here is displayed as default value when the control is initialized. "From" and "to" values are separated by semicolons.

**Set checkbox:** If the value of this field is set to *true* during a CheckEdit, the checkbox is set after initializing.

**Preallocation of text fields with "from" and "to" values (InputAsArray):** set value1;value2 to prepopulate the 'from' and 'to' fields during a text edit.

**Date fields:** In case of date fields, the field can be preallocated with an offset. If you set default values for date fields, you must absolutely specify the type of offset. The following offsets are possible:

- h (hours)
- d (days)
- w (weeks)
- m (months)
- y (years)

The 'to' value is always **relative to the 'from' value**. The default value is always a DateTime object. The presentation depends on the output format of the relevant field.

You can put "[" and "]" in front and at the end of the relevant value to specify the start and end of a period of time. Consequently, e.g. "[0d;0d]" means that 12:00:00 AM is entered in the 'from' field today and 11:59:59 PM is entered in the 'to' field today. "[-1w;0w]" means from Monday last week up to Sunday last week.

### Examples

Current date:

0d

From today to the day after tomorrow:

0d;2d

From today to one week from today:

0d;1w

From yesterday to tomorrow:

-1d;2d

From one year ago today to one year from today:

-1y;2y

**Year shortlists:** You can configure a year shortlist by **ControlDataSource** = YearList and **ControlDataSourceMode** = Script, or even by standard "Service-ControlDataSource". In this case, you can use the following default values:

- Current year: 0y and/or currentyear
- Last year: -1y
- Following year: 1y
- 4 years ago: -4y
- Year that was current 10 months ago: y-10m → this is mostly the case when the relevant year field is used in combination with a month shortlist.

If you want to preallocate two fields (**ShowSecondControl**), you have to separate both values by semicolon, e.g. -1y;1y

**Month shortlists:** You can use the following default values for a month shortlist:

- Current month: 0m
- Last month: -1m
- Following month: 1m
- 4 months ago: -4m

If you want to preallocate two fields (**ShowSecondControl**), you have to separate both values by semicolon, e.g. -1y;1y.

### 5.6.7 IsKey

The IsKey column is very important and should be occupied for all key columns of a service, since otherwise data records cannot be clearly identified. Columns including the value 'null' may NOT be defined as keys. The IsKey columns should be identical for all services (insert, update, delete, lock, unlock, copy). These entries are important, so it is best to verify them twice.

This field specifies the positioning of the cursor after an editing operation. If the positioning option OnKeyValue is selected, the client should only request one new data row after editing. You also use values that are marked **IsKey** as selection criteria. **IsKey** must also be set for *delete*, since this data record must be deleted from the view.

**IsKey** must also be indicated for *list*. If no sorting is given in the *list*, sorting takes place according to **IsKey** fields.

Every parameter which is **IsKey** MUST always be **IsMandatory**. This rule has two exceptions:

- List service
- Wrappers with composed keys.

### 5.6.8 ShowInGrid

Specifies whether the parameter is to be displayed in tables by default.

### 5.6.9 ShowInDetail

Specifies whether the parameter is to be displayed in detail views by default.

### 5.6.10 ShowInSearch

Specifies if the parameter is to be used as selection criterion (i.e. in selection panels) by default.

### 5.6.11 ColumnCategory

#### 5.6.11.1 General

In the tabular view, the client should provide the option to summarize the columns in the table to categories. You specify a language key that is displayed as title of the summarized columns.

### **5.6.11.2 Processing in the MOC client**

The ColumnCategory is used to assign the parameter to a "strip" in the grid (table view).

## **5.6.12 Category1, Category2, Category3**

### **5.6.12.1 General**

The client processes the columns Category1, Category2, Category3 in order to group fields in applications. The grouping can be performed via tabs or frames for a group of fields. You specify a language key that is displayed as title or label text of the grouped elements.

### **5.6.12.2 Processing in the MOC client**

Category1: Assigns the parameter to a tab in the detail view.

Category2: Grouping options for detail screens.

Category3: Currently not used.

## **5.6.13 TabOrder**

You specify the order of tabs for detail views.

## **5.6.14 ColumnOrder**

You specify the order of columns in tabular views.

## **5.6.15 ShowSecondControllnSearch**

### **5.6.15.1 General**

Specifies whether a second control is to be displayed (from/to). You can use this setting with selection criteria that include a value range via the operator CanBetween, e.g. "date from/to".

### **5.6.15.2 Processing in the MOC client**

The MOC provides two adjoining fields. The label text of the second field is automatically "to". If it is a field of "date" type, you can predefine a relative date for both fields.

## **5.6.16 SearchTabOrder**

Specifies the tab sequence for the selection panel.

## 5.6.17 SearchCategory1, SearchCategory2

### 5.6.17.1 General

The client processes the columns SearchCategory1 and SearchCategory2 in order to group fields in selection panels. The grouping can be performed via tabs or frames for a group of fields. You specify a language key that is displayed as title or label text of the grouped elements.

### 5.6.17.2 Processing in the MOC client

SearchCategory1: You allocate the parameter to a tab in the selection panel.

SearchCategory2: Grouping options for the selection panel.

## 5.6.18 ControlType

Use the ControlType to specify which control should be used for the relevant parameter. The client will map the abstract type onto a specific control class. If you do not specify a type, the client uses the data type to decide on the ControlType. Possible values for the ControlType:

**CheckEdit:** Selects a Boolean value (true/false) or multiple values if a reference to a data source is given.

**ColorEdit:** Selects a color value.

**ComboBoxEdit:** Combobox with selection of values from web service or data reference.

**DateTimeEdit:** Enter a date and/or a time.

**MemoEdit:** Enter an arbitrary text.

**RadioGroup:** Selects a Boolean value (true/false) or one of multiple values if a reference to a data source is given.

**TextEdit:** Standard text input. You can add a button opening a search dialog to this control, if you add a reference to a service in **ControlDataSource**. If you enter the name of a DataLogic in **ControlParameter** and if a mapping is included in **ControlDataSourceResult**, data will be requested upon leaving the control and return values will be mapped appropriately.

## 5.6.19 ControlTypeMode

### 5.6.19.1 General

Allows for controlling the input control.

**CheckEdit:** DualState (default), TriState, J;N;J (checked;unchecked;tristate)

**ColorEdit:** none

**ComboBoxEdit:** SingleEdit, Single, Multiple (multiple selection)

**DateTimeEdit:** Date (date display), Time (time display), DateTime, RelativeDate, RelativeDateTime

**MemoEdit:** none .

**RadioGroup:** SingleColumn, SingleRow

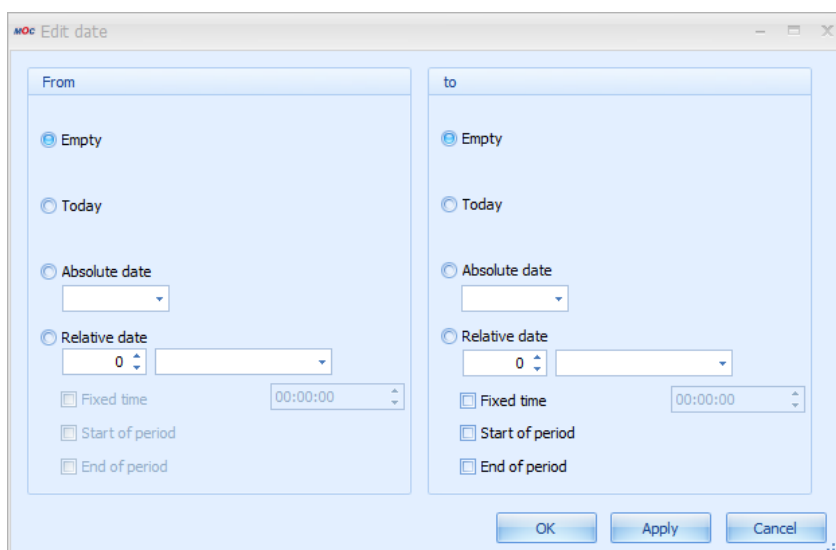
**TextEdit:**

- Empty: the search button is shown if a **ControlDataSource** is defined.
- "SearchButton": Search button is shown.
- "SearchButtonValidate": Search button is shown. If you enter an invalid value, an error is displayed.
- "OpenFileDialog": opens a file selection dialog.

## 5.6.19.2 Processing in the MOC client

If you use **DateTimeEdit** including the definition of a relative date (**ControlTypeMode:** RelativeDate or RelativeDateTime), you can enter a relative date.

If **ShowSecondControl** = true, you can predefine the complete relative value range. In this case, a button is displayed behind the second input control. You can use this button to open the following dialog:



Use this dialog to customize the values for **ClientDefaultValue** . The following entries are possible:

- **Empty:** no value is adopted



- **Today:** the current date is adopted
- **Absolute date:** you can select a fixed date value via a calendar control
- **Relative date:** you can select and adopt a date relative to the current date. In this context, "Start of period" means that you additionally go to the start of the selected period. Example: current date is 20-MAY-2010. If you select "- 1 month", 20-APR-2010 is adopted. If you also select "Start of period", the date is changed to 01-APR-2010. The same applies to "End of period". These settings are saved in the mpdvEdit or the selection profiles as **ClientDefaultValue**.

## 5.6.20 ControlParameter

See **ControlType** → **TextEdit**

## 5.6.21 ControlDataSource

Data source for the selection of values. The data source can be:

- **Web service** (configuration: **ControlType** = *ComboBoxEdit*, **ControlDataSourceMode** = *Lookup*, **ControlDataSource** = **Name** of a **ControlDataSource**. See also section "5.8 ControlDataSource")
- **ReferenceData** (configuration: **ControlType** = *ComboBoxEdit*, **ControlDataSourceMode** = *Reference*, **ControlDataSource** = **Type** of **ReferenceData**)
- **Search application** (configuration: **ControlType** = *TextEdit*, **ControlDataSourceMode** = *Lookup*, **ControlDataSource** = application name)
- **Script** (configuration: **ControlType** = *ComboBoxEdit*, **ControlDataSourceMode** = *Script*, **ControlDataSource** = Name of script)

## 5.6.22 ControlDataSourceMode

Data source mode (*Lookup*, *Reference* or *Script*).

## 5.6.23 ControlDataSourceParameter

Optional setting of parameters of a **ControlDataSource**. If you make settings here, these settings overwrite the settings in the **ControlDataSource**.

See also the description **ControlDataSource - Parameter**

## 5.6.24 ControlDataSourceResult

Optional setting of the result of a **ControlDataSource**. If you make settings here, these settings overwrite the settings in the **ControlDataSource**.

The settings in this field provide more options than the **Result** in the **ControlDataSource**:

Result columns are separated by semicolon. Field mapping:

- First entry: Value
- Second entry: Labeling
- Third entry: UnitLabel
- As of the fourth entry, the fields are mapped:
  - o Via acronym or semantic type.
  - o Field mapping: in **ControlDataSourceResult**, you can enter a mapping in the form "FieldName=ColumnFromResult" as from the fourth entry. For example, you can specify tool.id=resource.id in order to fill the field "tools.id" with the "resource.id" value from the search application. Several mappings are separated by ";" - spaces are not allowed.
  - o Asterisk mapping: Instead of mapping, you can also enter \* . Subsequently, all return columns of the search application are mapped. The mapping is performed as usual via ID or semantic type.

## 5.6.25 VisibleCondition

This value decides whether an input field is visible on the client. For customization, see EditableCondition.

## 5.6.26 EditableCondition

This value decides whether you can edit an input field on the client. There are three possibilities:

- Boolean value: In case of *TRUE* or *FALSE*, the field is always editable / non-editable.
- Binary expression:
  - o Field name must be the name of a field that is also located in the ControlPanel.
  - o Valid operators: =, <, >, <=, >=, <>, !=
  - o The value is written as a string and interpreted depending on the comparative field value.
  - o Field, operator and value must be separated by a space!
- Concatenation of binary expressions:
  - o You can concatenate an arbitrary number of binary expressions.
  - o You can use the operators "&&", "AND", "||", "OR" to link expressions.
  - o Here, too, all components of the conditions must be separated by a space.
  - o Priority of operators: "AND" or "&&" are evaluated first, then "OR" and "||".  
You cannot use brackets.
  - o Example: resource.id = 12345 && resource.costcenter = 20 || resource.id = 60610



The client assigns the default value of the property "ClientDefaultValue" to the field, if the result of an expression in the EditableCondition or the VisibleCondition changes from FALSE to TRUE. The client dynamically evaluates the expressions in the EditableCondition and the VisibleCondition, if the fields of the application change.

## 5.6.27 ScriptId

### 5.6.27.1 General

The ID of the script that is allocated to the parameter. If you set the ID, the relevant script is performed upon various events (at present EditValueChanged and Leave).

### 5.6.27.2 Processing in the MOC client

The method name of the script is ScriptId+EditValueChanged and/or ScriptId+Leave. The script can be included in any DLL that is read by the CodeManager.

## 5.7 Property

For the acronyms, properties include information on data types, input and output formats, display options, a name (that can be localized) and other settings specifying how ServiceParameters are displayed in the client. Each property has a system-wide unique acronym.

A number of settings exist in the ServiceParameterGui and in the properties. You normally use the properties to define how data is displayed on the client. You only fill the respective field in the ServiceParameterGui if you want to display specific services on the client in a way that is different to the settings in the properties. The ServiceParameterGui fields overwrite the property fields of the same name.

### 5.7.1 Acronym

Clear identification of the property across all domains.

### 5.7.2 WebServiceType

Describes the data type used to transfer the property between client and server. The currently supported WebServiceTypes are exclusively

- *binary*
- *boolean*
- *datetime*
- *decimal*
- *integer*

- string

Important: the types *\*date* and *\*time* are internal types which are not transferred.

### 5.7.3 NETType

The data type used by the client. If NETType is empty, the **WebServiceType** is used to automatically identify the data type used by the client. At present, NETType supports the following entries:

- **color:** Use *color* to convert the transferred integer into an RGB code. In this case, the conversion is implemented by the grid.
- **duration:** creates a duration from an integer.
- **image:** either creates an image from a transferred byte array or interprets a transferred string as image name. For example, the *maintenance.active.led* property is transferred as a string including the name of an icon.
- **preview:** Specifies that the contents in the client may be displayed as "preview" (similar to auto-preview outlook) (application e.g. in DevExpress grid).
- **timestamp:** Use *timestamp* to automatically create an additional column for date values in the client in order to process time and date separately.

### 5.7.4 SemanticType

Use semantic types to inherit semantic properties. The "order.id" is therefore used to identify orders (semantic meaning). The acronym *operation.order.id* includes such an order identification and therefore has the semantic type *order.id*. If an attribute of the property is not set (empty), the respective value from the semantic type is used for the processing in the client.

For example: You must set the semantic type if you want to adopt a value from a lookup screen in the field. For the *workplace* field, enter e.g. *resource.id* as semantic type in order to adopt the selected workplace from a search screen for workplaces. Refer to the description of the SyntacticType for further information on the priority used to specify the attributes of a Property, the SemanticType and the SyntacticType.

### 5.7.5 SyntacticType

You mainly use a syntactic type for a uniform presentation of the different properties. The syntactic type does not include any semantic content. For example: The properties *booking.begin\_ts* and *booking.shift.start\_ts* have different semantic meanings, but are presented in a uniform format that can be controlled centrally.

Syntactic types are used to control the characteristics of a *Property*: for example length, input and output screen, tooltip, label, etc. To select the valid value for a characteristic, the client proceeds as follows:

- If the characteristic (e.g. length) is set in Property, the client uses this value.
- Or: If a semantic type is available and the characteristic is set, the client uses this value.
- Or: If a syntactic type is available and the characteristic is set, the client uses this value.

Note:

- You must always enter a **description** for syntactic and semantic types.
- Syntactic types can reference other syntactic types so that "inheritance hierarchies" can be created.
- Create syntactic types as property of the *SyntacticType* domain.
- Semantic types are usually "real" properties of a "normal" domain that are used as semantic type at other places.

## 5.7.6 Label

### 5.7.6.1 General

The label includes a language key. Using this language key, the parameter on the user interface is identified (by default), e.g. as label text of a column header. Overwrites the value from the property configuration.

### 5.7.6.2 Processing in the MOC client

The label is displayed as label text of a field or a column title.

## 5.7.7 DefaultTooltip

Specifies the default tooltip for the property as language key.

## 5.7.8 UnitLabel

Text key for unit. The unit is displayed to the right of the input field.

## 5.7.9 OutputFormat

This field specifies the format that is used to display a value (e.g. for date or quantity values). If you do not enter an **InputFormat** in the repository, the MOC tries to develop an appropriate format from the **OutputFormat**. Enter the value **InputFormat** in the repository only if special masking is required. Find further details in section "5.7.12 Rules for the input/output formatting".

## 5.7.10 InputFormat

Equivalent to OutputFormat. You can enter a valid regular expression in the field InputFormat. Other entries that are not regular expressions are not permissible. Find further details in section 5.7.12.

## 5.7.11 Length

The client shows the control for this acronym in the specified width (i.e. the specified number of characters). With Length=0, the control uses the entire width available. If a width is specified but the space available is not sufficient, the control is cut off.

This field also specifies the number of characters that you can enter in an input field with **ControlType = TextEdit**, if no other **InputFormat** is specified.

## 5.7.12 Rules for the input/output formatting

### Overview

In the repository, you define the formatting of the data output and the input dialogs to edit data. The "Properties" of the different acronyms include an **OutputFormat** and an **InputFormat**. The **OutputFormat** defines formatting if you display a value.

**Important:** If you do not enter an **InputFormat** in the repository, the MOC uses the **OutputFormat** to generate an appropriate formatting. Enter the value InputFormat in the repository only if special masking is required.



In case of strings, you cannot enter the special characters asterisk (\*) and pipe (|), if you have not defined any input format. As you use these two special characters as separator and control character, they can cause problems if they are written in the database.



With strings, the maximum number of characters that you can enter is defined by the attribute **Length**, if no other input format is defined.

### Syntactic types

The Properties provide so-called "syntactic types" in order to make groups (similar to field types in Delphi). Syntactic types have the same properties as real properties. The real properties have a syntactic type. For example, if the **output format** of the syntactic type includes a value, this value is used wherever this syntactic type is entered.

### Example: Industrial minutes

The syntactic type "Durations" has the format {0:mpdv\_timespan}. With the different properties showing durations, "Durations" is entered in the column **SyntacticType** and no entries are made in the columns "output format" and "input format". When the property is read - and if no **output format** is available in the property - the format of the syntactic type is used.

If a system displays industrial minutes (no standard function!) and if the syntactic type "Durations" is specified, the **output format** is automatically changed from {0:mpdv\_timespan} to {0:mpdv\_industrialMinutes}. As a result, all formats including the syntactic type "Durations" are shown in industrial time units.

Times and durations are internally stored in the system as integer seconds. If you convert times or durations during input or output formatting to formats other than hours, minutes and seconds (HH:MM:SS), the conversion may not be possible without losses. For example, this applies to the use of the "mpdv\_calc" format and the classic industry minute display:



When converting from seconds to hours (division by 3600), decimal numbers with an infinite number of decimal places can occur, which inevitably have to be rounded when displayed on the client. Example: 20 minutes = 1200 seconds = 0.333333... hours. If the value is rounded to three decimal places, you calculate backward as follows:  $0.333 \times 3600 = 1198.8$  seconds. Depending on how the client rounds, the internal value is then no longer 1200 seconds, but 1999 or 1998 seconds.

If you use less than three decimal places, the conversion error gets even greater:

The system recorded a duration of 123 seconds. The client displays 0.03 hours. If you calculate backwards, the result is 108 seconds.

## Output formats

OutputFormat	Automatically created masking (input format)	Examples	Description
<b>Numeric data</b>			
f(number)	None	f3, f1	Numeric value without thousands separator. The number specifies the number of decimal places.
n(number)	None	n0, n2, n5	Numeric value with thousands separator. The number specifies the number of decimal places, even if the data type to be displayed is an integer type. In case of n0, no decimal places will be shown.
#.##) , #.(0)	None	#.####, #.0000	Arbitrary format
{0:mpdv_timespan}	None	2:33:30	MPDV format provider. Conversion of seconds to hh:mm:ss and vice-versa.

{0:mpdv_timespan_short}	None	2:33	MPDV format provider. Conversion of seconds to hh:mm and vice-versa.
{0:mpdv_timespan_minutes}	None	45	MPDV format provider. Conversion of seconds to minutes and vice-versa.
{0:mpdv_cycletime}	None	1:30:00	MPDV format provider. Hours per 1000 pieces. Conversion into seconds and vice versa.
{0:mpdv_te}	None	2.00	MPDV format provider. Hours per 1000 pieces. Conversion into seconds and vice versa.
<b>Strings</b>			
empty	[^*]]*		Illegal characters begin with ^. In this example * and   <b>* No limitation in length</b>
empty	[^*]]{0.10}		Illegal characters begin with ^. Max. length: 10 characters
empty	[0-9a-fA-F]		Allowed characters 0 through 9, a through f, A through F.
<b>Special formats</b>			
{0:mpdv_cycletime_sec_cycle}	None	29 sec/cycle	MPDV format provider. Seconds per cycle. Conversion into seconds per 1000.
{0:mpdv_IndustrialMinutes}	None	1.50	MPDV format provider. Conversion of seconds into industrial minutes and vice versa.
{0:mpdv_leadingzeros_order}	ORDER		You must combine this output format with the input format ORDER. The combination is used in the syntactic type "order_id". The basic settings are used to automatically specify the length.
{0:mpdv_leadingzeros_operation}	ORDER		You must combine this output format with the input format ORDER. The combination is used in the syntactic type "operation". The basic settings are used to automatically specify the length.
{0:mpdv_leadingzeros_sequence}	ORDER		You must combine this output format with the input format ORDER. The combination is used in the syntactic type "ordersequence_id". The basic settings are used to automatically specify the length.

## Input formats

The following definitions are available for the input format:

- Leave empty: The input format is implicitly defined using the output format. See table above.
- Use of logical input formats
- Use of regular expressions



## Logical input formats

To simplify the definition of input formats and limit the variety of entries in the repository, the logical input formats are provided. These input formats are permanently implemented in the client and can directly be used in the repository. Input formats are customized in the properties. But service parameters specify whether wildcards are allowed. For this reason, the input format actually used can vary depending on the allocated service.

In order to use logical input formats, define the name of the input format in the affected property in the repository. The following formats are currently available:

Name	Input format without wildcard	Input format with wildcard
CHARACTER	[^*][LENGTH]	[^][LENGTH]
NUMBER_N0	[0-9][LENGTH]	[0-9][LENGTH]
NUMBER_N1	[0-9][LENGTH]\R.[0-9]{0,1}	[0-9][LENGTH]\R.[0-9]{0,1}
NUMBER_N2	[0-9][LENGTH]\R.[0-9]{0,2}	[0-9][LENGTH]\R.[0-9]{0,2}
NUMBER_N3	[0-9][LENGTH]\R.[0-9]{0,3}	[0-9][LENGTH]\R.[0-9]{0,3}
NUMBER_N6	[0-9][LENGTH]\R.[0-9]{0,6}	[0-9][LENGTH]\R.[0-9]{0,6}
TIMESPAN_SHORT	[0-9][LENGTH]\R:[0-9]{2,2}	[0-9][LENGTH]\R:[0-9]{2,2}
TIMESPAN	[0-9][LENGTH]\R:[0-9]{2,2}\R:[0-9]{2,2}	[0-9][LENGTH]\R:[0-9]{2,2}\R:[0-9]{2,2}
ORDER	[0-9a-zA-Z.+][LENGTH]	[0-9a-zA-Z.+*][LENGTH]

The placeholder [LENGTH] is replaced with the configured field length at runtime. If the defined length is '0', an '\*' is entered. With the logical format "ORDER", the system automatically changes the [LENGTH] according to the basic settings when the output format changes.

## Input/Output formats including calculation

If you specify the **output format** *mpdv\_calc*, you can include calculations in the formatting. In the format, you can specify a divisor and multiplier and an identifier that specifies if a reciprocal is calculated. You can also specify the number of decimal places. The **OutputFormat** *mpdv\_calc* implicitly defines the **InputFormat**. If an input of values is made, the reciprocal value is calculated.

Example: "mpdv\_calc;MULT=5;DIV=2;INVERSE=false;FORMAT=n3" (the value is multiplied by 5, divided by 2, then the reciprocal is calculated and the result is displayed with 3 decimal places).

The input/output format including calculation is normally used for the display of cycle times or specifications of single pieces. In the database, these times are always saved in seconds per 1000 pieces. If an input/output format including calculation is used, you can convert the times to hours per 1000 pieces, minutes per piece or with reciprocal also to piece per hour.

## Overview of regular expressions

You can find a large amount of information on regular expressions using the search engines on the internet. In the following, the most important aspects are presented.

## Meta characters

Represent a range of characters.

Character	Description
.	Matches any character.
[ <i>aeiou</i> ]	Matches any single character included in the specified set of characters.
[^ <i>aeiou</i> ]	Matches any single character, which is not included in the specified set of characters.
[ <i>0-9a-fA-F</i> ]	Use of a hyphen (–) allows specification of contiguous character ranges.
\R.	Matches the decimal separator specified by the System.Globalization.NumberFormatInfo.NumberDecimalSeparator property of the current culture.
\R:	Matches the time separator specified by the DateTimeFormatInfo.TimeSeparator property of the current culture.

## Quantifier

Repetition, number of characters

Quantifier	Description	Samples
*	Specifies zero or more matches.	The "\w*" mask matches a string consisting of zero or more letter characters. It's equivalent to the "\w{0,}" mask.
+	Specifies one or more matches.	The "\w+" mask matches a string consisting of one or more letter characters. It's equivalent to the "\w{1,}" mask.
?	Specifies zero or one match.	The "\w?" mask matches zero or one letter character. It's equivalent to the "\w{0,1}" mask.
{ <i>n</i> }	Specifies exactly <i>n</i> matches.	The "\d{4}" mask matches exactly four digits.
{ <i>n</i> ,}	Specifies at least <i>n</i> matches.	The "\d{2,}" mask matches two or more digits.
{ <i>n</i> , <i>m</i> }	Specifies at least <i>n</i> , but no more than <i>m</i> , matches.	The "\d{1,3}" mask matches either one, or two, or three digits.

## Special characters

Special characters

Character	Description	Samples
	Alternation symbol. This can be used to implement a choice between two or more alternatives.	The "1 2 3" mask matches either "1" or "2" or "3". The "abc 123" mask matches either "abc" or "123". The "\d{2} \p{L}{2}" mask matches either two digits or two letters.
()	Grouping. You can use parentheses to create sub-expressions, or to limit the scope of the alternation.	The "(an ba)t" mask matches either "ant" or "bat". The "(net)+" mask matches "net", "netnet", "netnetnet", ... strings. Compare with the "net+" mask which matches the "net", "nett", "nettt", ... strings. The "(0 1)+" mask matches a string of indeterminate length, consisting of "0" and "1".

## Examples

Input 1..9999 => Input format for property : ([1-9][1-9][0-9][1-9][0-9][0-9][1-9][0-9][0-9][0-9])

Input 0..999 => Input format for property : ([0-9][1-9][0-9][0-9])

## Best practice: input of long string fields

The client identifies the width of an input field using the attribute **Length**. In case of long string fields with more than 20 characters, the layout can become confusing because these string fields use the complete width of the layout and are very long compared to other input fields. Very long string fields are cut off on the right-hand side, if the available space is not enough. To avoid this behavior, you can control the displayed field width regardless of the number of characters that you can enter.

- Use the attribute **Length** to specify the **width** of the input field.
- You can use a regular expression in the **InputFormat** to specify the **number of characters that you can enter**.

If you enter strings that are larger than the displayed field, the input field automatically scrolls horizontally.

Examples:

Attribute	Length	InputFormat	Effect
article.designation	50	.{0.250}	The field is displayed with a width of 50 characters. You can enter up to 250 characters.
operation.input_component_list	25	.{0.40}	The field is displayed with a width of 25 characters. You can enter up to 40 characters.

### 5.7.13 FillChar

Obsolete. This field must be left empty.

### 5.7.14 Calculation

Obsolete. This field must be left empty.

### 5.7.15 Further fields see ServiceParameterGui

For a description of the following fields, refer to the data types of the ServiceParameterGui:

ControlType, ControlTypeMode, ControlParameter, ControlDataSource, ControlDataSourceMode, ControlDataSourceParameter, ControlDataSourceResult, VisibleCondition and EditableCondition.

## 5.8 ControlDataSource

A ControlDataSource defines a data source that you can use to fill selection lists in controls, for example. These can be data logics (service requests) or reference values (see also ReferenceData).

Reference values are usually required to fill selection lists (and/or RadioGroups) with static contents.

You use data logics to request services that identify selection lists (or RadioGroups) dynamically. For example, these lists can include master data that are configured in the database.

The settings made in the columns **Parameter** and **Result** can be overwritten in a **Property** or **ServiceParameterGui**.

### 5.8.1 Name

Name of the ControlDataSource. The name should be composed of English terms clearly describing the data source. You usually use the camelCase notation.

### 5.8.2 Source

If the data source is a web service, this field contains the name of the client's data logic. You derive the data logic from the service name. To do so, remove the dot between domain and function and use a capital letter for the first letter of the function:

Service	Data logic
MDUser.list	MDUserList
MDUnits.list	MDUnitsList

In case of reference values, this field includes the **Type** of a **ReferenceData**.

### 5.8.3 Parameter

A list of parameters. The list does not include spaces, use semicolons to separate parameters. This field is only allowed in combination with web service data sources. A parameter can be allocated dynamically or permanently.

Permanent parameters appear as <acronym>=<value>, e.g.

"dialogconfiguration.type=AIPDEF;dialogconfiguration.type=AIPTNR".

Dynamic parameters are specified as a pair of <acronym1>=[<acronym2>]. e.g.

"resource.id=[resource.id];pdvprocessparameter.evaluation\_ts=[pdvsinglevalue.evaluation\_ts]"

The acronym in square brackets is replaced with the acronym values from the ControlPanel.

## 5.8.4 Columns

A list of requested columns. The list does not include spaces. To separate columns, semicolons are used. This is only permissible for web service data sources.

## 5.8.5 Result

You can enter 1-n acronyms separated by semicolon. The sequence used specifies the importance.

- Position 1 (Value): Name of acronym whose value is entered in the input field.
- Position 2 (ControlValue): Name of acronym whose value is displayed in the selection list. If you do not specify position 2, the acronym of position will be displayed.
- Position 3 (LabelValue): If you specify position 3, the value of the acronym is entered in the label field of the input field and also displayed in the selection list.
- Position 4-n: Use these positions to define additional return values, which are then used to update "dependent" controls in the client ("lookup").

Only with web service data sources:

Optional return columns of the data source, separated by semicolons. Without spaces. The return has the format <acronym>=<value> - for acronym pairs, the second acronym is therefore replaced with the result value (e.g. if you enter "operation.resource.id=resource.id", this results in "operation.resource.id=4711").

## 5.9 ReferenceData

Reference values are usually required to fill selection lists (and/or RadioGroups) with static contents. In contrast to values provided by web services, reference values are fixed and do not change. For this reason, reference values can be entered once in a list and are delivered in this form.

### 5.9.1 ref\_data\_key

The **ref\_data\_key** must be unambiguous for each entry. In special cases, this key is used in the source code (at least in the server).

Usually, the **ref\_data\_key** is composed of **type** + : + **db\_key**; this facilitates its allocation to type and key. An exception occurs if the **db\_key** includes a German expression. The **ref\_data\_key** must then be formed differently. For example, *pwdexclusion:person.firstname* is a super **ref\_data\_key** for the type *pwdexclusion.pwd* and **db\_key** *PNR.PVORNAME*.

## 5.9.2 Type

Use this field to summarize various ReferenceData entries to a list.

## 5.9.3 db\_key

The **db\_key** is the actual value that is selected in the list. This key identifies an entry unambiguously within a **Type**. You cannot freely select the key because the key is often transferred to services and can correspond to the content of a configuration identifier in the database, for example.

## 5.9.4 is\_default

The entry with this key is preallocated as default.

## 5.9.5 Designation

Text displayed in the selection list. A language key is specified.

## 5.9.6 sort\_key

Specifies the sequence that is used to display the entries in the selection list.

# 5.10 Authorization

The authorization mechanism

- protects applications and functions against unauthorized use on the client,
- hides fields or field groups on the GUI,
- prevents these fields from being edited.

## 5.10.1 Authorization type

Controls the type of authorization. Possible values:

- Acronym: enables the authorization of individual fields (properties)
- AcronymGroups: enables the authorization to group fields
- Application: enables the authorization of applications
- Functions: enables the authorization of functions which are e.g. requested from the application toolbar.

## **5.10.2 Authorization Context**

Context where the authorization is intended. If the field is left empty, authorization is always granted, irrespective of the context. You normally use this field to control the authorization of acronyms in the context of special services.

## **5.10.3 Authorization ID**

Identifies the object to be authorized, i.e. the name of the acronym or the ID of an application.

## **5.10.4 Authorization key**

The authorization key that is used to protect the object.

## **5.10.5 Authorization Designation**

(Optional) text description of the authorization.